

# Low Power Stereo Codec with Headphone Amp

The SGT5000 is a Low Power Stereo Codec with Headphone Amp from Freescal, and is designed to provide a complete audio solution for portable products needing LINEIN, mic-in, LINEOUT, headphone-out, and digital I/O. Deriving it's architecture from best in class, Freescal integrated products that are currently on the market. The SGT5000 is able to achieve ultra low power with very high performance and functionality, all in one of the smallest footprints available. Target markets include portable media players, GPS units, and smart phones. Features such as capless headphone design and an internal PLL, help lower overall system cost. This device is powered by SMARTMOS technology.

## Features

### Analog Inputs

- Stereo LINEIN - Support for external analog input
- Stereo LineIn - Codec bypass for low power
- MIC bias provided
- Programmable MIC gain
- ADC - 85 dB SNR (-60 dB input) and -73 dB THD+N ( $V_{DDA} = 1.8\text{ V}$ )

### Analog Outputs

- HP Output - Capless design
- HP Output - 45 mW max into 16 ohm load @ 3.3 V
- HP Output - 100 dB SNR (-60 dB input) and -80 dB THD+N ( $V_{DDA} = 1.8\text{ V}$ , 16 ohm load, DAC to headphone)
- LINEOUT - 100 dB SNR (-60 dB input) and -85 dB THD+N ( $V_{DDIO} = 3.3\text{ V}$ )

### Digital I/O

- I<sup>2</sup>S port to allow routing to Application Processor

### Integrated Digital Processing

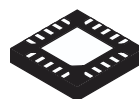
- Freescal surround, Freescal bass, tone control/ parametric equalizer/graphic equalizer clocking/control
- PLL allows input of an 8.0 MHz to 27 MHz system clock - standard audio clocks are derived from PLL

### Power Supplies

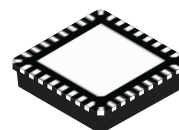
- Designed to operate from 1.62 to 3.6 volts

## SGT5000

### AUDIO CODEC



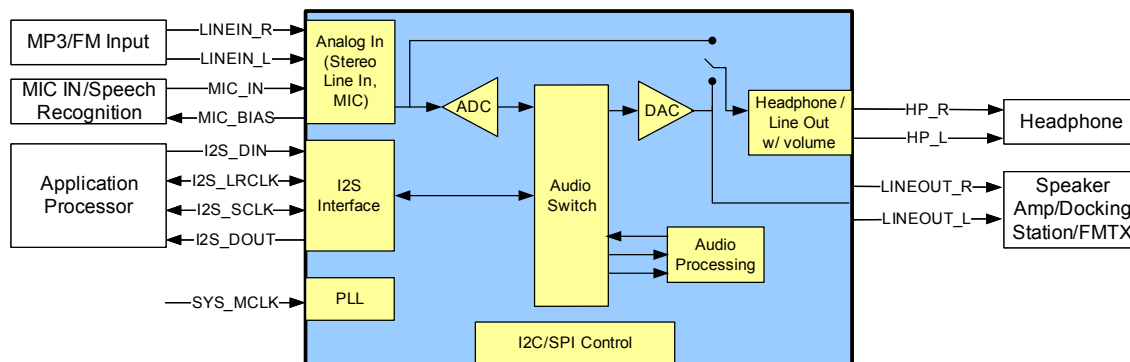
**PB-FREE**  
**98ARE10742D**  
**20-PIN QFN**



**PB-FREE**  
**98ARE10739D**  
**32-PIN QFN**

### ORDERING INFORMATION

| Device          | Temperature Range (T <sub>A</sub> ) | Package |
|-----------------|-------------------------------------|---------|
| SGT5000XNLA3/R2 | -40 to 85 °C                        | 20 QFN  |
| SGT5000XNAA3/R2 |                                     | 32 QFN  |



Note: SPI is not supported in the 3.0 mm x 3.0 mm 20-pin QFN package

**Figure 1. SGT5000 Simplified Application Diagram**

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## INTERNAL BLOCK DIAGRAM

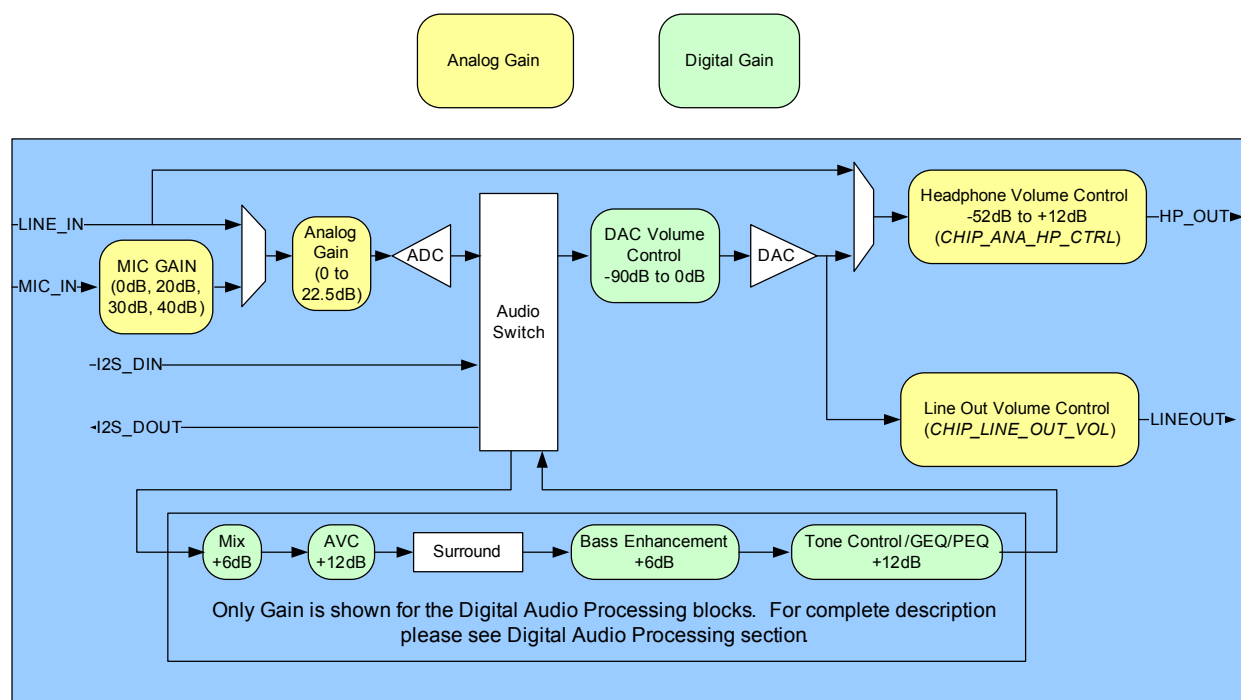


Figure 2. SGT5000 Simplified Internal Block Diagram

## PIN CONNECTIONS

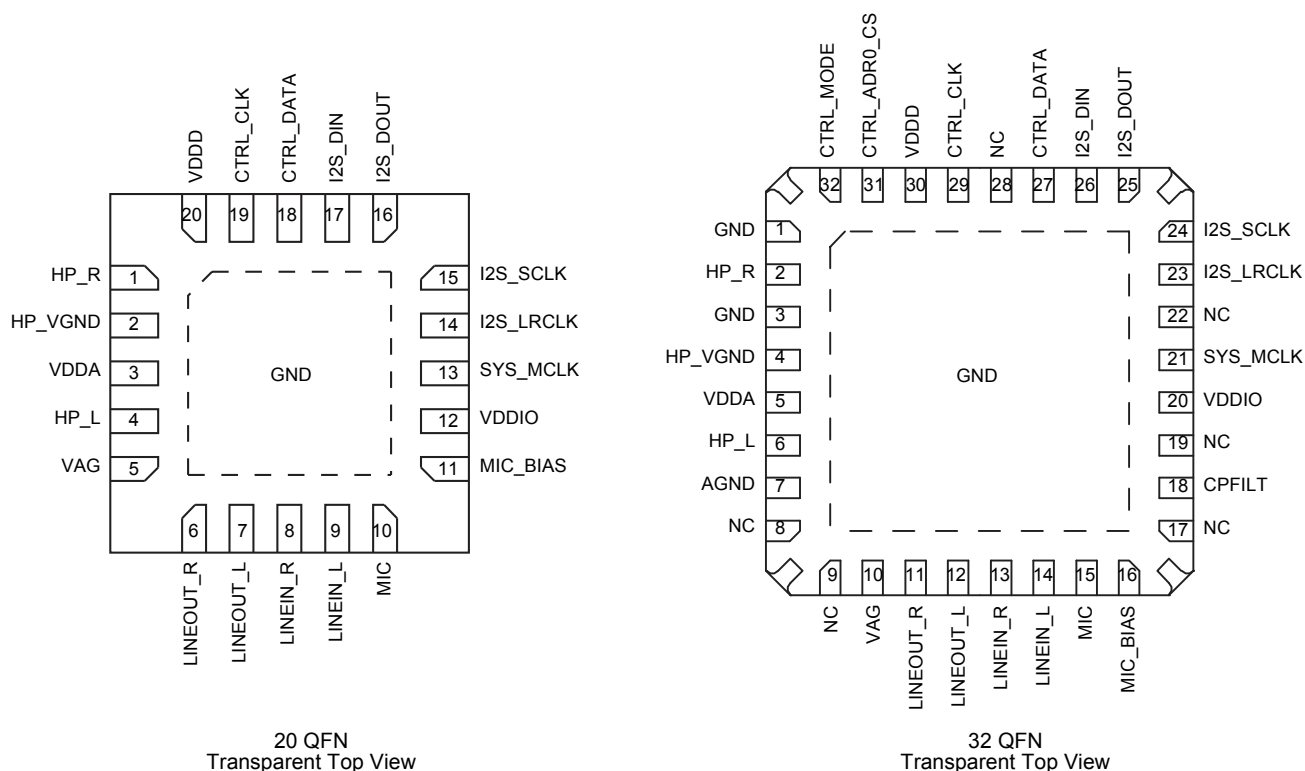


Figure 3. SGTL5000 Pin Connections

A functional description can be found in [Functional Description](#), beginning on [page 12](#).

Table 1. SGTL5000 Pin Definitions

| 20 Pin QFN | 32 Pin QFN           | Pin Name  | Pin Function  | Formal Name              | Definition                                                                                                                                                                                                        |
|------------|----------------------|-----------|---------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1          | 2                    | HP_R      | Analog        | Right headphone output   |                                                                                                                                                                                                                   |
| 2          | 4                    | HP_VGND   | Analog        | Headphone virtual ground | Do not connect HP_VGND to system ground, even when unused. This is a virtual ground (DC voltage) that should never connect to an actual "0 Volt ground". Use the widest, shortest trace possible for the HP_VGND. |
| 3          | 5                    | VDDA      | Power         | Analog voltage           |                                                                                                                                                                                                                   |
| 4          | 6                    | HP_L      | Analog        | Left headphone output    |                                                                                                                                                                                                                   |
| -          | 7                    | AGND      | Analog Ground | Ground                   |                                                                                                                                                                                                                   |
| -          | 8, 9, 17, 19, 22, 28 | NC        | No Connect    |                          |                                                                                                                                                                                                                   |
| 5          | 10                   | VAG       | Analog        | DAC VAG filter           |                                                                                                                                                                                                                   |
| 6          | 11                   | LINEOUT_R | Analog        | Right LINEOUT            |                                                                                                                                                                                                                   |
| 7          | 12                   | LINEOUT_L | Analog        | Left LINEOUT             |                                                                                                                                                                                                                   |
| 8          | 13                   | LINEIN_R  | Analog        | Right LINEIN             |                                                                                                                                                                                                                   |
| 9          | 14                   | LINEIN_L  | Analog        | Left LINEIN              |                                                                                                                                                                                                                   |

**Table 1. SGT5000 Pin Definitions (continued)**

| 20 Pin QFN | 32 Pin QFN   | Pin Name     | Pin Function | Formal Name                                                                                                                             | Definition                                                                                                                                                                                             |
|------------|--------------|--------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10         | 15           | MIC          | Analog       | Microphone input                                                                                                                        |                                                                                                                                                                                                        |
| 11         | 16           | MIC_BIAS     | Analog       | Mic bias                                                                                                                                |                                                                                                                                                                                                        |
| —          | 18           | CPFILT       | Analog       | Charge Pump Filter                                                                                                                      | The CPFILT cap value is 0.1 $\mu$ F. If both VDDIO and VDDA are $\leq 3.0$ V, the CPFILT pin must be connected to a 0.1 $\mu$ F cap to GND. If either is $> 3.0$ V, the CPFILT cap MUST NOT be placed. |
| 12         | 20           | VDDIO        | Power        | Digital I/O voltage                                                                                                                     |                                                                                                                                                                                                        |
| 13         | 21           | SYS_MCLK     | Digital      | System master clock                                                                                                                     |                                                                                                                                                                                                        |
| 14         | 23           | I2S_LRCLK    | Digital      | I <sup>2</sup> S frame clock                                                                                                            |                                                                                                                                                                                                        |
| 15         | 24           | I2S_SCLK     | Digital      | I <sup>2</sup> S bit clock                                                                                                              |                                                                                                                                                                                                        |
| 16         | 25           | I2S_DOUT     | Digital      | I <sup>2</sup> S data output                                                                                                            |                                                                                                                                                                                                        |
| 17         | 26           | I2S_DIN      | Digital      | I <sup>2</sup> S data input                                                                                                             |                                                                                                                                                                                                        |
| 18         | 27           | CTRL_DATA    | Digital      | I <sup>2</sup> C Mode: Serial Data (SDA); SPI Mode: Serial Data Input (MOSI)                                                            |                                                                                                                                                                                                        |
| 19         | 29           | CTRL_CLK     | Digital      | I <sup>2</sup> C Mode: Serial Clock (SCL); SPI Mode: Serial Clock (SCK)                                                                 |                                                                                                                                                                                                        |
| 20         | 30           | VDDD         | Digital      | Digital voltage                                                                                                                         | For new designs, connect VDDD to an external voltage source and to a 0.1 $\mu$ F capacitor to GND.                                                                                                     |
| -          | 31           | CTRL_ADR0_CS | Digital      | I <sup>2</sup> C Mode: I <sup>2</sup> C Address Select 0; SPI Mode: SPI Chip Select                                                     |                                                                                                                                                                                                        |
| -          | 32           | CTRL_MODE    | Digital      | Mode select for I <sup>2</sup> C or SPI; When pulled low the control mode is I <sup>2</sup> C, when pulled high the control mode is SPI |                                                                                                                                                                                                        |
| PAD        | 1, 3, 4, PAD | GND          | Ground       | Ground                                                                                                                                  | The PAD must be soldered to ground.<br>Star the ground pins of the chip, VAG ground, and all analog inputs/outputs to a single point, then to the ground plane.                                        |

## ELECTRICAL CHARACTERISTICS

## MAXIMUM RATINGS

**Table 2. Maximum Ratings**

Exceeding the absolute maximum ratings shown in the following table could cause permanent damage to the part and is not recommended. Normal operation is not guaranteed at the absolute maximum ratings, and extended exposure could affect long term reliability.

| Ratings                              | Symbol     | Value                     | Unit |
|--------------------------------------|------------|---------------------------|------|
| <b>ELECTRICAL RATINGS</b>            |            |                           |      |
| Maximum Digital Voltage              | $V_{DDD}$  | 1.98                      | V    |
| Maximum Digital I/O Voltage          | $V_{DDIO}$ | 3.6                       | V    |
| Maximum Analog Supply Voltage        | $V_{DDA}$  | 3.6                       | V    |
| Maximum voltage on any digital input |            | GND-0.3 to $V_{DDIO}+0.3$ | V    |
| Maximum voltage on any analog input  |            | GND-0.3 to $V_{DDA}+0.3$  | V    |
| <b>THERMAL RATINGS</b>               |            |                           |      |
| Storage Temperature                  | $T_{STG}$  | -55 to 125                | °C   |
| Operating Temperature<br>Ambient     | $T_A$      | -40 to 85                 | °C   |

**Table 3. Recommended Operating Conditions**

| Ratings                                                                                      | Symbol     | Value       | Unit |
|----------------------------------------------------------------------------------------------|------------|-------------|------|
| Digital Voltage (If supplied externally). External VDDD connection required for new designs. | $V_{DDD}$  | 1.1 to 2.0  | V    |
| Digital I/O Voltage                                                                          | $V_{DDIO}$ | 1.62 to 3.6 | V    |
| Analog Supply Voltage                                                                        | $V_{DDA}$  | 1.62 to 3.6 | V    |

**Table 4. Input/Output Electrical Characteristics**

Test Conditions unless otherwise noted:  $V_{DDIO} = 3.3\text{ V}$ ,  $V_{DDA} = 3.3\text{ V}$ ,  $T_A = 25\text{ }^{\circ}\text{C}$ , Slave mode,  $f_S = 48\text{ kHz}$ ,  $MCLK = 256 f_S$ , 24 bit input.

| Characteristic                                                                                             | Symbol | Min  | Typ        | Max            | Unit     |
|------------------------------------------------------------------------------------------------------------|--------|------|------------|----------------|----------|
| LINEIN Input Level (3.3 V VDDA)                                                                            |        | -    | -          | 2.83           | $V_{PP}$ |
| LINEIN Input Level (1.8 V VDDA)                                                                            |        | -    | -          | 1.60           | $V_{PP}$ |
| MIC Input Level (3.3 V VDDA)                                                                               |        | -    | -          | 2.83           | $V_{PP}$ |
| MIC Input Level (1.8 V VDDA)                                                                               |        | -    | -          | 1.60           | $V_{PP}$ |
| LINEOUT Output level<br>0 dBFS at 1.031 kHz 12S input, 1.8 V LINEOUT supply (normally VDDIO), 10 kOhm load |        | 1.46 | 1.52       | 1.68           | $V_{PP}$ |
| LINEOUT Output level<br>0 dBFS at 1.031 kHz 12S input, 3.3 V LINEOUT supply (normally VDDIO), 10 kOhm load |        | 2.53 | 2.61       | 3.11           | $V_{PP}$ |
| LINEIN Input Impedance                                                                                     |        | -    | 100        | -              | kOhm     |
| LINEOUT Output Impedance                                                                                   |        | -    | 320        | -              | Ohm      |
| LINEOUT Load                                                                                               |        | 10   | -          | -              | kOhm     |
| HP (headphone) Load                                                                                        |        | 16   | -          | -              | Ohm      |
| SYS_MCLK Input Voltage swing                                                                               |        | -0.3 | $V_{DDIO}$ | $V_{DDIO}+0.3$ | V        |
| SYS_MCLK Rise/Fall Time                                                                                    |        | 0.5  | -          | 10             | ns       |

## STATIC ELECTRICAL CHARACTERISTICS

**Table 5. Audio Performance 1**

Test Conditions unless otherwise noted:  $V_{DDIO} = 1.8\text{ V}$ ,  $V_{DDA} = 1.8\text{ V}$ ,  $T_A = 25\text{ }^{\circ}\text{C}$ , Slave mode,  $f_S = 48\text{ kHz}$ ,  $MCLK = 256 f_S$ , 24 bit input.

| Characteristic                                                        | Symbol | Min | Typ        | Max | Unit      |
|-----------------------------------------------------------------------|--------|-----|------------|-----|-----------|
| <b>AUDIO PERFORMANCE</b>                                              |        |     |            |     |           |
| LINEIN Input Level                                                    |        | -   | 0.57       | -   | $V_{RMS}$ |
| LINEIN Input Impedance                                                |        | 10  | -          | -   | kOhm      |
| <b>LINEIN -&gt; ADC -&gt; I<sup>2</sup>S OUT</b>                      |        |     |            |     |           |
| SNR (-60 dB input)                                                    |        | -   | 85         | -   | dB        |
| THD+N                                                                 |        | -   | -70        | -   | dB        |
| Frequency Response                                                    |        | -   | $\pm 0.11$ | -   | dB        |
| Channel Separation                                                    |        | -   | 79         | -   | dB        |
| <b>LINEIN -&gt; HEADPHONE_LINEOUT (CODEC BYPASS MODE)</b>             |        |     |            |     |           |
| SNR (-60 dB input)                                                    |        | -   | 98         | -   | dB        |
| THD+N (10 kOhm load)                                                  |        | -   | -87        | -   | dB        |
| THD+N (16 Ohm load)                                                   |        | -   | -87        | -   | dB        |
| Frequency Response                                                    |        | -   | $\pm 0.05$ | -   | dB        |
| Channel Separation (1.0 kHz)                                          |        |     | 82         |     | dB        |
| <b>I<sup>2</sup>S IN -&gt; DAC -&gt; LINEOUT</b>                      |        |     |            |     |           |
| Output Level                                                          |        | -   | 0.6        | -   | $V_{RMS}$ |
| SNR (-60 dB input)                                                    |        | -   | 95         | -   | dB        |
| THD+N                                                                 |        | -   | -85        | -   | dB        |
| Frequency Response                                                    |        | -   | $\pm 0.12$ | -   | dB        |
| <b>I<sup>2</sup>S IN -&gt; DAC -&gt; HEADPHONE OUT - 16 OHM LOAD</b>  |        |     |            |     |           |
| Output Power                                                          |        | -   | 17         | -   | mW        |
| SNR (-60 dB input)                                                    |        | -   | 100        | -   | dB        |
| THD+N                                                                 |        | -   | -80        | -   | dB        |
| Frequency Response                                                    |        | -   | $\pm 0.12$ | -   | dB        |
| <b>I<sup>2</sup>S IN -&gt; DAC -&gt; HEADPHONE OUT - 32 OHM LOAD</b>  |        |     |            |     |           |
| Output Power                                                          |        | -   | 10         | -   | mW        |
| SNR (-60 dB input)                                                    |        | -   | 95         | -   | dB        |
| THD+N                                                                 |        | -   | -86        | -   | dB        |
| Frequency Response                                                    |        | -   | $\pm 0.11$ | -   | dB        |
| <b>I<sup>2</sup>S IN -&gt; DAC -&gt; HEADPHONE OUT - 10 KOHM LOAD</b> |        |     |            |     |           |
| SNR (-60 dB input)                                                    |        | -   | 96         | -   | dB        |
| THD+N                                                                 |        | -   | -84        | -   | dB        |
| Frequency Response                                                    |        | -   | $\pm 0.11$ | -   | dB        |
| PSRR (200 mVp-p @ 1.0 kHz on VDDA)                                    |        | -   | 85         | -   | dB        |

**Table 6. Audio Performance 2**

Test Conditions unless otherwise noted:  $V_{DDIO} = 3.3\text{ V}$ ,  $V_{DDA} = 3.3\text{ V}$ ,  $T_A = 25^\circ\text{C}$ , Slave mode,  $f_S = 48\text{ kHz}$ ,  $MCLK = 256 f_S$ , 24 bit input. ADC tests were conducted with  $BIAS\_CTRL = -37.5\%$ , all other tests conducted with  $BIAS\_CTRL = -50\%$ .

| Characteristic                                                | Symbol | Min | Typ   | Max | Unit      |
|---------------------------------------------------------------|--------|-----|-------|-----|-----------|
| <b>AUDIO PERFORMANCE</b>                                      |        |     |       |     |           |
| LINEIN Input Level                                            |        | -   | 1.0   | -   | $V_{RMS}$ |
| LINEIN Input Impedance                                        |        | 10  | -     | -   | kOhm      |
| <b>LINEIN → ADC → I<sup>2</sup>S OUT</b>                      |        |     |       |     |           |
| SNR (-60 dB input)                                            |        | -   | 90    | -   | dB        |
| THD+N                                                         |        | -   | -72   | -   | dB        |
| Frequency Response                                            |        | -   | ±0.11 | -   | dB        |
| Channel Separation                                            |        | -   | 80    | -   | dB        |
| <b>LINEIN → HEADPHONE_LINEOUT (CODEC BYPASS MODE)</b>         |        |     |       |     |           |
| SNR (-60 dB input)                                            |        | -   | 102   | -   | dB        |
| THD+N (10 kOhm load)                                          |        | -   | -89   | -   | dB        |
| THD+N (16 Ohm load)                                           |        | -   | -87   | -   | dB        |
| Frequency Response                                            |        | -   | ±0.05 | -   | dB        |
| Channel Separation (1.0 kHz)                                  |        |     | 81    |     | dB        |
| <b>I<sup>2</sup>S IN → DAC → LINEOUT</b>                      |        |     |       |     |           |
| Output Level                                                  |        | -   | 1.0   | -   | $V_{RMS}$ |
| SNR (-60 dB input)                                            |        | -   | 100   | -   | dB        |
| THD+N                                                         |        | -   | -85   | -   | dB        |
| Frequency Response                                            |        | -   | ±0.12 | -   | dB        |
| <b>I<sup>2</sup>S IN → DAC → HEADPHONE OUT - 16 OHM LOAD</b>  |        |     |       |     |           |
| Output Power                                                  |        | -   | 58    | -   | mW        |
| SNR (-60 dB input)                                            |        | -   | 98    | -   | dB        |
| THD+N                                                         |        | -   | -86   | -   | dB        |
| Frequency Response                                            |        | -   | ±0.12 | -   | dB        |
| <b>I<sup>2</sup>S IN → DAC → HEADPHONE OUT - 32 OHM LOAD</b>  |        |     |       |     |           |
| Output Power                                                  |        | -   | 30    | -   | mW        |
| SNR (-60 dB input)                                            |        | -   | 100   | -   | dB        |
| THD+N                                                         |        | -   | -88   | -   | dB        |
| Frequency Response                                            |        | -   | ±0.11 | -   | dB        |
| <b>I<sup>2</sup>S IN → DAC → HEADPHONE OUT - 10 KOHM LOAD</b> |        |     |       |     |           |
| SNR (-60 dB input)                                            |        | -   | 97    | -   | dB        |
| THD+N                                                         |        | -   | -85   | -   | dB        |
| Frequency Response                                            |        | -   | ±0.11 | -   | dB        |
| PSRR (200 mVp-p @ 1.0 kHz on VDDA)                            |        | -   | 89    | -   | dB        |



## DYNAMIC ELECTRICAL CHARACTERISTICS

Table 7. Dynamic Electrical Characteristics

| Characteristic                                                                                                  | Symbol         | Min                | Typ                                            | Max | Unit    |
|-----------------------------------------------------------------------------------------------------------------|----------------|--------------------|------------------------------------------------|-----|---------|
| <b>POWER UP TIMING</b>                                                                                          |                |                    |                                                |     |         |
| Time from all supplies powered up and SYS_MCLK present to initial communication. See <a href="#">Figure 4</a> . | $t_{PC}$       | 1.0 <sup>(2)</sup> | –                                              | –   | $\mu$ s |
| <b>I<sup>2</sup>C BUS TIMING</b> <sup>(3)</sup> See <a href="#">Figure 5</a> .                                  |                |                    |                                                |     |         |
| I <sup>2</sup> C Serial Clock Frequency                                                                         | $f_{I2C\_CLK}$ | –                  | –                                              | 400 | kHz     |
| I <sup>2</sup> C Start condition hold time                                                                      | $t_{I2CSH}$    | 150                | –                                              | –   | ns      |
| I <sup>2</sup> C Stop condition setup time                                                                      | $t_{I2CSTSU}$  | 150                | –                                              | –   | ns      |
| I <sup>2</sup> C Data input setup time to rising edge of CTRL_CLK                                               | $t_{I2CDSU}$   | 125                | –                                              | –   | ns      |
| I <sup>2</sup> C Data input hold time from falling edge of CTRL_CLK (receiving data)                            | $t_{I2CDH}$    | 5.0                | –                                              | –   | ns      |
| I <sup>2</sup> C Data input hold time from falling edge of CTRL_CLK (driving data)                              | $t_{I2CDH}$    | 360                | –                                              | –   | ns      |
| I <sup>2</sup> C CTRL_CLK low time                                                                              | $t_{I2CCLKL}$  | 300                | –                                              | –   | ns      |
| I <sup>2</sup> C CTRL_CLK high time                                                                             | $t_{I2CCLKH}$  | 100                | –                                              | –   | ns      |
| <b>SPI BUS TIMING</b> <sup>(4)</sup> See <a href="#">Figure 6</a> .                                             |                |                    |                                                |     |         |
| SPI Serial Clock Frequency                                                                                      | $f_{SPI\_CLK}$ | –                  | –                                              | TBD | MHz     |
| SPI data input setup time                                                                                       | $t_{SPIDSU}$   | 10                 | –                                              | –   | ns      |
| SPI data input hold time                                                                                        | $t_{SPIDH}$    | 10                 | –                                              | –   | ns      |
| SPI CTRL_CLK low time                                                                                           | $t_{SPICLK L}$ | TBD                | –                                              | –   | ns      |
| SPI CTRL_CLK high time                                                                                          | $t_{SPICLK H}$ | TBD                | –                                              | –   | ns      |
| SPI clock to chip select                                                                                        | $t_{CCS}$      | 60                 | –                                              | –   | ns      |
| SPI chip select to clock                                                                                        | $t_{CSC}$      | 20                 | –                                              | –   | ns      |
| SPI chip select low                                                                                             | $t_{CSL}$      | 20                 | –                                              | –   | ns      |
| SPI chip select high                                                                                            | $t_{CSH}$      | 20                 |                                                |     | ns      |
| <b>SPECIFICATIONS AND TIMING FOR THE I<sup>2</sup>S PORT</b> <sup>(5)</sup> See <a href="#">Figure 7</a> .      |                |                    |                                                |     |         |
| Frequency of I <sup>2</sup> S_LRCLK                                                                             | $f_{LRCLK}$    | 8.0                | –                                              | 96  | kHz     |
| Frequency of I <sup>2</sup> S_SCLK                                                                              | $f_{SCLK}$     | –                  | 32*f <sub>LRCLK</sub><br>64*f <sub>LRCLK</sub> | –   | kHz     |
| I <sup>2</sup> S delay                                                                                          | $t_{I2S\_D}$   | –                  | –                                              | 10  | ns      |
| I <sup>2</sup> S setup time                                                                                     | $t_{I2S\_S}$   | 10                 | –                                              | –   | ns      |
| I <sup>2</sup> S hold time                                                                                      | $t_{I2S\_H}$   | 10                 | –                                              | –   | ns      |

## Notes

1. The SGTL5000 has an internal reset that is deasserted 8 SYS\_MCLK cycles after all power rails have been brought up. After this time, communication can start.
2. 1.0 $\mu$ s represents 8 SYS\_MCLK cycles at the minimum 8.0 MHz SYS\_MCLK.
3. This section provides timing for the SGTL5000 while in I<sup>2</sup>C mode (CTRL\_MODE = 0).
4. This section provides timing for the SGTL5000 while in SPI mode (CTRL\_MODE = 1)
5. The following are the specifications and timing for I<sup>2</sup>S port. The timing applies to all formats.

## TIMING DIAGRAMS

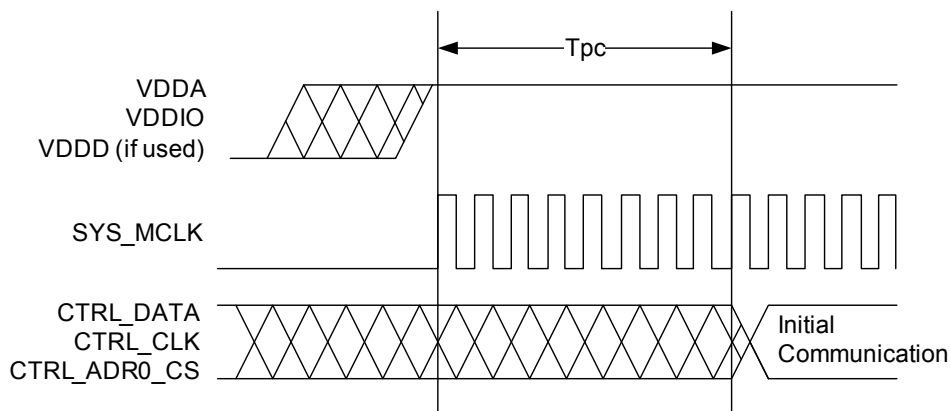


Figure 4. Power Up Timing

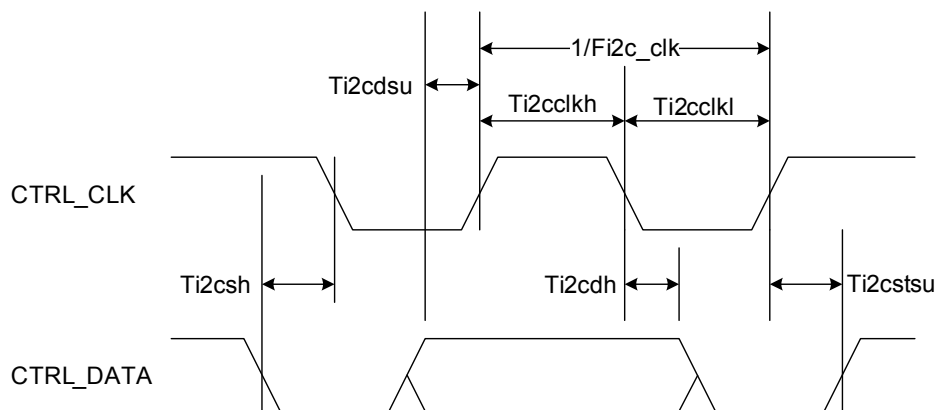


Figure 5. I<sup>2</sup>C Timing ( $CTRL\_MODE == 0$ )

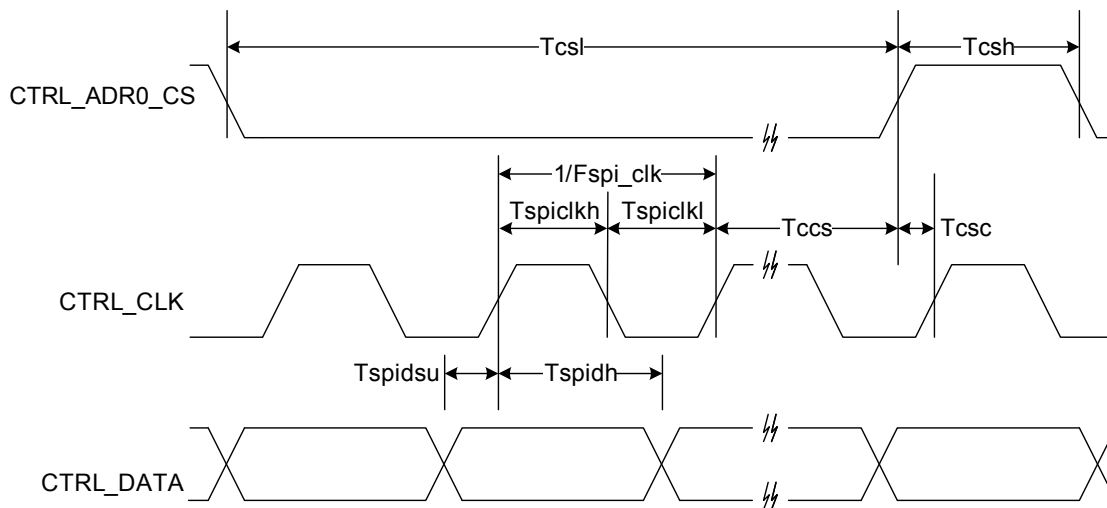


Figure 6. SPI Timing

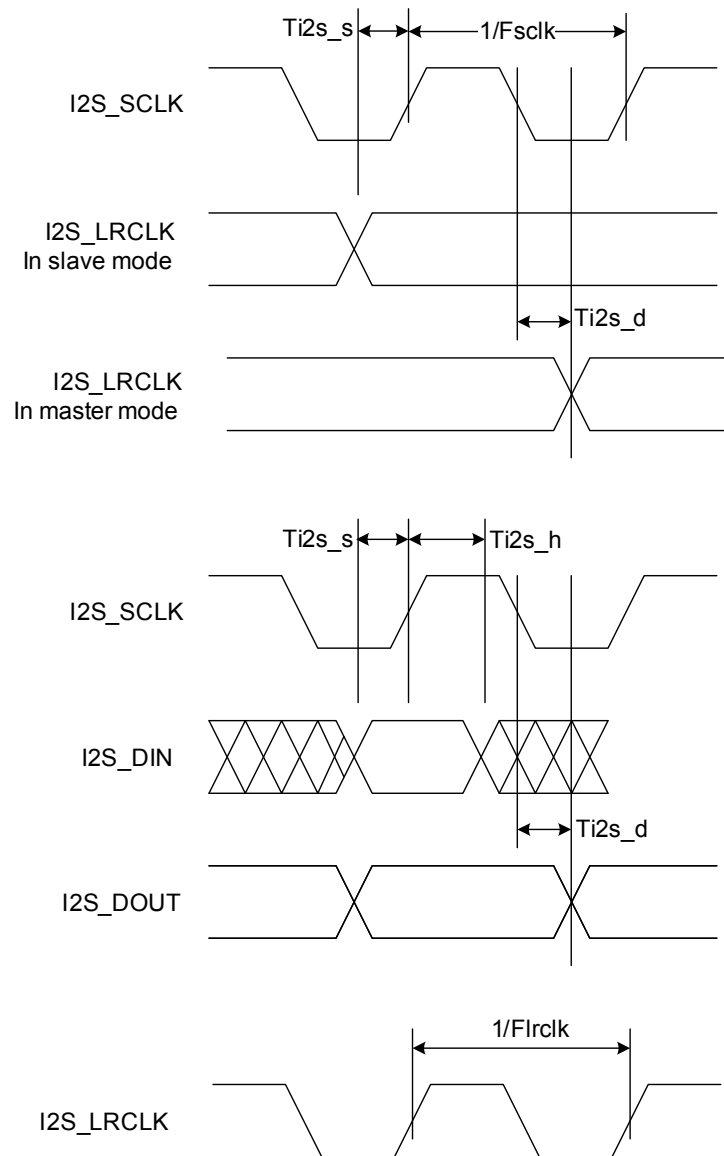


Figure 7. I<sup>2</sup>S Interface Timing

## FUNCTIONAL DESCRIPTION

### INTRODUCTION

The SGTL5000 is a low power stereo codec with integrated headphone amplifier. It is designed to provide a complete audio solution for portable products needing LINEIN, mic-in, LINEOUT, headphone-out, and digital I/O. Deriving its architecture from best in class Freescale integrated products that are currently on the market, the SGTL5000 is able to achieve ultra low power with very high performance and functionality, all in one of the smallest footprints available. Target markets include portable media players, GPS units and smart phones. Features such as capless headphone design and USB clocking mode (12 MHz SYS\_MCLK input) help lower overall system cost.

In summary, the SGTL5000 accepts the following inputs:

- Line input
- Microphone input, with mic bias
- Digital I<sup>2</sup>S input

In addition, the SGTL5000 supports the following outputs:

- Line output
- Headphone output
- Digital I<sup>2</sup>S output

The following digital audio processing is included to allow for product differentiation:

- Digital mixer
- Freescale Surround
- Freescale Bass Enhancement
- Tone Control, parametric equalizer, or graphic equalizer

The SGTL5000 can accept an external standard master clock at a multiple of the sampling frequency (i.e. 256\*Fs, 385\*Fs, 512\*Fs). In addition it can take non-standard frequencies and use the internal PLL to derive the audio clocks. The device supports 8.0 kHz, 11.025 kHz, 16 kHz, 22.05 kHz, 24 kHz, 32 kHz, 44.1kHz, 48 kHz, 96 kHz sampling frequencies.

### FUNCTIONAL INTERNAL BLOCK DESCRIPTION

#### SYSTEM BLOCK DIAGRAM W/ SIGNAL FLOW AND GAIN MAP

Figure 8 shows a block diagram that highlights the signal flow and gain map for the SGTL5000.

To guarantee against clipping, it is important that the gain in a signal path in addition to the signal level does not exceed 0 dB at any point.

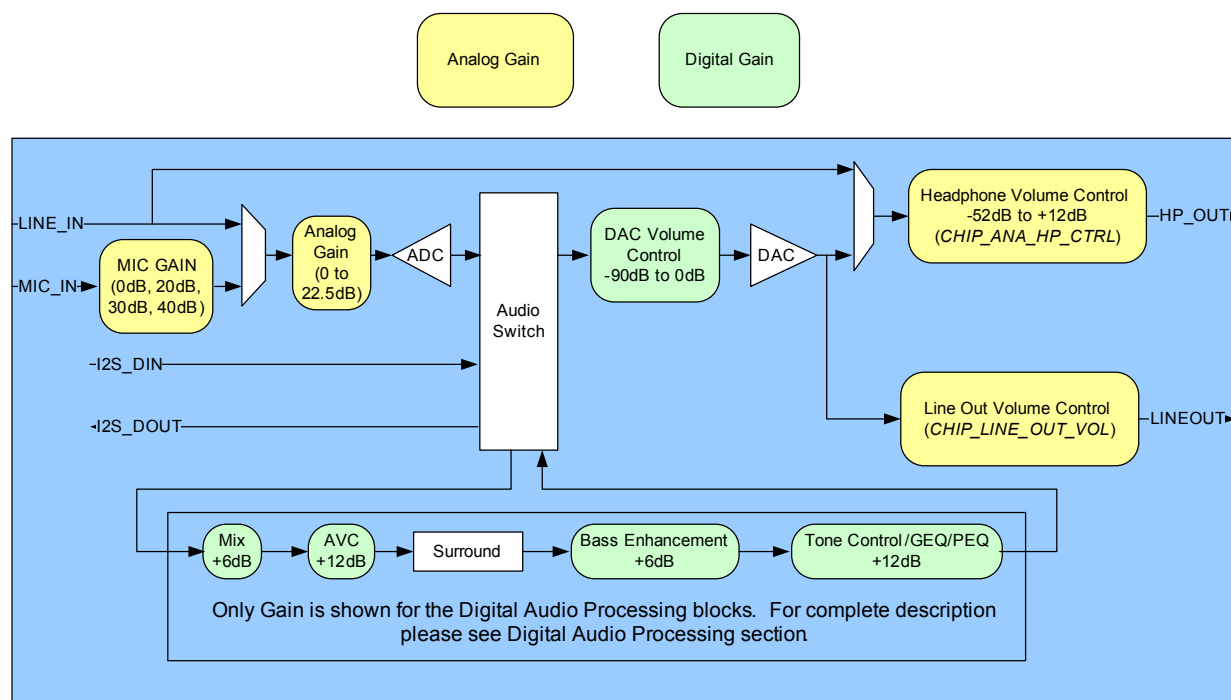


Figure 8. System Block Diagram, Signal Flow and Gain

## POWER

The SGT5000 has a flexible power architecture to allow the system designer to minimize power consumption and maximize performance at the lowest cost.

### External Power Supplies

The SGT5000 requires 2 external power supplies: VDDA and VDDIO. An optional third external power supply VDDD may be provided externally to achieve lower power. This external VDDD power supply is required for new designs. A description for the different power supplies is as follows:

- VDDA: This external power supply is used for the internal analog circuitry including ADC, DAC, LINE inputs, MIC inputs, headphone outputs and reference voltages. VDDA supply ranges are shown in [Maximum Ratings](#). A decoupling cap should be used on VDDA, as shown in the typical application diagrams in [Typical Applications](#).
- VDDIO: This external power supply controls the digital I/O levels as well as the output level of LINE outputs. VDDIO supply ranges are shown in [Maximum Ratings](#). A decoupling cap should be used on VDDIO as shown in the typical application diagrams in [Typical Applications](#).

Note that if VDDA and VDDIO are derived from the same voltage, a single decoupling capacitor can be used to minimize cost. This capacitor should be placed closest to VDDA.

- VDDD: This is a digital power supply that is used for internal digital circuitry. An external VDDD power supply is required for new designs. For lowest power, this supply can be driven at the lowest specified voltage given in [Maximum Ratings](#). If an external supply is used for VDDD, a decoupling capacitor is recommended, as shown in the typical applications diagram. VDDD supply ranges are shown in [Maximum Ratings](#) for when externally driven. If the system drives VDDD externally, an efficient switching supply should be used or no system power savings is realized.

### Internal Power Supplies

The SGT5000 has two exposed internal power supplies, VAG and charge pump.

- VAG is the internal voltage reference for the ADC and DAC. After startup the voltage of VAG should be set to VDDA/2 by writing `CHIP_REF_CTRL->VAG_VAL`. Refer to programming [Chip Powerup and Supply Configurations](#). The VAG pin should have an external filter capacitor as shown in the typical application diagram.
- Chargepump: This power supply is used for internal analog switches. If VDDA or VDDIO is greater than 2.7 V, this supply is automatically driven from the highest of

VDDIO and VDDA. If both VDDIO and VDDA are less than 3.1 V, then the user should turn on the charge pump function to create the charge pump rail from VDDIO by writing `CHIP_ANA_POWER->`

`VDDC_CHRGPMPOWERUP` register. Refer to programming [Chip Powerup and Supply Configurations](#).

- LINE\_OUT\_VAG is the line output voltage reference. It should be set to VDDIO/2 by writing `CHIP_LINE_OUT_CTRL->LO_VAGCTRL`.

### Power Schemes

The SGT5000 supports a flexible architecture and allows the system designer to minimize power or maximize BOM savings.

- For maximum cost savings, all supplies can be run at the same voltage.
- Alternatively for minimum power, the analog and digital supplies can be run at minimum voltage while driving the digital I/O voltage at the voltage needed by the system.
- To save power, independent supplies are provided for line outputs and headphone outputs. This allows for 1VRMS line outputs while using minimal headphone power.
- For best power, VDDA should be run at the lowest possible voltage required for the maximum headphone output level. For highest performance, VDDA should be run at 3.3 V. For most applications a lower voltage can be used for the best performance/power combination.

## RESET

The SGT5000 has an internal reset that is deasserted 8 SYS\_MCLKs after all power rails have been brought up. After this time communication can start. See [Dynamic Electrical Characteristics](#).

## CLOCKING

Clocking for the SGT5000 is provided by a system master clock input (SYS\_MCLK). SYS\_MCLK should be synchronous to the sampling rate (Fs) of the I<sup>2</sup>S port. Alternatively any clock between 8.0 and 27 Mhz can be provided on SYS\_MCLK and the SGT5000 can use an internal PLL to derive all internal and I<sup>2</sup>S clocks. This allows the system to use an available clock such as 12 MHz (common USB clock) for SYS\_MCLK to reduce overall system costs.

### Synchronous SYS\_MCLK input

The SGT5000 supports various combinations of SYS\_MCLK frequency and sampling frequency as shown in Table 8. Using a synchronous SYS\_MCLK allows for lower power as the internal PLL is not used.

**Table 8. Synchronous MCLK Rates and Sampling Frequencies**

| CLOCK                          | SUPPORTED RATES                                       | UNITS |
|--------------------------------|-------------------------------------------------------|-------|
| System Master Clock (SYS_MCLK) | 256, 384, 512                                         | Fs    |
| Sampling Frequency (Fs)        | 8, 11.025, 16, 22.05, 32, 44.1, 48, 96 <sup>(6)</sup> | kHz   |

Notes

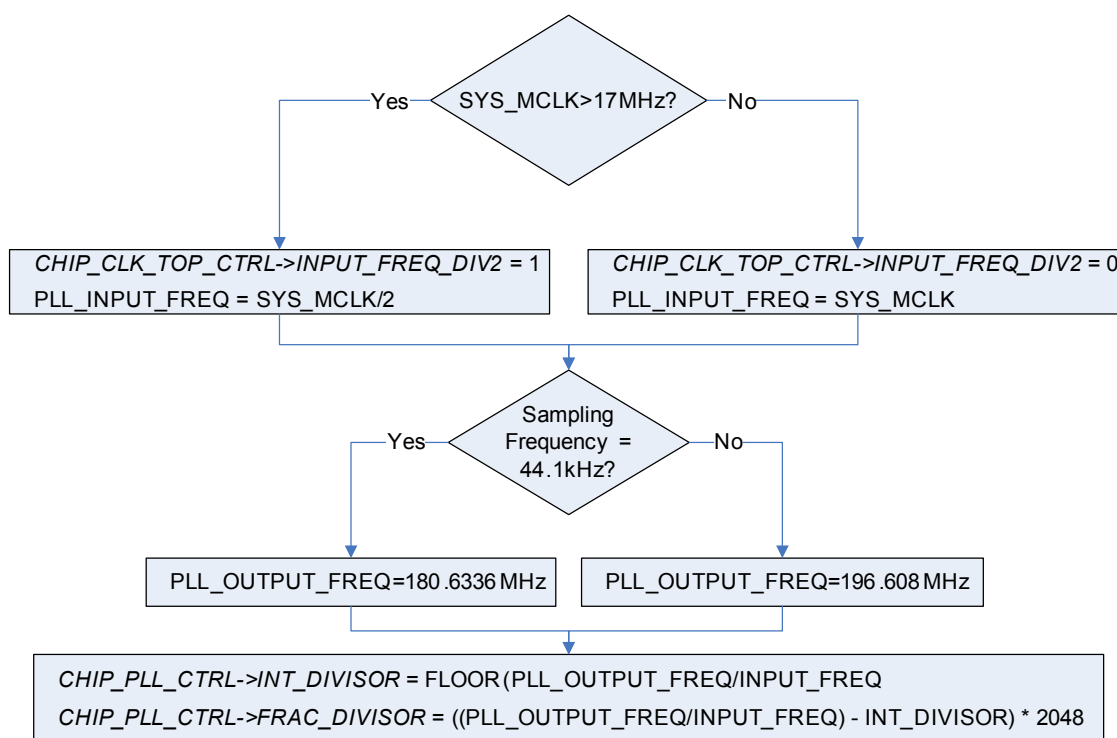
6. For a sampling frequency of 96 kHz, only 256 Fs SYS\_MCLK is supported

### Using the PLL - Asynchronous SYS\_MCLK input

An integrated PLL is provided in the SGT5000 that allows any clock from 8.0 to 27 MHz to be connected to SYS\_MCLK. This can help save system costs, as a clock available elsewhere in the system can be used to derive all audio clocks using the internal PLL. In this case, the clock input to SYS\_MCLK can be asynchronous with the sampling frequency needed in the system. For example, a 12 MHz

clock from the system processor could be used as the clock input to the SGT5000.

Three register fields need to be configured to properly use the PLL. They are *CHIP\_PLL\_CTRL->INT\_DIVISOR*, *CHIP\_PLL\_CTRL->FRAC\_DIVISOR* and *CHIP\_CLK\_TOP\_CTRL->INPUT\_FREQ\_DIV2*. [Figure 9](#) shows a flowchart that shows how to determine the values to program in the register fields.



**Figure 9. PLL Programming Flowchart**

For example, when a 12 MHz digital signal is placed on MCLK, for a 48 kHz frame clock

*CHIP\_CLK\_TOP\_CTRL->INPUT\_FREQ\_DIV2* = 0 //  
 SYS\_MCLK < 17 MHz

*CHIP\_PLL\_CTRL->INT\_DIVISOR* = FLOOR  
 (196.608 MHz/12 MHz) = 16 (decimal)

*CHIP\_PLL\_CTRL->FRAC\_DIVISOR* = ((196.608 MHz/  
 12 MHz) - 16) \* 2048 = 786 (decimal)

Refer to PLL programming [PLL Configuration](#).

### AUDIO SWITCH (SOURCE SELECT SWITCH)

The audio switch is the central routing block that controls the signal flow from input to output. Any single input can be routed to any single or multiple outputs.

Any signal can be routed to the Digital Audio Processor (DAP). The output of the DAP (an input to the audio switch) can in turn be routed to any physical output. The output of the DAP can not be routed into itself. Refer to [Digital Audio Processing](#), for DAP information and configuration.

It should be noted that the analog bypass from Line input to headphone output does not go through the audio switch.

To configure a route, the *CHIP\_SSS\_CTRL* register is used. Each output from the source select switch has its own register field that is used to select what input is routed to that output.

For example, to route the I<sup>2</sup>S digital input through the DAP and then out to the DAC (headphone) outputs write *SSS\_CTRL->DAP\_SELECT* to 0x1 (selects I2S\_IN) and *SSS\_CTRL->DAC\_SELECT* to 0x3 (selects DAP output).

## ANALOG INPUT BLOCK

The analog input block contains a stereo line input and a microphone input with mic bias. Either input can be routed to the ADC. The line input can also be configured to bypass the CODEC and be routed directly to the headphone output.

### Line Inputs

One stereo line input is provided for connection to line sources such as an FM radio or MP3 input.

The source should be connected to the left and right line inputs through series coupling capacitors. The suggested value is shown in the typical application diagram in [Typical Applications](#).

As detailed in [ADC](#), the line input can be routed to the ADC.

The line input can also be routed to the headphone output by writing *CHIP\_ANA\_CTRL->SELECT\_HP*. This selection bypasses the ADC and audio switch and routes the line input directly to the headphone output to enable a very low power pass through.

### Microphone Input

One mono microphone input is provided for uses such as voice recording.

Mic bias is provided. The mic bias is programmed with the *CHIP\_MIC\_CTRL->BIAS\_VOLT* register field. Values from 1.25 V to 3.00 V are supported in 0.25 V steps. Mic bias should be set less than 200 mV from VDDA, e.g. with VDDA at 1.70 V, Mic bias should be set no greater than 1.50 V.

The microphone should be connected through a series coupling capacitor. The suggested value is shown in the typical connection diagram.

The microphone has programmable gain through the *CHIP\_MIC\_CTRL->GAIN* register field. Values of 0 dB, +20 dB, +30 dB and +40 dB are available.

## ADC

The SGTL5000 contains an ADC, which takes its input from either the line input or a microphone. The register field *CHIP\_ANA\_CTRL->SELECT\_ADC* controls this selection. The output of the ADC feeds the audio switch.

The ADC has its own analog gain stage that provides 0 to +22.5 dB of gain in 1.5 dB steps. A bit is available that shifts this range down by 6.0 dB to effectively provide -6.0 dB to

+16.5 dB of gain. The ADC gain is controlled in the *CHIP\_ANA\_ADC\_CTRL* register.

The ADC has an available zero cross detect (ZCD) that prevents any volume change until a zero-volt crossing of the audio signal is detected. This helps in eliminating pop or other audio anomalies. If the ADC is to be used, the chip reference bias current should not be set to -50% when in 3.0 V mode.

## ANALOG OUTPUTS

The SGTL5000 contains a single stereo DAC that can be used to drive a headphone output and a line output. The DAC receives its input from the audio switch. The headphone output and the line output can be driven at the same time from the DAC.

The headphone output can also be driven directly by the line input bypassing the ADC and DAC for a very low power mode of operation.

The headphone output is powered by VDDA while the line output is powered by VDDIO. This allows the headphone output to be run at the lowest possible voltage while the line output can still meet line output level requirements.

### DAC

The DAC output is routed to the headphone and the dedicated line output.

The DAC output has a digital volume control from -90 dB to 0 dB in ~0.5 dB step sizes. This volume is shared among headphone output and line output. The register *CHIP\_DAC\_VOL* controls the DAC volume.

### Headphone

Stereo headphone outputs are provided which can be used to drive a headphone load or a line level output. The headphone output has its own independent analog volume control with a volume range of -52 dB to +12 dB in 0.5 dB step sizes. This volume control can be used in addition to the DAC volume control. For best performance the DAC volume control should be left at 0 dB until the headphone is brought to its lowest setting of -52 dB. The register *CHIP\_ANA\_HP\_CTRL* is used to control the headphone volume.

The headphone output has an independent mute that is controlled by the register field *CHIP\_ANA\_CTRL->MUTE\_HP*.

The line input is routed to the headphone output by writing *CHIP\_ANA\_CTRL->SELECT\_HP*. This selection bypasses the ADC and audio switch and routes the line input directly to the headphone output to enable a very low power pass through. When the line input is routed to the headphone output, only the headphone analog volume and mute affects the headphone output.

The headphone has an available zero cross detect (ZCD) which, as previously described, prevents any volume change until a zero-volt crossing of the audio signal is detected. This helps in eliminating pop or other audio anomalies.

### Line Outputs

The SGT5000 contains a stereo line output. The line output has a dedicated gain stage that can be used to adjust the output level. The *CHIP\_LINE\_OUT\_VOL* controls the line level output gain.

The line outputs also have a dedicated mute that is controlled by the register field *CHIP\_ANA\_CTRL->MUTE\_LO*.

The line out volume is intended as maximum output level adjustment. It is intended to be used to set the maximum output swing. It does not have the range of a typical volume control and does not have a zero cross detect (ZCD). However the DAC digital volume could be used if volume control is desired.



## FUNCTIONAL DEVICE OPERATION

### POWER CONSUMPTION

**Table 9. Power Consumption:  $V_{DDA}=1.8\text{ V}$ ,  $V_{DDIO}=1.8\text{ V}$**

| MODE                                                                                             | CURRENT CONSUMPTION (MA) |           |            | POWER (MW) |
|--------------------------------------------------------------------------------------------------|--------------------------|-----------|------------|------------|
|                                                                                                  | $V_{DDD}$                | $V_{DDA}$ | $V_{DDIO}$ |            |
| Playback ( $I^2S \rightarrow DAC \rightarrow \text{Headphone}$ )                                 | -                        | 2.54      | 0.9        | 6.19       |
| Playback with DAP ( $I^2S \rightarrow DAP \rightarrow DAC \rightarrow \text{Headphone}$ )        | -                        | 3.59      | 0.9        | 8.08       |
| Playback/Record ( $I^2S \rightarrow DAC \rightarrow \text{Headphone}$ , $ADC \rightarrow I^2S$ ) | -                        | 3.71      | 1.10       | 8.67       |
| Record ( $ADC \rightarrow I^2S$ )                                                                | -                        | 2.29      | 1.06       | 6.02       |
| Analog playback, CODEC bypassed ( $LINEIN \rightarrow HP$ )                                      | -                        | 1.48      | 0.89       | 4.27       |
| Standby, all analog power off                                                                    | -                        | 0.019     | 0.002      | 0.038      |
| Playback with PLL ( $I^2S \rightarrow DAC \rightarrow HP$ )                                      | -                        | 3.01      | 2.17       | 9.31       |

$V_{DDD}$  derived internally @ 1.2 V, slave mode except for PLL case, 32 ohm load on HP, Conditions: -100 dBFS signal

input, slave mode unless otherwise noted, paths tested as indicated, unused paths turned off.

**Table 10. Power Consumption:  $V_{DDA}=3.3\text{ V}$ ,  $V_{DDIO}=3.3\text{ V}$**

| MODE                                                                                             | CURRENT CONSUMPTION (MA) |           |            | POWER(MW) |
|--------------------------------------------------------------------------------------------------|--------------------------|-----------|------------|-----------|
|                                                                                                  | $V_{DDD}$                | $V_{DDA}$ | $V_{DDIO}$ |           |
| Playback ( $I^2S \rightarrow DAC \rightarrow \text{Headphone}$ )                                 | -                        | 3.45      | 0.067      | 11.60     |
| Playback with DAP ( $I^2S \rightarrow DAP \rightarrow DAC \rightarrow \text{Headphone}$ )        | -                        | 4.49      | 0.067      | 15.03     |
| Playback/Record ( $I^2S \rightarrow DAC \rightarrow \text{Headphone}$ , $ADC \rightarrow I^2S$ ) | -                        | 4.67      | 0.343      | 16.53     |
| Record ( $ADC \rightarrow I^2S$ )                                                                | -                        | 2.90      | 0.296      | 10.56     |
| Analog playback, CODEC bypassed ( $LINEIN \rightarrow HP$ )                                      | -                        | 1.91      | 0.039      | 6.43      |
| Standby, all analog power off                                                                    | -                        | 0.04      | 0.002      | 0.139     |
| Playback with PLL ( $I^2S \rightarrow DAC \rightarrow HP$ )                                      | -                        | 3.92      | 2.76       | 22.05     |

### DIGITAL INPUT & OUTPUT

One  $I^2S$  (Digital Audio) Port is provided which supports the following formats:  $I^2S$ , Left Justified, Right Justified, and PCM mode.

### $I^2S$ , Left Justified, and Right Justified Modes

$I^2S$ , Left Justified and Right Justified modes are stereo interface formats. The  $I2S\_SCLK$  frequency,  $I2S\_SCLK$  polarity,  $I2S\_DIN/DOUT$  data length, and  $I2S\_LRCLK$  polarity can all be changed through the  $CHIP\_I2S\_CTRL$  register. For  $I2S$ , Left Justified and Right Justified formats,

the left subframe should always be presented first regardless of the *CHIP\_I2S\_CTRL->LRPOL* setting.

The I2S\_LRCLK and I2S\_SCLK can be programmed as master (driven to an external target) or slave (driven from an external source). When the clocks are in slave mode, they must be synchronous to SYS\_MCLK. For this reason the SGT5000 can only operate in synchronous mode (see [Clocking](#)) while in I<sup>2</sup>S slave mode.

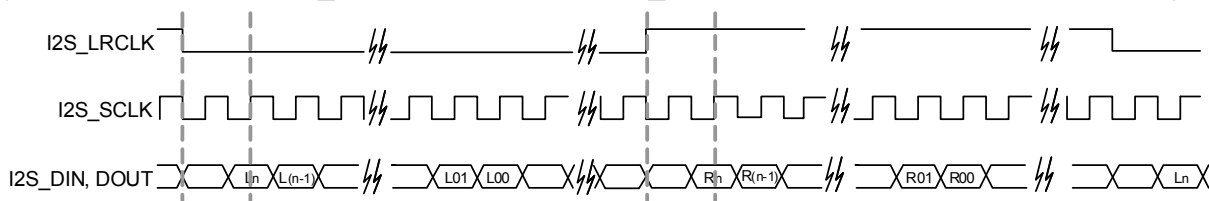
In master mode, the clocks are synchronous to SYS\_MCLK or the output of the PLL when the part is running in asynchronous mode.

[Figure 10](#) shows functional examples of different common digital interface formats and their associated register settings.

## I2S Format (n = bit length)

CHIP\_I2S0\_CTRL field values:

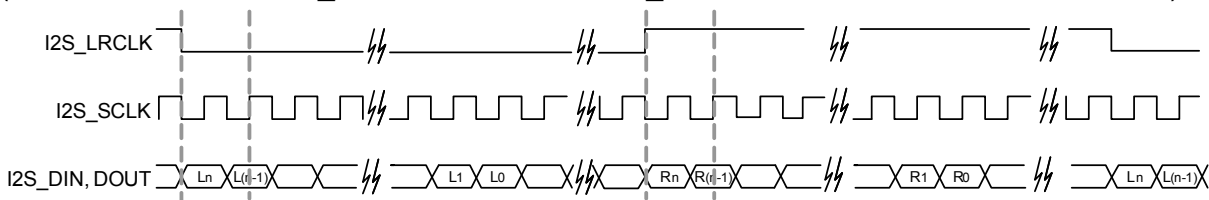
(SCLKFREQ = 0; SCLK\_INV = 0; DLEN = 1; I2S\_MODE = 0; LRALIGN = 0; LRPOL = 0)



## Left Justified Format (n = bit length)

CHIP\_I2S0\_CTRL field values:

(SCLKFREQ = 0; SCLK\_INV = 0; DLEN = 1; I2S\_MODE = 0; LRALIGN = 1; LRPOL = 0)



## Right Justified Format (n = bit length)

CHIP\_I2S0\_CTRL field values:

SCLKFREQ = 0; SCLK\_INV = 0; DLEN = 1; I2S\_MODE = 1; LRALIGN = 1; LRPOL = 0)

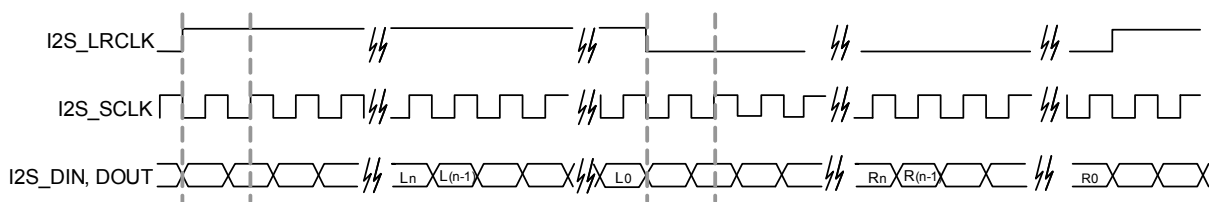


Figure 10. I<sup>2</sup>S Port Supported Formats

## PCM Mode

The I<sup>2</sup>S port can also be configured in PCM mode (also known as DSP mode). This mode is provided to allow connectivity to external devices such as Bluetooth modules. PCM mode differs from other interface formats presented in [I2S, Left Justified, and Right Justified Modes](#), in that the frame clock (I2S\_LRCLK) does not represent a different channel when high or low. Instead, it is a bit-wide pulse that marks the start of a frame. Data is aligned such that the left channel data is immediately followed by right channel data. Zero padding is filled in for the remaining bits. The data and

frame clock may be configured to clock in on the rising or falling edge of Bit Clock.

PCM Format A is a format in which the data word begins one SCLK bit following the I2S\_LRCLK transition, as in I<sup>2</sup>S Mode. PCM Format B is a format in which the data word begins after the I2S\_LRCLK transition, as in Left Justified.

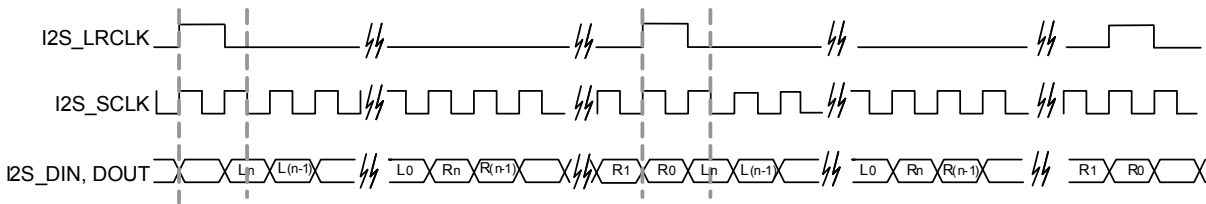
In slave mode, the pulse width of the I2S\_LRCLK does not matter. The pulse can range from one cycle high to all but one cycle high. In master mode, it is driven one cycle high.

[Figures 11](#) shows a functional drawing of the different formats in master mode.

## PCM Format A

CHIP\_I2S0\_CTRL = 0x01F4

(SCLKFREQ = 1; MS = 1; SCLK\_INV = 1; DLEN = 3; I2S\_MODE = 2; LRALIGN = 0)



## PCM Format B

CHIP\_I2S0\_CTRL = 0x01F6

(SCLKFREQ = 1; MS = 1; SCLK\_INV = 1; DLEN = 3; I2S\_MODE = 2; LRALIGN = 1)

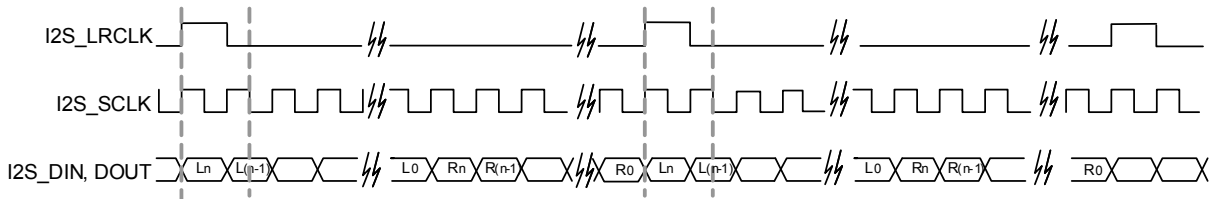


Figure 11. PCM Formats

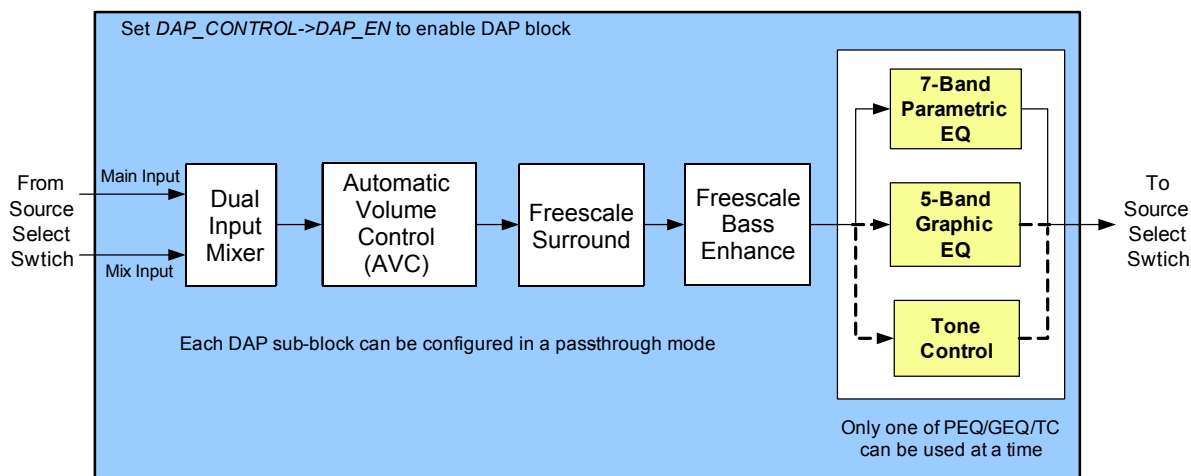
## DIGITAL AUDIO PROCESSING

The SGT5000 contains a digital audio processing block (DAP) connected to the source select switch. The digitized signal from the source select switch can be routed into the DAP block for audio processing. The DAP has the following 5 sub blocks:

- Dual Input Mixer

- Freescale Surround
- Freescale Bass Enhancement
- 7-Band Parameter EQ / 5-Band Graphic EQ / Tone Control (only one can be used at a time)
- Automatic Volume Control (AVC)

The block diagram in [Figure 12](#) shows the sequence in which the signal passes through these blocks.



**Figure 12. Digital Audio Processing Block Diagram**

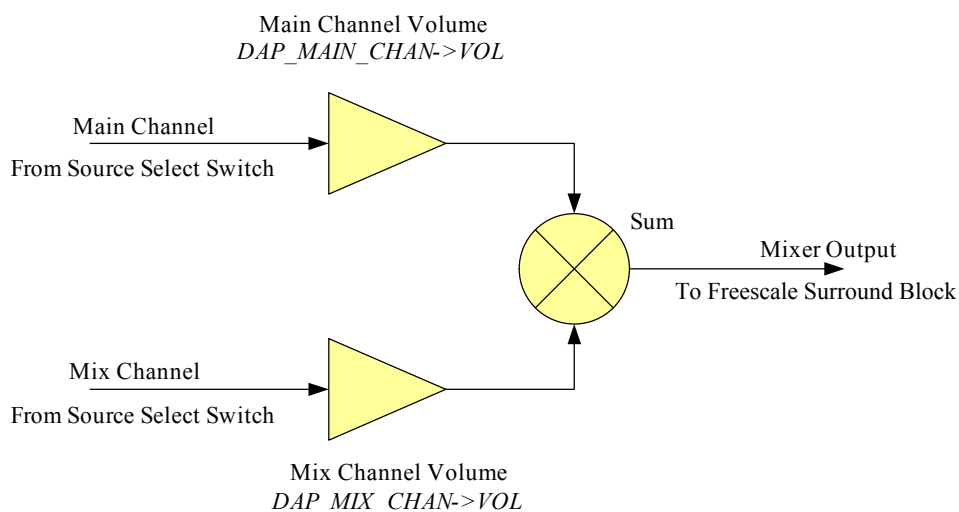
When the DAP block is added in the route, it must be enabled separately to get audio through. It is recommended to mute the outputs before enabling/disabling the DAP block to avoid any pops or clicks due to discontinuities in the output.

Refer to [Digital Audio Processor Configuration](#) for programming examples on how to enable/disable the DAP block.

Each sub-block of the DAP can be individually disabled if its processing is not required. The following sections describe the DAP sub-blocks and how to configure them.

#### Dual Input Mixer

The dual input digital mixer allows for two incoming streams from the source select switch as shown in [DAP - Dual Input Mixer](#).



**Figure 13. DAP - Dual Input Mixer**

The Dual Input Mixer can be enabled or configured in a pass-through mode (Main channel is passed through without any mixing). When enabled, the volume of the main and mix channels can be independently controlled before they are mixed together.

The volume range allowed on each channel is 0% to 200% of the incoming signal level. The default is 100% (same as input signal level) volume on the main input and 0% (muted) on the mix input.

Refer to [Dual Input Mixer](#) for programming examples on how to enable/disable the mixer and also to set the main and mix channel volume.

#### Freescale Surround

Freescale Surround is a royalty free virtual surround algorithm for stereo or mono inputs. It widens and deepens the sound stage of the music input.

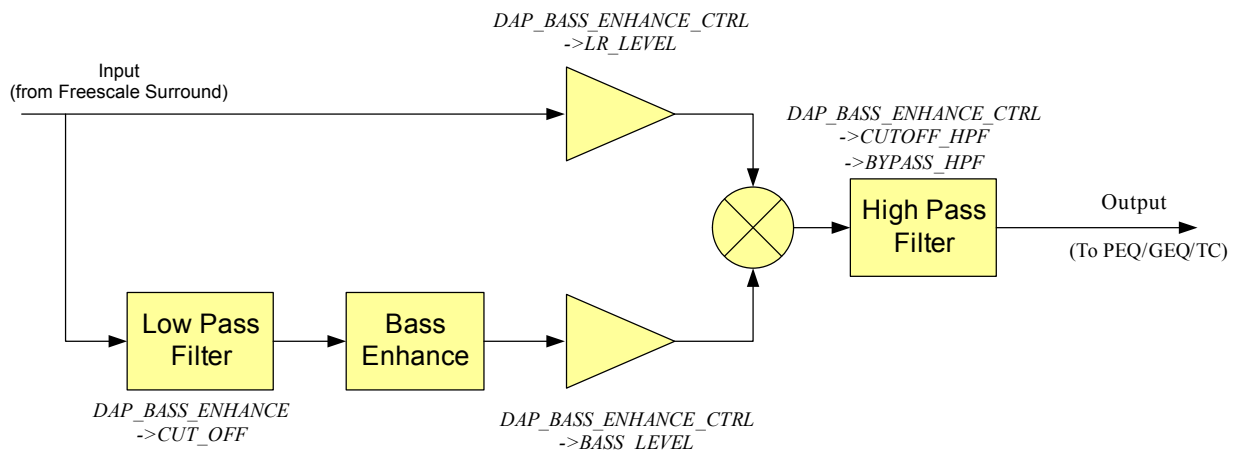


The Freescale Surround can be enabled or configured in pass-through mode (input is passed through without any processing). When enabling the Surround, mono or stereo input type must be selected based on the input signal. Surround width may be adjusted for the size of the sound stage.

Refer to [Freescale Surround](#) and [Freescale Surround On/Off](#) for a programming example on how to configure Surround width and how to enable/disable Surround.

### Freescale Bass Enhance

Freescale Bass Enhance is a royalty-free algorithm that enhances natural bass response of the audio. Bass Enhance extracts bass content from right and left channels, adds bass and mixes this back up with the original signal. An optional complementary high pass filter is provided after the mixer.



**Figure 14. DAP- Freescale Bass Enhance**

The Freescale Bass Enhance can be enabled or configured in pass-through mode (input is passed through without any processing).

The cutoff frequency of the low-pass filter (LPF) can be selected based on the speakers frequency response. The cutoff frequency of the low-pass and high-pass filters are selectable between 80 to 225 Hz. Also, the input signal and bass enhanced signal can be individually adjusted for level before the two signals are mixed.

Refer to [Freescale Bass Enhance](#) and [Bass Enhance On/Off](#) for a programming example on how to configure Bass Enhance and how to enable/disable this feature.

### 7-Band Parametric EQ / 5-Band Graphic EQ / Tone Control

One 7-band parametric equalizer (PEQ), one 5-band graphic equalizer (GEQ), and Tone Control (Bass and Treble

control) blocks are implemented as mutually exclusive blocks. Only one block can be used at a given time.

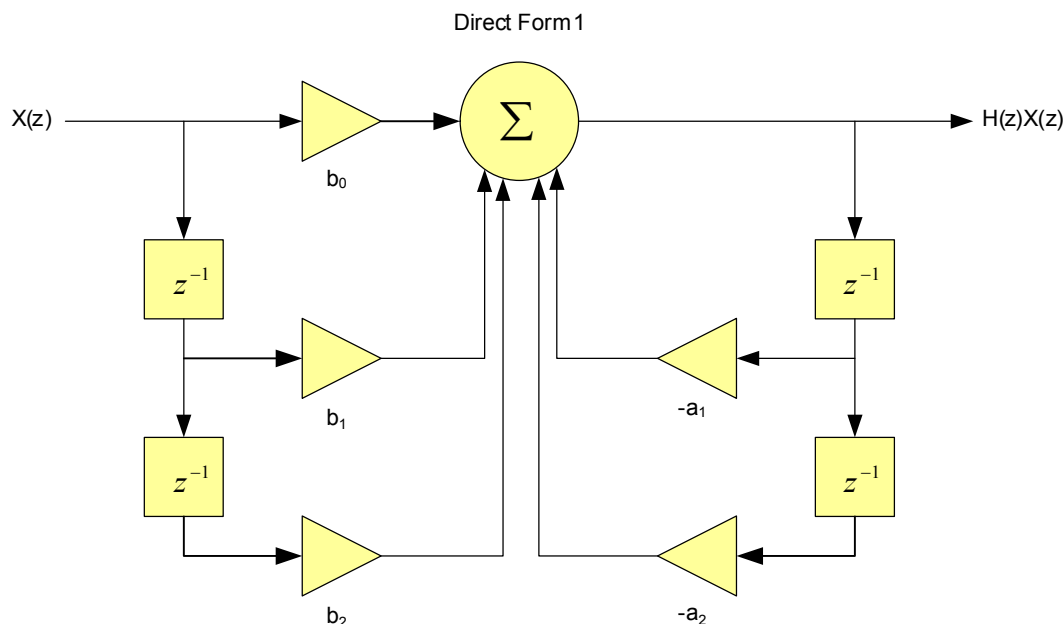
Refer to [7-Band Parametric EQ / 5-Band Graphic EQ / Tone Control](#) for a programming example that shows how to select the desired EQ mode.

### 7-Band Parametric EQ

The 7-band PEQ allows the designer to compensate for speaker response and to provide the ability to filter out resonant frequencies caused by the physical system design. The system designer can create custom EQ presets such as Rock, Speech, Classical, etc, which allows users the flexibility to customize their audio.

The 7-band PEQ is implemented using 7 cascaded second order IIR filters. All filters are implemented using programmable bi-quad filters. [Figure 15](#) shows the transfer function and Direct Form 1 of the five coefficient biquadratic filter.

$$H(z) = \frac{b_0 + b_1z^{-1} + b_2z^{-2}}{1 + a_1z^{-1} + a_2z^{-2}}$$



**Figure 15. 5-Coefficient Biquad Filter and Transfer Function**

If a band is enabled but is not being used (flat response), then a value of 0.5 should be put in  $b_0$  and all other coefficients should be set to 0.0. Note that the coefficients must be converted to hex values before writing to the registers. By default, all the filters are loaded with coefficients to give a flat response.

In order to create EQ presets such as Rock, Speech, Classical, etc, the coefficients must be calculated, converted to 20-bit hex values and written to the registers. Note that coefficients are sample-rate dependent and separate coefficients must be generated for different sample rates. Please contact Freescale for assistance with generating the coefficients.

Refer to [7-Band PEQ Preset Selection](#) for a programming example that shows how load the filter coefficients when the end-user changes the preset.

PEQ can be disabled (pass-through mode) by writing 0 to DAP\_AUDIO\_EQ->EN bits.

### 5-Band Graphic EQ

The 5-band graphic equalizer is implemented using 5 parallel second order IIR filters. All filters are implemented using biquad filters whose coefficients are programmed to set the bands at a specific frequency. The GEQ bands are fixed

at 115 Hz, 330 Hz, 990 Hz, 3000 Hz, and 9900 Hz. The volume on each band is independently adjustable in the range of +12 dB to -11.75 dB in 0.25 dB steps.

Refer to [5-Band GEQ Volume Change](#) for a programming example that shows how to change the GEQ volume.

### Tone Control

Tone control comprises treble and bass controls. The tone control is implemented as one 2nd order low pass filter (bass) and one 2nd order high pass filter (treble).

Refer to [Tone Control - Bass and Treble Change](#) for a programming example that shows how to change Bass and Treble values.

### Automatic Volume Control (AVC)

An Automatic Volume Control (AVC) block is provided to reduce loud signals and amplify low level signals for easier listening. The AVC is designed to compress audio when the measured level is above the programmed threshold or to expand the audio to the programmed threshold when the measured audio is below the threshold. The threshold level is programmable with an allowed range of 0 to -96 dB.

[Figure 16](#) shows the AVC block diagram and controls.

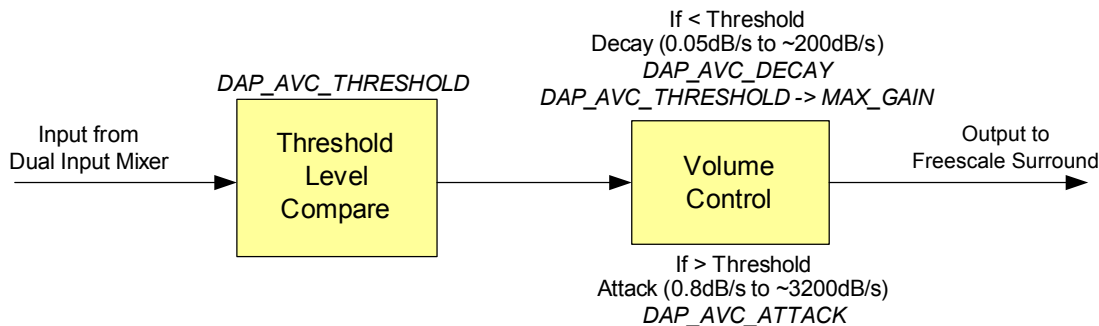


Figure 16. DAP AVC Block Diagram

When the measured audio level is below threshold, the AVC can apply a maximum gain of up to 12 dB. The maximum gain can be selected, either 0, 6, or 12 dB. When the maximum gain is set to 0 dB the AVC acts as a limiter. In this case the AVC only takes effect when the signal level is above the threshold.

The rate at which the incoming signal is attenuated down to the threshold is called the attack rate. Too high of an attack causes an unnatural sound as the input signal may be distorted. Too low of an attack may cause saturation of the output as the incoming signal is not compressed quickly enough. The attack rate is programmable with allowed range of 0.05 dB/s to 200 dB/s.

When the signal is below the threshold, AVC adjusts the volume up until either the threshold or the maximum gain is reached. The rate at which this volume is changed is called the decay rate. The decay rate is programmable with allowed range of 0.8 dB/s to 3200 dB/s. It is desirable to use very slow decay rate to avoid any distortion in the signal and prevent the AVC from entering a continuous attack-decay loop.

Refer to [Automatic Volume Control \(AVC\)](#) and [Automatic Volume Control \(AVC\) On/Off](#) for a programming example that shows how to configure AVC and how to enable/disable AVC respectively.

## CONTROL

The SGTL5000 supports both I<sup>2</sup>C and SPI control modes (note that SPI is not supported in the 20 QFN part). The CTRL\_MODE pin chooses which mode is used. When CTRL\_MODE is tied to ground, the control mode is I<sup>2</sup>C. When CTRL\_MODE is tied to VDDIO, the control mode is SPI.

Regardless of the mode, the control interface is used for all communication with the SGTL5000 including startup configuration, routing, volume, etc.

## I<sup>2</sup>C

The I<sup>2</sup>C port is implemented according to the I<sup>2</sup>C specification v2.0. The I<sup>2</sup>C interface is used to read and write all registers.

For the 32 QFN version of the SGTL5000, the I<sup>2</sup>C device address is 0n01010(R/W) where n is determined by CTRL\_ADR0\_CS and R/W is the read/write bit from the I<sup>2</sup>C protocol.

For the 20 QFN version of the SGTL5000 the I<sup>2</sup>C address is always 0001010(R/W).

The SGTL5000 is always the slave on all transactions, which means that an external master always drives CTRL\_CLK.

In general, an I<sup>2</sup>C transaction looks like the following.

All locations are accessed with a 16 bit address. Each location is 16 bits wide.

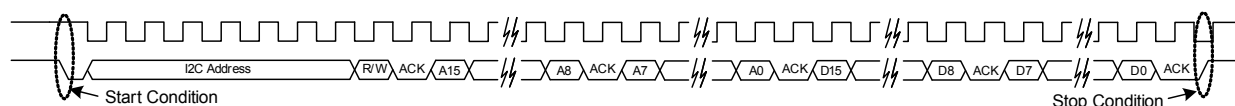
### Example I<sup>2</sup>C write

- Start condition
- Device address with the R/W bit cleared to indicate write
- Send two bytes for the 16 bit register address (most significant byte first)
- Send two bytes for the 16 bits of data to be written to the register (most significant byte first)
- Stop condition

### Example I<sup>2</sup>C read

- Start condition
- Device address with the R/W bit cleared to indicate write
- Send two bytes for the 16 bit register address (most significant byte first)
- Stop Condition followed by start condition (or a single restart condition)
- Device address with the R/W bit set to indicate read
- Read two bytes from the addressed register (most significant byte first)
- Stop condition

[Figure 17](#) shows the functional I<sup>2</sup>C timing diagram.



**Figure 17. Functional I<sup>2</sup>C Diagram**

The protocol has an auto increment feature. Instead of sending the stop condition after two bytes of data, the master may continue to send data byte pairs for writing, or it may send extra clocks for reading data byte pairs. In either case, the access address is incremented after every two bytes of data. A start or stop condition from the I<sup>2</sup>C master interrupts the current command. For reads, unless a new address is written, a new start condition with R/W=0 reads from the current address and continues to auto increment.

The following diagrams describe the different access formats. The gray fields are from the I<sup>2</sup>C master, and the white fields are the SGTL5000 responses. Data [n] corresponds to the data read from the address sent, data[n+1] is the data from the next register, and so on.

S = Start Condition

Sr = Restart Condition

A = Ack

N = Nack

P = Stop Condition

**Table 11. Write Single Location**

|   |                |       |   |             |   |             |   |             |   |             |   |   |
|---|----------------|-------|---|-------------|---|-------------|---|-------------|---|-------------|---|---|
| S | Device Address | W (0) | A | ADDR byte 1 | A | ADDR byte 0 | A | DATA byte 1 | A | DATA byte 0 | A | P |
|---|----------------|-------|---|-------------|---|-------------|---|-------------|---|-------------|---|---|

**Table 12. Write Auto increment**

|   |                |       |   |                   |   |                   |   |                 |   |                 |   |                   |   |                   |   |   |
|---|----------------|-------|---|-------------------|---|-------------------|---|-----------------|---|-----------------|---|-------------------|---|-------------------|---|---|
| S | Device Address | W (0) | A | start ADDR byte 1 | A | start ADDR byte 0 | A | DATA [n] byte 1 | A | DATA [n] byte 0 | A | DATA [n+1] byte 1 | A | DATA [n+1] byte 0 | A | P |
|---|----------------|-------|---|-------------------|---|-------------------|---|-----------------|---|-----------------|---|-------------------|---|-------------------|---|---|

**Table 13. Read Single Location**

|   |                |       |   |             |   |             |   |    |                |       |   |             |   |             |   |   |
|---|----------------|-------|---|-------------|---|-------------|---|----|----------------|-------|---|-------------|---|-------------|---|---|
| S | Device Address | W (0) | A | ADDR byte 1 | A | ADDR byte 0 | A | Sr | Device Address | R (1) | A | DATA byte 1 | A | DATA byte 0 | N | P |
|---|----------------|-------|---|-------------|---|-------------|---|----|----------------|-------|---|-------------|---|-------------|---|---|

**Table 14. Read Auto increment**

|   |                |       |   |                   |   |                   |   |    |                |       |   |                 |   |                 |   |                   |   |                   |   |   |
|---|----------------|-------|---|-------------------|---|-------------------|---|----|----------------|-------|---|-----------------|---|-----------------|---|-------------------|---|-------------------|---|---|
| S | Device Address | W (0) | A | start ADDR byte 1 | A | start ADDR byte 0 | A | Sr | Device Address | R (1) | A | DATA [n] byte 1 | A | DATA [n] byte 0 | A | DATA [n+1] byte 1 | A | DATA [n+1] byte 0 | N | P |
|---|----------------|-------|---|-------------------|---|-------------------|---|----|----------------|-------|---|-----------------|---|-----------------|---|-------------------|---|-------------------|---|---|

**Table 15. Read Continuing Auto increment**

|   |                |   |   |                   |   |                   |   |                   |   |                   |   |   |
|---|----------------|---|---|-------------------|---|-------------------|---|-------------------|---|-------------------|---|---|
| S | Device Address | R | A | DATA [n+2] byte 1 | A | DATA [n+2] byte 0 | A | DATA [n+3] byte 1 | A | DATA [n+3] byte 0 | N | P |
|---|----------------|---|---|-------------------|---|-------------------|---|-------------------|---|-------------------|---|---|

## SPI

Serial Peripheral Interface (SPI) is a communications protocol supported by the SGTL5000 (not supported in the 20 QFN package). The SGTL5000 is always a slave. The CTRL\_ADR0\_CS is used as the slave select (SS) when the master wants to select the SGTL5000 for communication. CTRL\_CLK is connected to master's SCLK and CTRL\_DATA

is connected to master's MOSI line. The part only supports SPI write operations and does not support read operations.

[Figure 18](#) shows the functional timing diagram of the SPI communication protocol as supported by the SGTL5000 chip. Note that on the rising edge of the SS, the chip latches to the previous 32 bits of data. It interprets the latest 16-bits as register value and the 16-bits preceding it as register address.



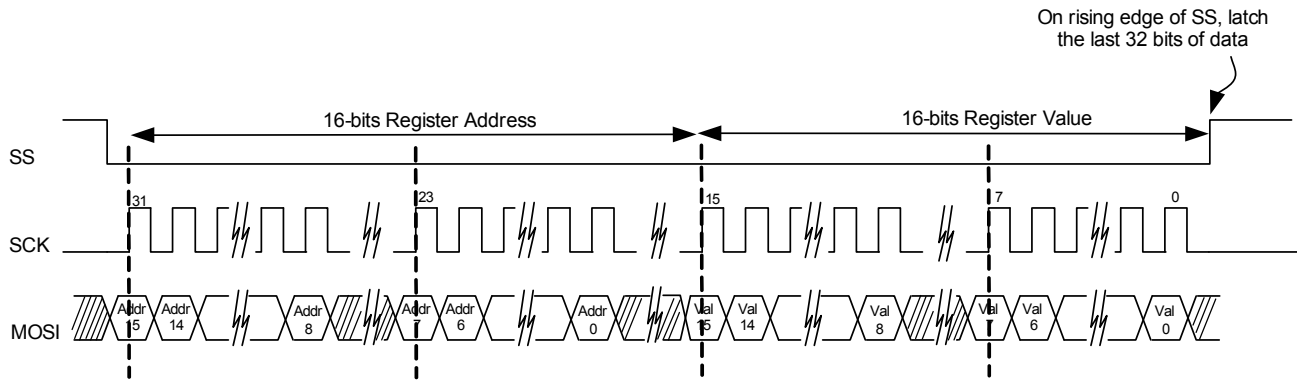


Figure 18. Functional Timing Diagram of SPI Protocol

## PROGRAMMING EXAMPLES

This section provides programming examples showing how to configure the chip. The registers can be written/read by using I<sup>2</sup>C communication protocol. The chip also supports

SPI communication protocol (not supported in the 20 QFN package), but only register write operation is supported.

### PROTOTYPE FOR READING AND WRITING A REGISTER

The generic register read write prototype is used throughout this section, as shown by the following. The I<sup>2</sup>C or SPI implementation is specific to the I<sup>2</sup>C/SPI hardware used in the system.

```
// This prototype writes a value to the entire register. All
// bit-fields of the register will be written.
Write REGISTER REGISTERVALUE
// This prototype writes a value only to the bit-field specified.
// In the actual implementation, the other bit-fields should be
// masked to prevent them from being written. Also, the
// actual implementation should left-shift the BITFIELDVALUE
// by appropriate number to match the starting bit location of
// the BITFIELD.
Modify REGISTER -> BITFIELD, BITFIELDVALUE //Bitfield
Location
// Example implementation
// Modify DAP_EN (bit 0) bit to value 1 to enable DAP block
Modify(DAP_CONTROL_REG, 0xFFFE, 1 <<
DAP_EN_STARTBIT);
// Example Implementation of Modify
void Modify(unsigned short usRegister,
            unsigned short usClearMask,
            unsigned short usSetValue)
{
    unsigned short usData;
    // 1) Read current value
    ReadRegister(usRegister, &usData);
    // 2) Clear out old bits
    usData = usData & usClearMask;
    // 3) set new bit values
    usData = usData | usSetValue;
    // 4) Write out new value created
    WriteRegister(usRegister, usData);
}
```

### CHIP CONFIGURATION

All outputs (LINEOUT, HP\_OUT, I2S\_OUT) are muted by default on power up. To avoid any pops/clicks, the outputs should remain muted during these chip configuration steps. Refer to [Volume Control](#) for volume and mute control.

#### Initialization

#### Chip Powerup and Supply Configurations

After the power supplies for the chip are turned on, the following initialization sequence should be followed. Please note that certain steps may be optional or different values may need to be written based on the power supply voltage

used and desired configuration. The initialization sequence below assumes VDDIO = 3.3 V and VDDA = 1.8 V.

```
//----- Power Supply Configuration-----
// NOTE: This next 2 Write calls is needed ONLY if VDDD is
// internally driven by the chip
// Configure VDDD level to 1.2V (bits 3:0)
Write CHIP_LINREG_CTRL 0x0008
// Power up internal linear regulator (Set bit 9)
Write CHIP_ANA_POWER 0x7260
// NOTE: This next Write call is needed ONLY if VDDD is
// externally driven
// Turn off startup power supplies to save power (Clear bit 12 and
13)
Write CHIP_ANA_POWER 0x4260
// NOTE: The next 2 Write calls is needed only if both VDDA and
// VDDIO power supplies are less than 3.1V.
// Enable the internal oscillator for the charge pump (Set bit 11)
Write CHIP_CLK_TOP_CTRL 0x0800
// Enable charge pump (Set bit 11)
Write CHIP_ANA_POWER 0x4A60
// NOTE: The next 2 modify calls is only needed if both VDDA and
// VDDIO are greater than 3.1 V
// Configure the charge pump to use the VDDIO rail (set bit 5 and
bit 6)
Write CHIP_LINREG_CTRL 0x006C
//--- Reference Voltage and Bias Current Configuration---
// NOTE: The value written in the next 2 Write calls is dependent
// on the VDDA voltage value.
// Set ground, ADC, DAC reference voltage (bits 8:4). The value
should
// be set to VDDA/2. This example assumes VDDA = 1.8 V.
VDDA/2 = 0.9 V.
// The bias current should be set to 50% of the nominal value (bits
3:1)
Write CHIP_REF_CTRL 0x004E
// Set LINEOUT reference voltage to VDDIO/2 (1.65 V) (bits 5:0)
and bias current (bits 11:8) to the recommended value of 0.36 mA
for 10 kOhm load with 1.0 nF capacitance
Write CHIP_LINE_OUT_CTRL 0x0322
//-----Other Analog Block Configurations-----
// Configure slow ramp up rate to minimize pop (bit 0)
Write CHIP_REF_CTRL 0x004F
// Enable short detect mode for headphone left/right
// and center channel and set short detect current trip level
// to 75 mA
Write CHIP_SHORT_CTRL 0x1106
// Enable Zero-cross detect if needed for HP_OUT (bit 5) and ADC
(bit 1)
```

```

Write CHIP_ANA_CTRL      0x0133
//-----Power up Inputs/Outputs/Digital Blocks-----
// Power up LINEOUT, HP, ADC, DAC
Write CHIP_ANA_POWER 0x6AFF
// Power up desired digital blocks
// I2S_IN (bit 0), I2S_OUT (bit 1), DAP (bit 4), DAC (bit 5),
// ADC (bit 6) are powered on
Write CHIP_DIG_POWER    0x0073
//-----Set LINEOUT Volume Level-----
// Set the LINEOUT volume level based on voltage reference
// (VAG)
// values using this formula
// Value = (int)(40*log(VAG_VAL/LO_VAGCNTRL) + 15)
// Assuming VAG_VAL and LO_VAGCNTRL is set to 0.9 V and
// 1.65 V respectively, the // left LO vol (bits 12:8) and right LO
// volume (bits 4:0) value should be set // to 5
Write CHIP_LINE_OUT_VOL 0x0505

```

### System MCLK and Sample Clock

```

// Configure SYS_FS clock to 48 kHz
// Configure MCLK_FREQ to 256*Fs
Modify CHIP_CLK_CTRL->SYS_FS 0x0002 // bits 3:2
Modify CHIP_CLK_CTRL->MCLK_FREQ 0x0000 // bits 1:0
// Configure the I2S clocks in master mode
// NOTE: I2S LRCLK is same as the system sample clock
Modify CHIP_I2S_CTRL->MS 0x0001 // bit 7

```

### PLL Configuration

These programming steps are needed only when the PLL is used. Refer to [Using the PLL - Asynchronous SYS\\_MCLK input](#) for details on when to use the PLL.

To avoid any pops/clicks, the outputs should be muted during these chip configuration steps. Refer to [Volume Control](#) for volume and mute control.

```

// Power up the PLL
Modify CHIP_ANA_POWER->PLL_POWERUP 0x0001 // bit 10
Modify CHIP_ANA_POWER->VCOAMP_POWERUP 0x0001 // bit 8
// NOTE: This step is required only when the external SYS_MCLK
// is above 17 MHz. In this case the external SYS_MCLK clock
// must be divided by 2
Modify CHIP_CLK_TOP_CTRL->INPUT_FREQ_DIV2 0x0001 // bit 3
Sys_MCLK_Input_Freq = Sys_MCLK_Input_Freq/2;
// PLL output frequency is different based on the sample clock
// rate used.
if (Sys_Fs_Rate == 44.1 kHz)
    PLL_Output_Freq = 180.6336 MHz
else
    PLL_Output_Freq = 196.608 MHz
// Set the PLL dividers
Int_Divisor = floor(PLL_Output_Freq/Sys_MCLK_Input_Freq)
Frac_Divisor = ((PLL_Output_Freq/Sys_MCLK_Input_Freq) -
Int_Divisor)*2048
Modify CHIP_PLL_CTRL->INT_DIVISOR Int_Divisor // bits 15:11

```

```

Modify CHIP_PLL_CTRL->FRAC_DIVISOR Frac_Divisor // bits
10:0

```

### Input/Output Routing

To avoid any pops/clicks, the outputs should be muted during these chip configuration steps. Refer to [Volume Control](#) for volume and mute control.

A few example routes are shown below:

```

// Example 1: I2S_IN -> DAP -> DAC -> LINEOUT, HP_OUT
// Route I2S_IN to DAP
Modify CHIP_SSS_CTRL->DAP_SELECT 0x0001 // bits 7:6
// Route DAP to DAC
Modify CHIP_SSS_CTRL->DAC_SELECT 0x0003 // bits 5:4
// Select DAC as the input to HP_OUT
Modify CHIP_ANA_CTRL->SELECT_HP 0x0000 // bit 6
// Example 2: MIC_IN -> ADC -> I2S_OUT
// Set ADC input to MIC_IN
Modify CHIP_ANA_CTRL->SELECT_ADC 0x0000 // bit 2
// Route ADC to I2S_OUT
Modify CHIP_SSS_CTRL->I2S_SELECT 0x0000 // bits 1:0
// Example 3: LINEIN -> HP_OUT
// Select LINEIN as the input to HP_OUT
Modify CHIP_ANA_CTRL->SELECT_HP 0x0001 // bit 6

```

### DIGITAL AUDIO PROCESSOR CONFIGURATION

To avoid any pops/clicks, the outputs should be muted during these chip configuration steps. Refer to [Volume Control](#) for volume and mute control.

```

// Enable DAP block
// NOTE: DAP will be in a pass-through mode if none of DAP
// sub-blocks are enabled.
Modify DAP_CONTROL->DAP_EN 0x0001 // bit 0

```

### Dual Input Mixer

These programming steps are needed only if dual input mixer feature is used.

```

// Enable Dual Input Mixer
Modify DAP_CONTROL->MIX_EN 0x0001 // bit 4
// NOTE: This example assumes mix level of main and mix
// channels as 100% and 50% respectively
// Configure main channel volume to 100% (No change from input
// level)
Write DAP_MAIN_CHAN 0x4000
// Configure mix channel volume to 50% (attenuate the mix
// input level by half)
Write DAP_MIX_CHAN 0x4000

```

### Freescall Surround

The Freescale Surround on/off function is typically controlled by the end-user. End-user driven programming steps are shown in [End-user Driven Chip Configuration](#).

The default WIDTH\_CONTROL of 4 should be appropriate for most applications. This optional programming step shows how to configure a different width value.

```
// Configure the surround width
// (0x0 = Least width, 0x7 = Most width). This example shows
// a width setting of 5
Modify DAP_SGTL_SURROUND->WIDTH_CONTROL 0x0005
// bits 6:4
```

### Freescape Bass Enhance

The Freescape Bass Enhance on/off function is typically controlled by the end-user. End-user driven programming steps are shown in [End-user Driven Chip Configuration](#).

The default LR\_LEVEL value of 0x0005 results in no change in the input signal level and BASS\_LEVEL value of 0x001F adds some harmonic boost to the main signal. The default settings should work for most applications. This optional programming step shows how to configure a different value.

```
// Gain up the input signal level
Modify DAP_BASS_ENHANCE_CTRL->LR_LEVEL 0x0002
// bits 7:4
// Add harmonic boost
Modify DAP_BASS_ENHANCE_CTRL->BASS_LEVEL 0x003F;
// bits 6:0
```

### 7-Band Parametric EQ / 5-Band Graphic EQ / Tone Control

Only one audio EQ block can be used at a given time. The pseudocode in this section shows how to select each block.

Some parameters of the audio EQ are typically controlled by the end-user. End-user driven programming steps are shown in [End-user Driven Chip Configuration](#).

```
// 7-Band PEQ Mode
// Select 7-Band PEQ mode and enable 7 PEQ filters
Write DAP_AUDIO_EQ 0x0001
Write DAP_PEQ 0x0007
// Tone Control mode
Write DAP_AUDIO_EQ 0x0002
// 5-Band GEQ Mode
Write DAP_AUDIO_EQ 0x0003
```

### Automatic Volume Control (AVC)

The AVC on/off function is typically controlled by the end-user. End-user driven programming steps are shown in [End-user Driven Chip Configuration](#).

The default configuration of the AVC should work for most applications. However, the following example shows how to change the configuration if needed.

```
// Configure threshold to -18dB
Write DAP_AVC_THRESHOLD 0x0A40
// Configure attack rate to 16dB/s
Write DAP_AVC_ATTACK 0x0014
// Configure decay rate to 2dB/s
```

Write DAP\_AVC\_DECAY 0x0028

## I<sup>2</sup>S CONFIGURATION

By default the I<sup>2</sup>S port on the chip is configured for 24-bits of data in I<sup>2</sup>S format with SCLK set for 64\*Fs. This can be modified by setting various bit-fields in the CHIP\_I2S\_CTRL register.

## VOLUME CONTROL

The outputs should be unmuted after all the configuration is complete.

```
//----- Input Volume Control-----
// Configure ADC left and right analog volume to desired default.
// Example shows volume of 0dB
Write CHIP_ANA_ADC_CTRL 0x0000
// Configure MIC gain if needed. Example shows gain of 20dB
Modify CHIP_MIC_CTRL->GAIN 0x0001
// bits 1:0
//----- Volume and Mute Control-----
// Configure HP_OUT left and right volume to minimum, unmute
// HP_OUT and ramp the volume up to desired volume.
Write CHIP_ANA_HP_CTRL 0x7F7F
Modify CHIP_ANA_CTRL->MUTE_HP 0x0000
// bit 5
// Code assumes that left and right volumes are set to same value
// So it only uses the left volume for the calculations
usCurrentVolLeft = 0x7F;
usNewVolLeft = usNewVol & 0xFF;
usNumSteps = usNewVolLeft - usCurrentVolLeft;
if (usNumSteps == 0) return;
// Ramp up
for (int i = 0; i < usNumSteps; i++)
{
    ++usCurrentVolLeft;
    usCurrentVol = (usCurrentVolLeft << 8) | (usCurrentVolLeft);
    Write CHIP_ANA_HP_CTRL usCurrentVol;
}
// LINEOUT and DAC volume control
Modify CHIP_ANA_CTRL->MUTE_LO 0x0000
// bit 8
// Configure DAC left and right digital volume. Example shows
// volume of 0dB
Write CHIP_DAC_VOL 0x3C3C
Modify CHIP_ADCDAC_CTRL->DAC_MUTE_LEFT 0x0000
// bit 2
Modify CHIP_ADCDAC_CTRL->DAC_MUTE_RIGHT 0x0000
// bit 3
// Unmute ADC
Modify CHIP_ANA_CTRL->MUTE_ADC 0x0000
// bit 0
```

## END-USER DRIVEN CHIP CONFIGURATION

End-users control features like volume up/down, and audio EQ parameters such as Bass and Treble. This requires programming the chip without introducing any pops/clicks or any other disturbance to the output. This section shows examples on how to program these features.

### VOLUME AND MUTE CONTROL

Refer to **Volume Control** for examples on how to program volume when end-user changes the volume or mutes/unmutes the output. Note that the DAC volume ramp is automatically handled by the chip.

### 7-BAND PEQ PRESET SELECTION

This programming example shows how to load the filter coefficients when the end-user changes PEQ presets such as Rock, Speech, Classical etc.

```
// Load the 5 coefficients for each band and write them to
// appropriate filter address. Repeat this for all enabled
// filters (this example shows 7 filters)
for (i = 0; i < 7; i++)
{
    // Note that each 20-bit coefficient is broken into 16-bit MSB
    // (unsigned short usXXMSB) and 4-bit LSB (unsigned short
    // usXXLSB)
    Write DAP_COEF_WR_B0_LSB usB0MSB[i]
    Write DAP_COEF_WR_B0_MSB usB0LSB[i]
    Write DAP_COEF_WR_B1_LSB usB1MSB[i]
    Write DAP_COEF_WR_B1_MSB usB1LSB[i]
    Write DAP_COEF_WR_B2_LSB usB2MSB[i]
    Write DAP_COEF_WR_B2_MSB usB2LSB[i]
    Write DAP_COEF_WR_A1_LSB usA1MSB[i]
    Write DAP_COEF_WR_A1_MSB usA1LSB[i]
    Write DAP_COEF_WR_A2_LSB usA2MSB[i]
    Write DAP_COEF_WR_A2_MSB usA2LSB[i]
    // Set the index of the filter (bits 7:0) and load the
    // coefficients
    Modify DAP_FILTER_COEF_ACCESS->INDEX (0x0101 + i)
    // bit 8
}
```

### 5-BAND GEQ VOLUME CHANGE

This programming example shows how to program the GEQ volume when end-user changes the volume on any of the 5 bands.

GEQ volume should be ramped in 0.5 dB steps in order to avoid any pops. The example assumes that volume is ramped on Band 0. Other bands can be programmed similarly.

```
// Read current volume set on Band 0
usCurrentVol = Read DAP_AUDIO_EQ_BASS_BAND0
// Convert the new volume to hex value
usNewVol = 4*dNewVolDb + 47;
// Calculate the number of steps
```

```
usNumSteps = abs(usNewVol - usCurrentVol);
if (usNumSteps == 0) return;
for (int i = 0; i++; usNumSteps)
{
    if (usNewVol > usCurrentVol)
        ++usCurrentVol;
    else
        --usCurrentVol;
    Write DAP_AUDIO_EQ_BASS_BAND0 usCurrentVol;
}
```

### tone control - Bass and Treble Change

This programming example shows how to program the Tone Control Bass and Treble when end-user changes it on the fly.

Tone Control Bass and Treble volume should be ramped in 0.5 dB steps in order to avoid any pops. The example assumes that Treble is changed to a new value. Bass can be programmed similarly.

```
// Read current Treble value
usCurrentVal = Read DAP_AUDIO_EQ_TREBLE_BAND4
// Convert the new Treble value to hex value
usNewVol = 4*dNewVolDb + 47;
// Calculate the number of steps
usNumSteps = abs(usNewVal - usCurrentVal);
if (usNumSteps == 0) return;
for (int i = 0; i++; usNumSteps)
{
    if (usNewVal > usCurrentVal)
        ++usCurrentVal;
    else
        --usCurrentVal;
    Write DAP_AUDIO_EQ_TREBLE_BAND4 usCurrentVal;
}
```

### FREESCALE SURROUND ON/OFF

This programming example shows how to program the Surround when end-user turns it on/off on their device.

The Surround width should be ramped up to highest value before enabling/disabling the Surround to avoid any pops.

```
// Read current Surround width value
// WIDTH_CONTROL bits 6:4
usOriginalVal = (Read DAP_SGTL_SURROUND >> 4) &&
0x0003;
usNextVal = usOriginalVal;
// Ramp up the width to maximum value of 7
for (int i = 0; i++; (7 - usOriginalVal))
{
    ++usNextVal;
    Modify DAP_SGTL_SURROUND->WIDTH_CONTROL
    usNextVal;
}
// Enable (To disable, write 0x0000) Surround
```

```
// SELECT bits 1:0
Modify DAP_SGTL_SURROUND->SELECT 0x0003;
// Ramp down the width to original value
for (int i = 0; i++; (7 - usOriginalVal))
{
    --usNextVal;
    Modify DAP_SGTL_SURROUND->WIDTH_CONTROL
    usNextVal;
}
```

## BASS ENHANCE ON/OFF

This programming example shows how to program the Bass Enhance on/off when end-user turns it on/off on their device.

The Bass level should be ramped down to the lowest Bass before Bass Enhance feature is turned on/off.

```
// Read current Bass level value
// BASS_LEVEL bits 6:0
usOriginalVal = Read DAP_BASS_ENHANCE_CTRL &&
0x007F;
usNextVal = usOriginalVal;
// Ramp Bass level to lowest bass (lowest bass = 0x007F)
usNumSteps = abs(0x007F - usOriginalVal);
for (int i = 0; i++; usNumSteps)
```

```
{
    ++usNextVal;
    Modify DAP_BASS_ENHANCE_CTRL->BASS_LEVEL
    usNextVal;
}
// Enable (To disable, write 0x0000) Bass Enhance
// EN bit 0
Modify DAP_BASS_ENHANCE->EN 0x0001;
// Ramp Bass level back to original value
for (int i = 0; i++; usNumSteps)
{
    --usNextVal;
    Modify DAP_BASS_ENHANCE_CTRL->BASS_LEVEL
    usNextVal;
}
```

## AUTOMATIC VOLUME CONTROL (AVC) ON/OFF

This programming example shows how to program the AVC on/off when end-user turns it on/off on their device.

```
// Enable AVC (To disable, write 0x0000)
Modify DAP_AVC_CTRL->EN 0x0001
// bit 0
Register description
CHIP_ID      0x0000
```

**Table 16. CHIP\_ID 0x0000**

| 15     | 14     | 13 | 12 | 11    | 10                                                           | 9 | 8 | 7     | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|--------|--------|----|----|-------|--------------------------------------------------------------|---|---|-------|---|---|---|---|---|---|---|
| PARTID |        |    |    |       |                                                              |   |   | REVID |   |   |   |   |   |   |   |
| BITS   | FIELD  |    | RW | RESET | DEFINITION                                                   |   |   |       |   |   |   |   |   |   |   |
| 15:8   | PARTID |    | RO | 0xA0  | SGTL5000 Part ID<br>0xA0 - 8 bit identifier for SGTL5000     |   |   |       |   |   |   |   |   |   |   |
| 7:0    | REVID  |    | RO | 0x00  | SGTL5000 Revision ID<br>0xHH - revision number for SGTL5000. |   |   |       |   |   |   |   |   |   |   |

**Table 17. CHIP\_DIG\_POWER 0x0002**

| 15   | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6           | 5           | 4           | 3    | 2 | 1 | 0               |                |
|------|----|----|----|----|----|---|---|---|-------------|-------------|-------------|------|---|---|-----------------|----------------|
| RSVD |    |    |    |    |    |   |   |   | ADC_POWERUP | DAC_POWERUP | DAP_POWERUP | RSVD |   |   | I2S_OUT_POWERUP | I2S_IN_POWERUP |

| BITS | FIELD       | RW | RESET | DEFINITION                                                                             |
|------|-------------|----|-------|----------------------------------------------------------------------------------------|
| 15:7 | RSVD        | RO | 0x0   | Reserved                                                                               |
| 6    | ADC_POWERUP | RW | 0x0   | Enable/disable the ADC block, both digital and analog<br>0x0 = Disable<br>0x1 = Enable |

| BITS | FIELD           | RW | RESET | DEFINITION                                                                             |
|------|-----------------|----|-------|----------------------------------------------------------------------------------------|
| 5    | DAC_POWERUP     | RW | 0x0   | Enable/disable the DAC block, both analog and digital<br>0x0 = Disable<br>0x1 = Enable |
| 4    | DAP_POWERUP     | RW | 0x0   | Enable/disable the DAP block<br>0x0 = Disable<br>0x1 = Enable                          |
| 3:2  | RSVD            | RW | 0x0   | Reserved                                                                               |
| 1    | I2S_OUT_POWERUP | RW | 0x0   | Enable/disable the I2S data output<br>0x0 = Disable<br>0x1 = Enable                    |
| 0    | I2S_IN_POWERUP  | RW | 0x0   | Enable/disable the I2S data input<br>0x0 = Disable<br>0x1 = Enable                     |

**Table 18. CHIP\_CLK\_CTRL 0x0004**

| 15   | 14 | 13        | 12 | 11 | 10    | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 8 | 7 | 6 | 5         | 4 | 3      | 2 | 1         | 0 |
|------|----|-----------|----|----|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|-----------|---|--------|---|-----------|---|
| RSVD |    |           |    |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   |   | RATE_MODE |   | SYS_FS |   | MCLK_FREQ |   |
| BITS |    | FIELD     |    | RW | RESET | DEFINITION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |   |   |           |   |        |   |           |   |
| 15:6 |    | RSVD      |    | RO | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |   |   |           |   |        |   |           |   |
| 5:4  |    | RATE_MODE |    | RW | 0x0   | Sets the sample rate mode. MCLK_FREQ is still specified relative to the rate in SYS_FS<br>0x0 = SYS_FS specifies the rate<br>0x1 = Rate is 1/2 of the SYS_FS rate<br>0x2 = Rate is 1/4 of the SYS_FS rate<br>0x3 = Rate is 1/6 of the SYS_FS rate                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |   |   |           |   |        |   |           |   |
| 3:2  |    | SYS_FS    |    | RW | 0x2   | Sets the internal system sample rate<br>0x0 = 32 kHz<br>0x1 = 44.1 kHz<br>0x2 = 48 kHz<br>0x3 = 96 kHz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |   |           |   |        |   |           |   |
| 1:0  |    | MCLK_FREQ |    | RW | 0x0   | Identifies incoming SYS_MCLK frequency and if the PLL should be used<br>0x0 = 256*Fs<br>0x1 = 384*Fs<br>0x2 = 512*Fs<br>0x3 = Use PLL<br><br>The 0x3 (Use PLL) setting must be used if the SYS_MCLK is not a standard multiple of Fs (256, 384, or 512). This setting can also be used if SYS_MCLK is a standard multiple of Fs. Before this field is set to 0x3 (Use PLL), the PLL must be powered up by setting CHIP_ANA_POWER->PLL_POWERUP and CHIP_ANA_POWER->VCOAMP_POWERUP. Also, the PLL dividers must be calculated based on the external MCLK rate and CHIP_PLL_CTRL register must be set (see CHIP_PLL_CTRL register description details on how to calculate the divisors). |   |   |   |           |   |        |   |           |   |



**Table 19. CHIP\_I2S\_CTRL 0x0006**

| 15   | 14 | 13 | 12 | 11 | 10 | 9 | 8        | 7  | 6        | 5    | 4 | 3        | 2 | 1       | 0     |
|------|----|----|----|----|----|---|----------|----|----------|------|---|----------|---|---------|-------|
| RSVD |    |    |    |    |    |   | SCLKFREQ | MS | SCLK_INV | DLEN |   | I2S_MODE |   | LRALIGN | LRPOL |

| BITS | FIELD    | RW | RESET | DEFINITION                                                                                                                                                                                                                                                                               |
|------|----------|----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:9 | RSVD     | RO | 0x0   | Reserved                                                                                                                                                                                                                                                                                 |
| 8    | SCLKFREQ | RW | 0x0   | Sets frequency of I2S_SCLK when in master mode (MS=1). When in slave mode (MS=0), this field must be set appropriately to match SCLK input rate.<br>0x0 = 64Fs<br>0x1 = 32Fs - Not supported for RJ mode (I2S_MODE = 1)                                                                  |
| 7    | MS       | RW | 0x0   | Configures master or slave of I2S_LRCLK and I2S_SCLK. 0x0 = Slave: I2S_LRCLK and I2S_SCLK are inputs<br>0x1 = Master: I2S_LRCLK and I2S_SCLK are outputs<br>NOTE: If the PLL is used (CHIP_CLK_CTRL->MCLK_FREQ==0x3), the SGTL5000 must be a master of the I <sup>2</sup> S port (MS==1) |
| 6    | SCLK_INV | RW | 0x0   | Sets the edge that data (input and output) is clocked in on for I2S_SCLK<br>0x0 = data is valid on rising edge of I2S_SCLK<br>0x1 = data is valid on falling edge of I2S_SCLK                                                                                                            |
| 5:4  | DLEN     | RW | 0x1   | I <sup>2</sup> S data length<br>0x0 = 32 bits (only valid when SCLKFREQ=0), not valid for Right Justified Mode<br>0x1 = 24 bits (only valid when SCLKFREQ=0)<br>0x2 = 20 bits<br>0x3 = 16 bits                                                                                           |
| 3:2  | I2S_MODE | RW | 0x0   | Sets the mode for the I <sup>2</sup> S port<br>0x0 = I <sup>2</sup> S mode or Left Justified (Use LRALIGN to select)<br>0x1 = Right Justified Mode<br>0x2 = PCM Format A/B<br>0x3 = RESERVED                                                                                             |
| 1    | LRALIGN  | RW | 0x0   | I2S_LRCLK Alignment to data word. Not used for Right Justified mode<br>0x0 = Data word starts 1 I2S_SCLK delay after I2S_LRCLK transition (I <sup>2</sup> S format, PCM format A)<br>0x1 = Data word starts after I2S_LRCLK transition (left justified format, PCM format B)             |
| 0    | LRPOL    | RW | 0x0   | I2S_LRCLK Polarity when data is presented.<br>0x0 = I2S_LRCLK = 0 - Left, 1 - Right<br>1x0 = I2S_LRCLK = 0 - Right, 1 - Left<br>The left subframe should be presented first regardless of the setting of LRPOL.                                                                          |



**Table 20. CHIP\_SSS\_CTRL 0x000A**

| 15   | 14             | 13         | 12         | 11   | 10         | 9              | 8 | 7          | 6 | 5          | 4 | 3    | 2 | 1          | 0 |
|------|----------------|------------|------------|------|------------|----------------|---|------------|---|------------|---|------|---|------------|---|
| RSVD | DAP_MIX_LRSWAP | DAP_LRSWAP | DAC_LRSWAP | RSVD | I2S_LRSWAP | DAP_MIX_SELECT |   | DAP_SELECT |   | DAC_SELECT |   | RSVD |   | I2S_SELECT |   |

| BITS | FIELD          | RW | RESET | DEFINITION                                                                                                           |
|------|----------------|----|-------|----------------------------------------------------------------------------------------------------------------------|
| 15   | RSVD           | RW | 0x0   | Reserved                                                                                                             |
| 14   | DAP_MIX_LRSWAP | RW | 0x0   | DAP Mixer Input Swap<br>0x0 = Normal Operation<br>0x1 = Left and Right channels for the DAP MIXER Input are swapped. |
| 13   | DAP_LRSWAP     | RW | 0x0   | DAP Input Swap<br>0x0 = Normal Operation<br>0x1 = Left and Right channels for the DAP Input are swapped              |
| 12   | DAC_LRSWAP     | RW | 0x0   | DAC Input Swap<br>0x0 = Normal Operation<br>0x1 = Left and Right channels for the DAC are swapped                    |
| 11   | RSVD           | RW | 0x0   | Reserved                                                                                                             |
| 10   | I2S_LRSWAP     | RW | 0x0   | I2S_DOUT Swap<br>0x0 = Normal Operation<br>0x1 = Left and Right channels for the I2S_DOUT are swapped                |
| 9:8  | DAP_MIX_SELECT | RW | 0x0   | Select data source for DAP mixer<br>0x0 = ADC<br>0x1 = I2S_IN<br>0x2 = Reserved<br>0x3 = Reserved                    |
| 7:6  | DAP_SELECT     | RW | 0x0   | Select data source for DAP<br>0x0 = ADC<br>0x1 = I2S_IN<br>0x2 = Reserved<br>0x3 = Reserved                          |
| 5:4  | DAC_SELECT     | RW | 0x1   | Select data source for DAC<br>0x0 = ADC<br>0x1 = I2S_IN<br>0x2 = Reserved<br>0x3 = DAP                               |
| 3:2  | RSVD           | RW | 0x0   | Reserved                                                                                                             |
| 1:0  | I2S_SELECT     | WO | 0x0   | Select data source for I2S_DOUT<br>0x0 = ADC<br>0x1 = I2S_IN<br>0x2 = Reserved<br>0x3 = DAP                          |

**Table 21. CHIP\_ADCDAC\_CTRL 0x000E**

| 15   | 14 | 13                 | 12                | 11   | 10 | 9           | 8             | 7    | 6 | 5 | 4 | 3              | 2             | 1              | 0              |
|------|----|--------------------|-------------------|------|----|-------------|---------------|------|---|---|---|----------------|---------------|----------------|----------------|
| RSVD |    | VOL_BUSY_DAC_RIGHT | VOL_BUSY_DAC_LEFT | RSVD |    | VOL_RAMP_EN | VOL_EXPO_RAMP | RSVD |   |   |   | DAC_MUTE_RIGHT | DAC_MUTE_LEFT | ADC_HPF_FREEZE | ADC_HPF_BYPASS |

| BITS  | FIELD              | RW | RESET | DEFINITION                                                                                                                                                                                                                                                          |
|-------|--------------------|----|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:14 | RSVD               | RO | 0x0   | Reserved                                                                                                                                                                                                                                                            |
| 13    | VOL_BUSY_DAC_RIGHT | RO | 0x0   | Volume Busy DAC Right<br>0x0 = Ready<br>0x1 = Busy - This indicates the channel has not reached its programmed volume/mute level                                                                                                                                    |
| 12    | VOL_BUSY_DAC_LEFT  | RO | 0x0   | Volume Busy DAC Left<br>0x0 = Ready<br>0x1 = Busy - This indicates the channel has not reached its programmed volume/mute level                                                                                                                                     |
| 11:10 | RSVD               | RO | 0x0   | Reserved                                                                                                                                                                                                                                                            |
| 9     | VOL_RAMP_EN        | RW | 0x1   | Volume Ramp Enable<br>0x0 = Disables volume ramp. New volume settings take immediate effect without a ramp<br>0x1 = Enables volume ramp<br>This field affects DAC_VOL. The volume ramp effects both volume settings and mute. When set to 1 a soft mute is enabled. |
| 8     | VOL_EXPO_RAMP      | RW | 0x0   | Exponential Volume Ramp Enable<br>0x0 = Linear ramp over top 4 volume octaves<br>0x1 = Exponential ramp over full volume range<br>This bit only takes effect if VOL_RAMP_EN is 1.                                                                                   |
| 7:4   | RSVD               | RW | 0x0   | Reserved                                                                                                                                                                                                                                                            |
| 3     | DAC_MUTE_RIGHT     | RW | 0x1   | DAC Right Mute<br>0x0 = Unmute<br>0x1 = Muted<br>If VOL_RAMP_EN = 1, this is a soft mute.                                                                                                                                                                           |
| 2     | DAC_MUTE_LEFT      | RW | 0x1   | DAC Left Mute<br>0x0 = Unmute<br>0x1 = Muted<br>If VOL_RAMP_EN = 1, this is a soft mute.                                                                                                                                                                            |
| 1     | ADC_HPF_FREEZE     | RW | 0x0   | ADC High Pass Filter Freeze<br>0x0 = Normal operation<br>0x1 = Freeze the ADC high-pass filter offset register. The offset continues to be subtracted from the ADC data stream.                                                                                     |
| 0     | ADC_HPF_BYPASS     | RW | 0x0   | ADC High Pass Filter Bypass<br>0x0 = Normal operation<br>0x1 = Bypassed and offset not updated                                                                                                                                                                      |

**Table 22. CHIP\_DAC\_VOL 0x0010**

|               |               |    |    |       |                                                                                                                                                                                                                                                                                           |   |   |              |   |   |   |   |   |   |   |
|---------------|---------------|----|----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--------------|---|---|---|---|---|---|---|
| 15            | 14            | 13 | 12 | 11    | 10                                                                                                                                                                                                                                                                                        | 9 | 8 | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| DAC_VOL_RIGHT |               |    |    |       |                                                                                                                                                                                                                                                                                           |   |   | DAC_VOL_LEFT |   |   |   |   |   |   |   |
| BITS          | FIELD         |    | RW | RESET | DEFINITION                                                                                                                                                                                                                                                                                |   |   |              |   |   |   |   |   |   |   |
| 15:8          | DAC_VOL_RIGHT |    | RW | 0x3C  | DAC Right Channel Volume<br>Set the Right channel DAC volume with 0.5017 dB steps from 0 to -90 dB<br>0x3B and less = Reserved<br>0x3C = 0 dB<br>0x3D = -0.5 dB<br>0xF0 = -90 dB<br>0xFC and greater = Muted<br>If VOL_RAMP_EN = 1, there is an automatic ramp to the new volume setting. |   |   |              |   |   |   |   |   |   |   |
| 7:0           | DAC_VOL_LEFT  |    | RW | 0x3C  | DAC Left Channel Volume<br>Set the Left channel DAC volume with 0.5017 dB steps from 0 to -90 dB<br>0x3B and less = Reserved<br>0x3C = 0 dB<br>0x3D = -0.5 dB<br>0xF0 = -90 dB<br>0xFC and greater = Muted<br>If VOL_RAMP_EN = 1, there is an automatic ramp to the new volume setting.   |   |   |              |   |   |   |   |   |   |   |

**Table 23. CHIP\_PAD\_STRENGTH 0x0014**

| 15    | 14        | 13    | 12    | 11 | 10    | 9                                                                                                                                                                                                                                                                                                                                                                                                               | 8 | 7        | 6 | 5        | 4 | 3         | 2 | 1        | 0 |       |       |       |       |
|-------|-----------|-------|-------|----|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------|---|----------|---|-----------|---|----------|---|-------|-------|-------|-------|
| RSVD  |           |       |       |    |       | I2S_LRCLK                                                                                                                                                                                                                                                                                                                                                                                                       |   | I2S_SCLK |   | I2S_DOUT |   | CTRL_DATA |   | CTRL_CLK |   |       |       |       |       |
| BITS  | FIELD     |       |       | RW | RESET | DEFINITION                                                                                                                                                                                                                                                                                                                                                                                                      |   |          |   |          |   |           |   |          |   |       |       |       |       |
| 15:14 | RSVD      |       |       | RW | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                                                        |   |          |   |          |   |           |   |          |   |       |       |       |       |
| 9:8   | I2S_LRCLK |       |       | RW | 0x1   | <div>I<sup>2</sup>S LRCLK Pad Drive Strength</div> <div>Sets drive strength for output pads per the table below.</div> <table><tr><td>VDDIO</td><td>1.8 V</td><td>2.5 V</td><td>3.3 V</td></tr></table> <div>0x0 = Disable</div> <div>0x1 =            1.66 mA    2.87 mA    4.02 mA</div> <div>0x2 =            3.33 mA    5.74 mA    8.03 mA</div> <div>0x3 =            4.99 mA    8.61 mA    12.05 mA</div> |   |          |   |          |   |           |   |          |   | VDDIO | 1.8 V | 2.5 V | 3.3 V |
| VDDIO | 1.8 V     | 2.5 V | 3.3 V |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                 |   |          |   |          |   |           |   |          |   |       |       |       |       |
| 7:6   | I2S_SCLK  |       |       | RW | 0x1   | <div>I<sup>2</sup>S SCLK Pad Drive Strength</div> <div>Sets drive strength for output pads per the table below.</div> <table><tr><td>VDDIO</td><td>1.8 V</td><td>2.5 V</td><td>3.3 V</td></tr></table> <div>0x0 = Disable</div> <div>0x1 =            1.66 mA    2.87 mA    4.02 mA</div> <div>0x2 =            3.33 mA    5.74 mA    8.03 mA</div> <div>0x3 =            4.99 mA    8.61 mA    12.05 mA</div>  |   |          |   |          |   |           |   |          |   | VDDIO | 1.8 V | 2.5 V | 3.3 V |
| VDDIO | 1.8 V     | 2.5 V | 3.3 V |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                 |   |          |   |          |   |           |   |          |   |       |       |       |       |

| BITS | FIELD     | RW | RESET | DEFINITION                                                                                                                                                                                                                                                                                           |
|------|-----------|----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5:4  | I2S_DOUT  | RW | 0x1   | I <sup>2</sup> S DOUT Pad Drive Strength<br>Sets drive strength for output pads per the table below.<br>VDDIO      1.8 V      2.5 V      3.3 V<br>0x0 = Disable<br>0x1 =      1.66 mA    2.87 mA    4.02 mA<br>0x2 =      3.33 mA    5.74 mA    8.03 mA<br>0x3 =      4.99 mA    8.61 mA    12.05 mA |
| 3:2  | CTRL_DATA | RW | 0x3   | I <sup>2</sup> C DATA Pad Drive Strength<br>Sets drive strength for output pads per the table below.<br>VDDIO      1.8 V      2.5 V      3.3 V<br>0x0 = Disable<br>0x1 =      1.66 mA    2.87 mA    4.02 mA<br>0x2 =      3.33 mA    5.74 mA    8.03 mA<br>0x3 =      4.99 mA    8.61 mA    12.05 mA |
| 1:0  | CTRL_CLK  | RW | 0x3   | I <sup>2</sup> C CLK Pad Drive Strength<br>Sets drive strength for output pads per the table below.<br>VDDIO      1.8 V      2.5 V      3.3 V<br>0x0 = Disable<br>0x1 =      1.66 mA    2.87 mA    4.02 mA<br>0x2 =      3.33 mA    5.74 mA    8.03 mA<br>0x3 =      4.99 mA    8.61 mA    12.05 mA  |

Table 24. CHIP\_ANA\_ADC\_CTRL 0x0020

| 15   | 14 | 13 | 12 | 11 | 10 | 9 | 8            | 7             | 6 | 5 | 4 | 3            | 2 | 1 | 0 |
|------|----|----|----|----|----|---|--------------|---------------|---|---|---|--------------|---|---|---|
| RSVD |    |    |    |    |    |   | ADC_VOL_M6DB | ADC_VOL_RIGHT |   |   |   | ADC_VOL_LEFT |   |   |   |

| BITS | FIELD        | RW | RESET | DEFINITION                                                                                                                                                                                        |
|------|--------------|----|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:9 | RSVD         | RO | 0x0   | Reserved                                                                                                                                                                                          |
| 8    | ADC_VOL_M6DB | RW | 0x0   | <p>ADC Volume Range Reduction</p> <p>This bit shifts both right and left analog ADC volume range down by 6.0 dB.</p> <p>0x0 = No change in ADC range</p> <p>0x1 = ADC range reduced by 6.0 dB</p> |

| BITS | FIELD         | RW | RESET | DEFINITION                                                                                                                                                                                                              |
|------|---------------|----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:4  | ADC_VOL_RIGHT | RW | 0x0   | ADC Right Channel Volume<br>Right channel analog ADC volume control in 1.5.0 dB steps.<br>0x0 = 0 dB<br>0x1 = +1.5 dB<br>...<br>0xF = +22.5 dB<br>This range is -6.0 dB to +16.5 dB if <i>ADC_VOL_M6DB</i> is set to 1. |
| 3:0  | ADC_VOL_LEFT  | RW | 0x0   | ADC Left Channel Volume<br>Left channel analog ADC volume control in 1.5 dB steps.<br>0x0 = 0 dB<br>0x1 = +1.5 dB<br>...<br>0xF = +22.5 dB<br>This range is -6.0 dB to +16.5 dB if <i>ADC_VOL_M6DB</i> is set to 1.     |

**Table 25. CHIP\_ANA\_HP\_CTRL 0x0022**

|      |              |    |    |       |                                                                                                                                                                          |   |   |      |             |   |   |   |   |   |   |
|------|--------------|----|----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|------|-------------|---|---|---|---|---|---|
| 15   | 14           | 13 | 12 | 11    | 10                                                                                                                                                                       | 9 | 8 | 7    | 6           | 5 | 4 | 3 | 2 | 1 | 0 |
| RSVD | HP_VOL_RIGHT |    |    |       |                                                                                                                                                                          |   |   | RSVD | HP_VOL_LEFT |   |   |   |   |   |   |
| BITS | FIELD        |    | RW | RESET | DEFINITION                                                                                                                                                               |   |   |      |             |   |   |   |   |   |   |
| 15   | RSVD         |    | RO | 0x0   | Reserved                                                                                                                                                                 |   |   |      |             |   |   |   |   |   |   |
| 14:8 | HP_VOL_RIGHT |    | RW | 0x18  | Headphone Right Channel Volume<br>Right channel headphone volume control with 0.5 dB steps.<br>0x00 = +12 dB<br>0x01 = +11.5 dB<br>0x18 = 0 dB<br>...<br>0x7F = -51.5 dB |   |   |      |             |   |   |   |   |   |   |
| 7    | RSVD         |    | RO | 0x0   | Reserved                                                                                                                                                                 |   |   |      |             |   |   |   |   |   |   |
| 6:0  | HP_VOL_LEFT  |    | RW | 0x18  | Headphone Left Channel Volume<br>Left channel headphone volume control with 0.5 dB steps.<br>0x00 = +12 dB<br>0x01 = +11.5 dB<br>0x18 = 0 dB<br>...<br>0x7F = -51.5 dB   |   |   |      |             |   |   |   |   |   |   |

[Table 26](#) is an analog control register that includes mutes, input selects, and zero-cross-detectors for the ADC, headphone, and LINEOUT.

**Table 26. 7.0.0.11. CHIP\_ANA\_CTRL 0x0024**

| 15   | 14 | 13 | 12 | 11 | 10 | 9 | 8       | 7    | 6         | 5         | 4       | 3    | 2          | 1          | 0        |
|------|----|----|----|----|----|---|---------|------|-----------|-----------|---------|------|------------|------------|----------|
| RSVD |    |    |    |    |    |   | MUTE_LO | RSVD | SELECT_HP | EN_ZCD_HP | MUTE_HP | RSVD | SELECT_ADC | EN_ZCD_ADC | MUTE_ADC |

| BITS | FIELD      | RW | RESET | DEFINITION                                                                                         |
|------|------------|----|-------|----------------------------------------------------------------------------------------------------|
| 15:9 | RSVD       | RO | 0x0   | Reserved                                                                                           |
| 8    | MUTE_LO    | RW | 0x1   | LINEOUT Mute<br>0x0 = Unmute<br>0x1 = Mute                                                         |
| 7    | RSVD       | RO | 0x0   | Reserved                                                                                           |
| 6    | SELECT_HP  | RW | 0x0   | Select the headphone input.<br>0x0 = DAC<br>0x1 = LINEIN                                           |
| 5    | EN_ZCD_HP  | RW | 0x0   | Enable the headphone zero cross detector (ZCD)<br>0x0 = HP ZCD disabled<br>0x1 = HP ZCD enabled    |
| 4    | MUTE_HP    | RW | 0x1   | Mute the headphone outputs<br>0x0 = Unmute<br>0x1 = Mute                                           |
| 3    | RSVD       | RO | 0x0   | Reserved                                                                                           |
| 2    | SELECT_ADC | RW | 0x0   | Select the ADC input.<br>0x0 = Microphone<br>0x1 = LINEIN                                          |
| 1    | EN_ZCD_ADC | RW | 0x0   | Enable the ADC analog zero cross detector (ZCD)<br>0x0 = ADC ZCD disabled<br>0x1 = ADC ZCD enabled |
| 0    | MUTE_ADC   | RW | 0x1   | Mute the ADC analog volume<br>0x0 = Unmute<br>0x1 = Mute                                           |

The [Table 27, CHIP\\_LINREG\\_CTRL 0x0026](#) register controls the VDDD linear regulator and the charge pump.

**Table 27. CHIP\_LINREG\_CTRL 0x0026**

| 15   | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6             | 5              | 4    | 3             | 2 | 1 | 0 |
|------|----|----|----|----|----|---|---|---|---------------|----------------|------|---------------|---|---|---|
| RSVD |    |    |    |    |    |   |   |   | VDDC_MAN_ASSN | VDDC_ASSN_OVRD | RSVD | D_PROGRAMMING |   |   |   |

| BITS | FIELD          | RW | RESET | DEFINITION                                                                                                                                                                                                                                                                                                                                                                             |
|------|----------------|----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:7 | RSVD           | RO | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                               |
| 6    | VDDC_MAN_ASSN  | RW | 0x0   | Determines chargepump source when VDDC_ASSN_OVRD is set.<br>0x0 = VDDA<br>0x1 = VDDIO                                                                                                                                                                                                                                                                                                  |
| 5    | VDDC_ASSN_OVRD | RW | 0x0   | Charge pump Source Assignment Override<br>0x0 = Charge pump source is automatically assigned based on higher of VDDA and VDDIO<br><br>0x1 = the source of charge pump is manually assigned by VDDC_MAN_ASSN<br><br>If VDDIO and VDDA are both the same and greater than 3.1 V, VDDC_ASSN_OVRD and VDDC_MAN_ASSN should be used to manually assign VDDIO as the source for charge pump. |
| 4    | RSVD           | RW | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                               |
| 3:0  | D_PROGRAMMING  | RW | 0x0   | Sets the VDDD linear regulator output voltage in 50 mV steps. Must clear the LINREG_SIMPLE_POWERUP and STARTUP_POWERUP bits in the 0x0030 register after power-up, for this setting to produce the proper VDDD voltage.<br><br>0x0 = 1.60<br><br>0xF = 0.85                                                                                                                            |

The [Table 28. CHIP\\_REF\\_CTRL 0x0028](#) register controls the bandgap reference bias voltage and currents.

**Table 28. CHIP\_REF\_CTRL 0x0028**

| 15   | 14      | 13 | 12 | 11 | 10    | 9                                                                                                                                                                                                                                                                                                                                                                                                                        | 8       | 7 | 6 | 5 | 4 | 3         | 2 | 1 | 0         |
|------|---------|----|----|----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---|---|---|---|-----------|---|---|-----------|
| RSVD |         |    |    |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                          | VAG_VAL |   |   |   |   | BIAS_CTRL |   |   | SMALL_POP |
|      |         |    |    |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                          |         |   |   |   |   |           |   |   |           |
|      |         |    |    |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                          |         |   |   |   |   |           |   |   |           |
|      |         |    |    |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                          |         |   |   |   |   |           |   |   |           |
| BITS | FIELD   |    |    | RW | RESET | DEFINITION                                                                                                                                                                                                                                                                                                                                                                                                               |         |   |   |   |   |           |   |   |           |
| 15:9 | RSVD    |    |    | RO | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                 |         |   |   |   |   |           |   |   |           |
| 8:4  | VAG_VAL |    |    | RW | 0x0   | Analog Ground Voltage Control<br><br>These bits control the analog ground voltage in 25 mV steps. This should usually be set to VDDA/2 or lower for best performance (maximum output swing at minimum THD). This VAG reference is also used for the DAC and ADC voltage reference. So changing this voltage scales the output swing of the DAC and the output signal of the ADC.<br><br>0x00 = 0.800 V<br>0x1F = 1.575 V |         |   |   |   |   |           |   |   |           |

| BITS | FIELD     | RW | RESET | DEFINITION                                                                                                                                                                                                                                                                                                                                        |
|------|-----------|----|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3:1  | BIAS_CTRL | RW | 0x0   | <p>Bias control</p> <p>These bits adjust the bias currents for all of the analog blocks. By lowering the bias current a lower quiescent power is achieved. It should be noted that this mode can affect performance by 3-4 dB.</p> <p>0x0 = Nominal</p> <p>0x1-0x3=+12.5%</p> <p>0x4=-12.5%</p> <p>0x5=-25%</p> <p>0x6=-37.5%</p> <p>0x7=-50%</p> |
| 0    | SMALL_POP | RW | 0x0   | <p>VAG Ramp Control</p> <p>Setting this bit slows down the VAG ramp from ~200 to ~400 ms to reduce the startup pop, but increases the turn on/off time.</p> <p>0x0 = Normal VAG ramp</p> <p>0x1 = Slow down VAG ramp</p>                                                                                                                          |

The [Table 29. CHIP\\_MIC\\_CTRL 0x002A](#) register controls the microphone gain and the internal microphone biasing circuitry.

**Table 29. CHIP\_MIC\_CTRL 0x002A**

| 15    | 14 | 13            | 12 | 11 | 10    | 9             | 8                                                                                                                                                                                                                                                                    | 7    | 6         | 5 | 4 | 3    | 2 | 1    | 0 |
|-------|----|---------------|----|----|-------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------|---|---|------|---|------|---|
| RSVD  |    |               |    |    |       | BIAS_RESISTOR |                                                                                                                                                                                                                                                                      | RSVD | BIAS_VOLT |   |   | RSVD |   | GAIN |   |
| BITS  |    | FIELD         |    | RW | RESET |               | DEFINITION                                                                                                                                                                                                                                                           |      |           |   |   |      |   |      |   |
| 15:10 |    | RSVD          |    | RO | 0x0   |               | Reserved                                                                                                                                                                                                                                                             |      |           |   |   |      |   |      |   |
| 9:8   |    | BIAS_RESISTOR |    | RW | 0x0   |               | MIC Bias Output Impedance Adjustment<br>Controls an adjustable output impedance for the microphone bias. If this is set to zero the micbias block is powered off and the output is highZ.<br>0x0 = Powered off<br>0x1 = 2.0 kohm<br>0x2 = 4.0 kohm<br>0x3 = 8.0 kohm |      |           |   |   |      |   |      |   |
| 7     |    | RSVD          |    | RO | 0x0   |               | Reserved                                                                                                                                                                                                                                                             |      |           |   |   |      |   |      |   |
| 6:4   |    | BIAS_VOLT     |    | RW | 0x0   |               | MIC Bias Voltage Adjustment<br>Controls an adjustable bias voltage for the microphone bias amp in 250 mV steps. This bias voltage setting should be no more than VDDA-200 mV for adequate power supply rejection.<br>0x0 = 1.25 V<br>...<br>0x7 = 3.00 V             |      |           |   |   |      |   |      |   |
| 3:2   |    | RSVD          |    | RO | 0x0   |               | Reserved                                                                                                                                                                                                                                                             |      |           |   |   |      |   |      |   |
| 1:0   |    | GAIN          |    | RW | 0x0   |               | MIC Amplifier Gain<br>Sets the microphone amplifier gain. At 0 dB setting the THD can be slightly higher than other paths- typically around ~65 dB. At other gain settings the THD are better.<br>0x0 = 0 dB<br>0x1 = +20 dB<br>0x2 = +30 dB<br>0x3 = +40 dB         |      |           |   |   |      |   |      |   |



**Table 30. CHIP\_LINE\_OUT\_CTRL 0x002C**

| 15    | 14          | 13 | 12 | 11          | 10    | 9                                                                                                                                                                                                                                                            | 8 | 7    | 6 | 5           | 4 | 3 | 2 | 1 | 0 |
|-------|-------------|----|----|-------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|---|-------------|---|---|---|---|---|
| RSVD  |             |    |    | OUT_CURRENT |       |                                                                                                                                                                                                                                                              |   | RSVD |   | LO_VAGCNTRL |   |   |   |   |   |
| BITS  | FIELD       |    |    | RW          | RESET | DEFINITION                                                                                                                                                                                                                                                   |   |      |   |             |   |   |   |   |   |
| 15:12 | RSVD        |    |    | RO          | 0x0   | Reserved                                                                                                                                                                                                                                                     |   |      |   |             |   |   |   |   |   |
| 11:8  | OUT_CURRENT |    |    | RW          | 0x0   | Controls the output bias current for the LINEOUT amplifiers. The nominal recommended setting for a 10 kohm load with 1.0 nF load cap is 0x3. There are only 5 valid settings.<br>0x0=0.18 mA, 0x1=0.27 mA, 0x3=0.36 mA, 0x7=0.45 mA, 0xF=0.54 mA             |   |      |   |             |   |   |   |   |   |
| 7:6   | RSVD        |    |    | RO          | 0x0   | Reserved                                                                                                                                                                                                                                                     |   |      |   |             |   |   |   |   |   |
| 5:0   | LO_VAGCNTRL |    |    | RW          | 0x0   | LINEOUT Amplifier Analog Ground Voltage<br>Controls the analog ground voltage for the LINEOUT amplifiers in 25 mV steps. This should usually be set to VDDIO/2.<br>0x00 = 0.800 V<br>...<br>0x1F = 1.575 V<br>...<br>0x23 = 1.675 V<br>0x24-0x3F are invalid |   |      |   |             |   |   |   |   |   |

**Table 31. CHIP\_LINE\_OUT\_VOL 0x002E**

| 15    | 14           | 13 | 12           | 11    | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 9 | 8 | 7    | 6 | 5 | 4           | 3 | 2 | 1 | 0 |
|-------|--------------|----|--------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|------|---|---|-------------|---|---|---|---|
| RSVD  |              |    | LO_VOL_RIGHT |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |   | RSVD |   |   | LO_VOL_LEFT |   |   |   |   |
|       |              |    |              |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |   |      |   |   |             |   |   |   |   |
| BITS  | FIELD        |    | RW           | RESET | DEFINITION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |      |   |   |             |   |   |   |   |
| 15:13 | RSVD         |    | RO           | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |   |      |   |   |             |   |   |   |   |
| 12:8  | LO_VOL_RIGHT |    | RW           | 0x4   | LINEOUT Right Channel Volume<br>Controls the right channel LINEOUT volume in 0.5 dB steps. Higher codes have more attenuation. See programming information for Left channel.                                                                                                                                                                                                                                                                                                                                                                                                                 |   |   |      |   |   |             |   |   |   |   |
| 7:5   | RSVD         |    | RO           | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |   |      |   |   |             |   |   |   |   |
| 4:0   | LO_VOL_LEFT  |    | RW           | 0x4   | LINEOUT Left Channel Output Level<br>The LO_VOL_LEFT is used to normalize the output level of the left line output to full scale based on the values used to set LINE_OUT_CTRL -> LO_VAGCNTRL and CHIP_REF_CTRL -> VAG_VAL. In general this field should be set to:<br>$40 * \log((VAG\_VAL)/(LO\_VAGCNTRL)) + 15$<br><a href="#">Table 32</a> shows suggested values based on typical VDDIO and VDDA voltages.<br>After setting to the nominal voltage, this field can be used to adjust the output level in +/-0.5 dB increments by using values higher or lower than the nominal setting. |   |   |      |   |   |             |   |   |   |   |

**Table 32. LINEOUT Output Level Values**

| VDDA  | VAG_VAL | VDDIO | LO_VAGCNTRL | LO_VOL_* |
|-------|---------|-------|-------------|----------|
| 1.8 V | 0.9     | 3.3 V | 1.55        | 0x06     |
| 1.8 V | 0.9     | 1.8 V | 0.9         | 0x0F     |
| 3.3 V | 1.55    | 1.8 V | 0.9         | 0x19     |
| 3.3 V | 1.55    | 3.3 V | 1.55        | 0x0F     |

The [Table 33. CHIP\\_ANA\\_POWER 0x0030](#) register contains all of the power down controls for the analog blocks. The only other power-down controls are BIAS\_RESISTOR in

the MIC\_CTRL register and the EN\_ZCD control bits in ANA\_CTRL.

**Table 33. CHIP\_ANA\_POWER 0x0030**

| 15   | 14       | 13                    | 12              | 11                    | 10          | 9                | 8              | 7           | 6        | 5              | 4                 | 3           | 2                         | 1           | 0               |
|------|----------|-----------------------|-----------------|-----------------------|-------------|------------------|----------------|-------------|----------|----------------|-------------------|-------------|---------------------------|-------------|-----------------|
| RSVD | DAC_MONO | LINREG_SIMPLE_POWERUP | STARTUP_POWERUP | VDDC_CHRGMPMP_POWERUP | PLL_POWERUP | LINREG_D_POWERUP | VCOAMP_POWERUP | VAG_POWERUP | ADC_MONO | REFTOP_POWERUP | HEADPHONE_POWERUP | DAC_POWERUP | CAPLESS_HEADPHONE_POWERUP | ADC_POWERUP | LINEOUT_POWERUP |

| BITS | FIELD                 | RW | RESET | DEFINITION                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------|-----------------------|----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | RSVD                  | RW | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 14   | DAC_MONO              | RW | 0x1   | While DAC_POWERUP is set, this allows the DAC to be put into left only mono operation for power savings.<br>0x0 = Mono (left only)<br>0x1 = Stereo                                                                                                                                                                                                                                                                            |
| 13   | LINREG_SIMPLE_POWERUP | RW | 0x1   | Power up the simple (low power) digital supply regulator. After reset, this bit can be cleared IF VDDD is driven externally OR the primary digital linreg is enabled with LINREG_D_POWERUP<br>0x0 = Power down<br>0x1 = Power up                                                                                                                                                                                              |
| 12   | STARTUP_POWERUP       | RW | 0x1   | Power up the circuitry needed during the power up ramp and reset. After reset this bit can be cleared if VDDD is coming from an external source.<br>0x0 = Power down<br>0x1 = Power up                                                                                                                                                                                                                                        |
| 11   | VDDC_CHRGMPMP_POWERUP | RW | 0x0   | Power up the VDDC charge pump block. If neither VDDA or VDDIO is 3.0 V or larger this bit should be cleared before analog blocks are powered up.<br>0x0 = Power down<br>0x1 = Power up<br><br>Note that for charge pump to function, either the PLL must be powered on and programmed correctly (refer to CHIP_CLK_CTRL->MCLK_FREQ description) or the internal oscillator (set CLK_TOP_CTRL->ENABLE_INT_OSC) must be enabled |
| 10   | PLL_POWERUP           | RW | 0x0   | PLL Power Up<br>0x0 = Power down<br>0x1 = Power up<br><br>When cleared, the PLL is turned off. This must be set before CHIP_CLK_CTRL -> MCLK_FREQ is programmed to 0x3. The CHIP_PLL_CTRL register must be configured correctly before setting this bit.                                                                                                                                                                      |
| 9    | LINREG_D_POWERUP      | RW | 0x0   | Power up the primary VDDD linear regulator.<br>0x0 = Power down<br>0x1 = Power up                                                                                                                                                                                                                                                                                                                                             |

| BITS | FIELD                     | RW | RESET | DEFINITION                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------|---------------------------|----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8    | VCOAMP_POWERUP            | RW | 0x0   | Power up the PLL VCO amplifier.<br>0x0 = Power down<br>0x1 = Power up                                                                                                                                                                                                                                                                                                                                                        |
| 7    | VAG_POWERUP               | RW | 0x0   | Power up the VAG reference buffer. Setting this bit starts the power up ramp for the headphone and LINEOUT. The headphone (and/or LINEOUT) powerup should be set BEFORE clearing this bit. When this bit is cleared the power-down ramp is started. The headphone (and/or LINEOUT) powerup should stay set until the VAG is fully ramped down (200 to 400 ms after clearing this bit).<br>0x0 = Power down<br>0x1 = Power up |
| 6    | ADC_MONO                  | RW | 0x1   | While ADC_POWERUP is set, this allows the ADC to be put into left only mono operation for power savings. This mode is useful when only using the microphone input.<br>0x0 = Mono (left only)<br>0x1 = Stereo                                                                                                                                                                                                                 |
| 5    | REFTOP_POWERUP            | RW | 0x1   | Power up the reference bias currents<br>0x0 = Power down<br>0x1 = Power up<br>This bit can be cleared when the part is a sleep state to minimize analog power.                                                                                                                                                                                                                                                               |
| 4    | HEADPHONE_POWERUP         | RW | 0x0   | Power up the headphone amplifiers<br>0x0 = Power down<br>0x1 = Power up                                                                                                                                                                                                                                                                                                                                                      |
| 3    | DAC_POWERUP               | RW | 0x0   | Power up the DACs<br>0x0 = Power down<br>0x1 = Power up                                                                                                                                                                                                                                                                                                                                                                      |
| 2    | CAPLESS_HEADPHONE_POWERUP | RW | 0x0   | Power up the capless headphone mode<br>0x0 = Power down<br>0x1 = Power up                                                                                                                                                                                                                                                                                                                                                    |
| 1    | ADC_POWERUP               | RW | 0x0   | Power up the ADCs<br>0x0 = Power down<br>0x1 = Power up                                                                                                                                                                                                                                                                                                                                                                      |
| 0    | LINEOUT_POWERUP           | RW | 0x0   | Power up the LINEOUT amplifiers<br>0x0 = Power down<br>0x1 = Power up                                                                                                                                                                                                                                                                                                                                                        |

The [Table 34. CHIP\\_PLL\\_CTRL 0x0032](#) register may only be changed after reset, and before PLL\_POWERUP is set.

**Table 34. CHIP\_PLL\_CTRL 0x0032**

| 15          | 14           | 13 | 12 | 11 | 10           | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|----|----|----|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|
| INT_DIVISOR |              |    |    |    | FRAC_DIVISOR |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |   |   |   |   |   |   |   |   |
| BITS        | FIELD        |    |    | RW | RESET        | DEFINITION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |   |   |
| 15:11       | INT_DIVISOR  |    |    | RW | 0xA          | <p>This is the integer portion of the PLL divisor. To determine the value of this field, use the following calculation:</p> <p>INT_DIVISOR = FLOOR(PLL_OUTPUT_FREQ/INPUT_FREQ)</p> <p>PLL_OUTPUT_FREQ = 180.6336 MHz if System sample rate = 44.1 kHz</p> <p>else</p> <p>PLL_OUTPUT_FREQ = 196.608 MHz if System sample rate!= 44.1 kHz</p> <p>INPUT_FREQ = Frequency of the external MCLK provided if CHIP_CLK_TOP_CTRL-&gt;INPUT_FREQ_DIV2 = 0x0</p> <p>else</p> <p>INPUT_FREQ = (Frequency of the external MCLK provided/2) If</p> <p>CHIP_CLK_TOP_CTRL-&gt;INPUT_FREQ_DIV2 = 0x1</p>                     |   |   |   |   |   |   |   |   |   |
| 10:0        | FRAC_DIVISOR |    |    | RW | 0x0          | <p>This is the fractional portion of the PLL divisor. To determine the value of this field, use the following calculation:</p> <p>FRAC_DIVISOR = ((PLL_OUTPUT_FREQ/INPUT_FREQ) - INT_DIVISOR)*2048</p> <p>PLL_OUTPUT_FREQ = 180.6336 MHz if System sample rate = 44.1 kHz</p> <p>else</p> <p>PLL_OUTPUT_FREQ = 196.608 MHz if System sample rate!= 44.1 kHz</p> <p>INPUT_FREQ = Frequency of the external MCLK provided if CHIP_CLK_TOP_CTRL-&gt;INPUT_FREQ_DIV2 = 0x0</p> <p>else</p> <p>INPUT_FREQ = (Frequency of the external MCLK provided/2) If</p> <p>CHIP_CLK_TOP_CTRL-&gt;INPUT_FREQ_DIV2 = 0x1</p> |   |   |   |   |   |   |   |   |   |

[Table 35. CHIP\\_CLK\\_TOP\\_CTRL 0x0034](#) has the miscellaneous controls for the clock block.

**Table 35. CHIP\_CLK\_TOP\_CTRL 0x0034**

| 15   | 14 | 13 | 12 | 11             | 10   | 9 | 8 | 7 | 6 | 5 | 4 | 3               | 2    | 1 | 0 |
|------|----|----|----|----------------|------|---|---|---|---|---|---|-----------------|------|---|---|
| RSVD |    |    |    | ENABLE_INT_OSC | RSVD |   |   |   |   |   |   | INPUT_FREQ_DIV2 | RSVD |   |   |

| BITS  | FIELD          | RW | RESET | DEFINITION                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|----------------|----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:12 | RESERVED       | RO | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 11    | ENABLE_INT_OSC | RW | 0x0   | Setting this bit enables an internal oscillator to be used for the zero cross detectors, the short detect recovery, and the charge pump. This allows the I <sup>2</sup> S clock to be shut off while still operating an analog signal path. This bit can be kept on when the I <sup>2</sup> S clock is enabled, but the I <sup>2</sup> S clock is more accurate so it is preferred to clear this bit when I <sup>2</sup> S is present. |
| 10:4  | RSVD           | RW | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                               |

| BITS | FIELD           | RW | RESET | DEFINITION                                                                                                                                                                                                               |
|------|-----------------|----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3    | INPUT_FREQ_DIV2 | RW | 0x0   | SYS_MCLK divider before PLL input<br>0x0 = pass through<br>0x1 = SYS_MCLK is divided by 2 before entering PLL<br>This must be set when the input clock is above 17 Mhz. This has no effect when the PLL is powered down. |
| 2:0  | RSVD            | RW | 0x0   | Reserved                                                                                                                                                                                                                 |

Status bits for analog blocks are found in [Table 36.](#)  
[CHIP\\_ANA\\_STATUS 0x0036](#)

**Table 36. CHIP\_ANA\_STATUS 0x0036**

| 15   | 14 | 13 | 12 | 11 | 10 | 9           | 8          | 7    | 6 | 5 | 4             | 3    | 2 | 1 | 0 |
|------|----|----|----|----|----|-------------|------------|------|---|---|---------------|------|---|---|---|
| RSVD |    |    |    |    |    | LRSHORT_STS | CSHORT_STS | RSVD |   |   | PLL_IS_LOCKED | RSVD |   |   |   |

| BITS  | FIELD         | RW | RESET | DEFINITION                                                                                                                                    |
|-------|---------------|----|-------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 15:10 | RSVD          | RO | 0x0   | Reserved                                                                                                                                      |
| 9     | LRSHORT_STS   | RO | 0x0   | This bit is high whenever a short is detected on the left or right channel headphone drivers.<br>0x0 = Normal<br>0x1 = Short detected         |
| 8     | CSHORT_STS    | RO | 0x0   | This bit is high whenever a short is detected on the capless headphone common/ center channel driver.<br>0x0 = Normal<br>0x1 = Short detected |
| 7:5   | RSVD          | RO | 0x0   | Reserved                                                                                                                                      |
| 4     | PLL_IS_LOCKED | RO | 0x0   | This bit goes high after the PLL is locked.<br>0x0 = PLL is not locked<br>0x1 = PLL is locked                                                 |
| 3:0   | RSVD          | RO | 0x0   | Reserved                                                                                                                                      |

[Table 37. CHIP\\_ANA\\_TEST1 0x0038](#) and [Table 38.](#)  
[CHIP\\_ANA\\_TEST2 0x003A](#) register controls are intended  
only for debug.

**Table 37. CHIP\_ANA\_TEST1 0x0038**

| 15          | 14 | 13        | 12 | 11         | 10 | 9 | 8          | 7                  | 6           | 5                | 4          | 3             | 2           | 1             | 0        |
|-------------|----|-----------|----|------------|----|---|------------|--------------------|-------------|------------------|------------|---------------|-------------|---------------|----------|
| HP_IALL_ADJ |    | HP_I1_ADJ |    | HP_ANTIPOP |    |   | HP_CLASSAB | HP_HOLD_GND_CENTER | HP_HOLD_GND | VAG_DOUB_CURRENT | VAG_CLASSA | TM_ADCIN_TOHP | TM_HPCOMMON | TM_SELECT_MIC | TESTMODE |

| BITS  | FIELD              | RW | RESET | DEFINITION                                                                                                                                                                                             |
|-------|--------------------|----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:14 | HP_IALL_ADJ        | RW | 0x0   | These bits control the overall bias current of the headphone amplifier (all stages including first and output stage).<br>0x0=nominal, 0x1=-50%, 0x2=+50%, 0x3=-40%                                     |
| 13:12 | HP_I1_ADJ          | RW | 0x0   | These bits control the bias current for the first stage of the headphone amplifier.<br>0x0=nominal, 0x1=-50%, 0x2=+100%, 0x3=+50%                                                                      |
| 11:9  | HP_ANTIPOP         | RW | 0x0   | These bits control the headphone output current in classA mode and also the pull-down strength while powering off. These bits normally are not needed.                                                 |
| 8     | HP_CLASSAB         | RW | 0x1   | This defaults high. When this bit is high the headphone is in classAB mode. ClassA mode would normally not be used.                                                                                    |
| 7     | HP_HOLD_GND_CENTER | RW | 0x1   | This defaults high. When this bit is high and the capless headphone center channel is powered off, the output is tied to ground. This is the preferred mode of operation for best antipop performance. |
| 6     | HP_HOLD_GND        | RW | 0x1   | This defaults high. When this bit is high and the headphone is powered off, the output is tied to ground. This is the preferred mode of operation for best antipop performance.                        |
| 5     | VAG_DOUB_CURRENT   | RW | 0x0   | Double the VAG output current when in classA mode.                                                                                                                                                     |
| 4     | VAG_CLASSA         | RW | 0x0   | Turn off the classAB output current for the VAG buffer. The classA current is limited so this may cause clipping in some modes.                                                                        |
| 3     | TM_ADCIN_TOHP      | RW | 0x0   | Put ADCmux output onto the headphone output pin. Must remove headphone load and any external headphone compensation for this mode.                                                                     |
| 2     | TM_HPCOMMON        | RW | 0x0   | Enable headphone common to be used in ADCmux for testing                                                                                                                                               |
| 1     | TM_SELECT_MIC      | RW | 0x0   | Enable the mic-adc-dac-HP path                                                                                                                                                                         |
| 0     | TESTMODE           | RW | 0x0   | Enable the analog test mode paths                                                                                                                                                                      |

Table 38. CHIP\_ANA\_TEST2 0x003A

| 15   | 14              | 13    | 12           | 11             | 10                | 9                       | 8                      | 7              | 6            | 5           | 4          | 3                       | 2                      | 1         | 0             |
|------|-----------------|-------|--------------|----------------|-------------------|-------------------------|------------------------|----------------|--------------|-------------|------------|-------------------------|------------------------|-----------|---------------|
| RSVD | LINEOUT_TO_VDDA | SPARE | MONOMODE_DAC | VCO_TUNE_AGAIN | LO_PASS_MASTERVAG | INVERT_DAC_SAMPLE_CLOCK | INVERT_DAC_DATA_TIMING | DAC_EXTEND_RTZ | DAC_DOUBLE_I | DAC_DIS_RTZ | DAC_CLASSA | INVERT_ADC_SAMPLE_CLOCK | INVERT_ADC_DATA_TIMING | ADC_LESSI | ADC_DITHEROFF |

| BITS | FIELD           | RW | RESET | DEFINITION                                                                                                                                                                         |
|------|-----------------|----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | RSVD            | RO | 0x0   | Reserved                                                                                                                                                                           |
| 14   | LINEOUT_TO_VDDA | RW | 0x0   | Changes the LINEOUT amplifier power supply from VDDIO to VDDA. Typically LINEOUT should be on the higher power supply. This bit is useful when VDDA is ~3.3 V and VDDIO is ~1.8 V. |
| 13   | SPARE           | RW | 0x0   | Spare registers to analog.                                                                                                                                                         |
| 12   | MONOMODE_DAC    | RW | 0x0   | Copy the left channel DAC data to the right channel. This allows both left and right to play from MONO dac data.                                                                   |
| 11   | VCO_TUNE_AGAIN  | RW | 0x0   | When toggled high then low forces the PLL VCO to retune the number of inverters in the ring oscillator loop.                                                                       |

| BITS | FIELD                       | RW | RESET | DEFINITION                                                                                                                                                                |
|------|-----------------------------|----|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10   | LO_PASS_MASTERVAG           | RW | 0x0   | Tie the main analog VAG to the LINEOUT VAG. This can improve SNR for the LINEOUT when both are the same voltage.                                                          |
| 9    | INVERT_DAC_SAMPL<br>E_CLOCK | RW | 0x0   | Change the clock edge used for the DAC output sampling.                                                                                                                   |
| 8    | INVERT_DAC_DATA_<br>TIMING  | RW | 0x0   | Change the clock edge used for the digital to analog DAC data crossing.                                                                                                   |
| 7    | DAC_EXTEND_RTZ              | RW | 0x0   | Extend the return-to-zero time for the DAC.                                                                                                                               |
| 6    | DAC_DOUBLE_I                | RW | 0x0   | Double the output current of the DAC amplifier when it is in classA mode.                                                                                                 |
| 5    | DAC_DIS_RTZ                 | RW | 0x0   | Turn off the return-to-zero in the DAC. In mode cases, this hurts the SNDR of the DAC.                                                                                    |
| 4    | DAC_CLASSA                  | RW | 0x0   | Turn off the classAB mode in the DAC amplifier. This mode should normally not be used. The output current is not high enough to support a full scale signal in this mode. |
| 3    | INVERT_ADC_SAMPL<br>E_CLOCK | RW | 0x0   | Change the clock edge used for the ADC sampling.                                                                                                                          |
| 2    | INVERT_ADC_DATA_<br>TIMING  | RW | 0x0   | Change the clock edge used for the analog to digital ADC data crossing                                                                                                    |
| 1    | ADC_LESSI                   | RW | 0x0   | Drops ADC bias currents by 20%                                                                                                                                            |
| 0    | ADC_DITHEROFF               | RW | 0x0   | Turns off the ADC dithering.                                                                                                                                              |

The [Table 39. CHIP\\_SHORT\\_CTRL 0x003C](#) register contains controls for the headphone short detectors.

**Table 39. CHIP\_SHORT\_CTRL 0x003C**

| 15   | 14      | 13 | 12 | 11   | 10      | 9 | 8 | 7    | 6       | 5 | 4 | 3       | 2 | 1       | 0 |
|------|---------|----|----|------|---------|---|---|------|---------|---|---|---------|---|---------|---|
| RSVD | LVLADJR |    |    | RSVD | LVLADJL |   |   | RSVD | LVLADJC |   |   | MODE_LR |   | MODE_CM |   |

| BITS  | FIELD   | RW | RESET | DEFINITION                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|---------|----|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | RSVD    | RO | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 14:12 | LVLADJR | RW | 0x0   | These bits adjust the sensitivity of the right channel headphone short detector in 25 mA steps. This trip point can vary by ~30% over process so leave plenty of guard band to avoid false trips. This short detect trip point is also effected by the bias current adjustments made by CHIP_REF_CTRL -> BIAS_CTRL and by CHIP_ANA_TEST1 -> HP_IALL_ADJ.<br>0x3=25 mA<br>0x2=50 mA<br>0x1=75 mA<br>0x0=100 mA<br>0x4=125 mA<br>0x5=150 mA<br>0x6=175 mA<br>0x7=200 mA |
| 11    | RSVD    | RO | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| BITS | FIELD   | RW | RESET | DEFINITION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|---------|----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10:8 | LVLADJL | RW | 0x0   | <p>These bits adjust the sensitivity of the left channel headphone short detector in 25 mA steps. This trip point can vary by ~30% over process so leave plenty of guard band to avoid false trips. This short detect trip point is also effected by the bias current adjustments made by CHIP_REF_CTRL -&gt; BIAS_CTRL and by CHIP_ANA_TEST1 -&gt; HP_IALL_ADJ.</p> <p>0x3=25 mA<br/>0x2=50 mA<br/>0x1=75 mA<br/>0x0=100 mA<br/>0x4=125 mA<br/>0x5=150 mA<br/>0x6=175 mA<br/>0x7=200 mA</p>                                                                                                                                                                                       |
| 7    | RSVD    | RO | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6:4  | LVLADJC | RW | 0x0   | <p>These bits adjust the sensitivity of the capless headphone center channel short detector in 50 mA steps. This trip point can vary by ~30% over process so leave plenty of guard band to avoid false trips. This short detect trip point is also effected by the bias current adjustments CHIP_REF_CTRL -&gt; BIAS_CTRL and by CHIP_ANA_TEST1 -&gt; HP_IALL_ADJ.</p> <p>0x3=50 mA<br/>0x2=100 mA<br/>0x1=150 mA<br/>0x0=200 mA<br/>0x4=250 mA<br/>0x5=300 mA<br/>0x6=350 mA<br/>0x7=400 mA</p>                                                                                                                                                                                   |
| 3:2  | MODE_LR | RW | 0x0   | <p>These bits control the behavior of the short detector for the capless headphone central channel driver. This mode should be set prior to powering up the headphone amplifier. When a short is detected the amplifier output switches to classA mode internally to avoid excessive currents.</p> <p>0x0 = Disable short detector, reset short detect latch, software view non-latched short signal<br/>0x1 = Enable short detector and reset the latch at timeout (every ~50 ms)<br/>0x2 = This mode is not used/invalid<br/>0x3 = Enable short detector with only manual reset (have to return to 0x0 to reset the latch)</p>                                                   |
| 1:0  | MODE_CM | RW | 0x0   | <p>These bits control the behavior of the short detector for the capless headphone central channel driver. This mode should be set prior to powering up the headphone amplifier. When a short is detected the amplifier output switches to classA mode internally to avoid excessive currents.</p> <p>0x0 = Disable short detector, reset short detect latch, software view non-latched short signal<br/>0x1 = Enable short detector and reset the latch at timeout (every ~50 ms)<br/>0x2 = Enable short detector and auto reset when output voltage rises (preferred mode)<br/>0x3 = Enable short detector with only manual reset (have to return to 0x0 to reset the latch)</p> |



**Table 40. DAP\_CONTROL 0x0100**

| 15   | 14     | 13 | 12 | 11 | 10    | 9                                                                                                                                                                                                                | 8 | 7 | 6 | 5 | 4      | 3    | 2 | 1 | 0      |
|------|--------|----|----|----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|--------|------|---|---|--------|
| RSVD |        |    |    |    |       |                                                                                                                                                                                                                  |   |   |   |   | MIX_EN | RSVD |   |   | DAP_EN |
| BITS | FIELD  |    |    | RW | RESET | DEFINITION                                                                                                                                                                                                       |   |   |   |   |        |      |   |   |        |
| 15:5 | RSVD   |    |    | RO | 0x0   | Reserved                                                                                                                                                                                                         |   |   |   |   |        |      |   |   |        |
| 4    | MIX_EN |    |    | RW | 0x0   | Enable/Disable the DAP mixer path<br>0x0 = Disable<br>0x1 = Enable<br>When enabled, DAP_EN must also be enabled to use the mixer.                                                                                |   |   |   |   |        |      |   |   |        |
| 3:1  | RSVD   |    |    | RO | 0x0   | Reserved                                                                                                                                                                                                         |   |   |   |   |        |      |   |   |        |
| 0    | DAP_EN |    |    | RW | 0x0   | Enable/Disable digital audio processing (DAP)<br>0x0 = Disable. When disabled, no audio passes through.<br>0x1 = Enable. When enabled, audio can pass through DAP even if none of the DAP functions are enabled. |   |   |   |   |        |      |   |   |        |

**Table 41. DAP\_PEQ 0x0102**

| 15   | 14    | 13 | 12 | 11 | 10    | 9                                                                                                                                                                                                             | 8 | 7 | 6 | 5 | 4 | 3 | 2  | 1 | 0 |
|------|-------|----|----|----|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|----|---|---|
| RSVD |       |    |    |    |       |                                                                                                                                                                                                               |   |   |   |   |   |   | EN |   |   |
|      |       |    |    |    |       |                                                                                                                                                                                                               |   |   |   |   |   |   |    |   |   |
| BITS | FIELD |    |    | RW | RESET | DEFINITION                                                                                                                                                                                                    |   |   |   |   |   |   |    |   |   |
| 15:3 | RSVD  |    |    | RO | 0x0   | Reserved                                                                                                                                                                                                      |   |   |   |   |   |   |    |   |   |
| 2:0  | EN    |    |    | RW | 0x0   | Set to Enable the PEQ filters<br>0x0 = Disabled<br>0x1 = 1 Filter Enabled<br>0x2 = 2 Filters Enabled<br>.....<br>0x7 = Cascaded 7 Filters<br>DAP_AUDIO_EQ->EN bit must be set to 1 in order to enable the PEQ |   |   |   |   |   |   |    |   |   |

**Table 42. DAP\_BASS\_ENHANCE 0x0104**

| 15   | 14         | 13 | 12 | 11 | 10    | 9                                                                                         | 8          | 7    | 6      | 5 | 4 | 3    | 2 | 1 | 0  |
|------|------------|----|----|----|-------|-------------------------------------------------------------------------------------------|------------|------|--------|---|---|------|---|---|----|
| RSVD |            |    |    |    |       |                                                                                           | BYPASS_HPF | RSVD | CUTOFF |   |   | RSVD |   |   | EN |
|      |            |    |    |    |       |                                                                                           |            |      |        |   |   |      |   |   |    |
|      |            |    |    |    |       |                                                                                           |            |      |        |   |   |      |   |   |    |
|      |            |    |    |    |       |                                                                                           |            |      |        |   |   |      |   |   |    |
|      |            |    |    |    |       |                                                                                           |            |      |        |   |   |      |   |   |    |
| BITS | FIELD      |    |    | RW | RESET | DEFINITION                                                                                |            |      |        |   |   |      |   |   |    |
| 15:9 | RSVD       |    |    | RO | 0x0   | Reserved                                                                                  |            |      |        |   |   |      |   |   |    |
| 8    | BYPASS_HPF |    |    | RW | 0x0   | Bypass high pass filter<br>0x0 = Enable high pass filter<br>0x1 = Bypass high pass filter |            |      |        |   |   |      |   |   |    |
| 7    | RSVD       |    |    | RO | 0x0   | Reserved                                                                                  |            |      |        |   |   |      |   |   |    |

| BITS | FIELD  | RW | RESET | DEFINITION                                                                                                                           |
|------|--------|----|-------|--------------------------------------------------------------------------------------------------------------------------------------|
| 6:4  | CUTOFF | RW | 0x4   | Set cut-off frequency<br>0x0 = 80 Hz<br>0x1 = 100 Hz<br>0x2 = 125 Hz<br>0x3 = 150 Hz<br>0x4 = 175 Hz<br>0x5 = 200 Hz<br>0x6 = 225 Hz |
| 3:1  | RSVD   | RO | 0x0   | Reserved                                                                                                                             |
| 0    | EN     | RW | 0x0   | Enable/Disable Bass Enhance<br>0x0 = Disable<br>0x1 = Enable                                                                         |

**Table 43. DAP\_BASS\_ENHANCE\_CTRL 0x0106**

|      |    |          |    |    |    |   |   |      |            |   |   |   |   |   |   |
|------|----|----------|----|----|----|---|---|------|------------|---|---|---|---|---|---|
| 15   | 14 | 13       | 12 | 11 | 10 | 9 | 8 | 7    | 6          | 5 | 4 | 3 | 2 | 1 | 0 |
| RSVD |    | LR_LEVEL |    |    |    |   |   | RSVD | BASS_LEVEL |   |   |   |   |   |   |

| BITS  | FIELD      | RW | RESET | DEFINITION                                                                                               |
|-------|------------|----|-------|----------------------------------------------------------------------------------------------------------|
| 15:14 | RSVD       | RO | 0x0   | Reserved                                                                                                 |
| 13:8  | LR_LEVEL   | RW | 0x5   | Left/Right Mix Level Control<br>0x00= +6.0 dB for Main Channel<br>.....<br>0x3F= Least L/R Channel Level |
| 7     | RSVD       | RO | 0x0   |                                                                                                          |
| 6:0   | BASS_LEVEL | RW | 0x1f  | Bass Harmonic Level Control<br>0x00= Most Harmonic Boost<br>.....<br>0x7F=Least Harmonic Boost           |

**Table 44. DAP\_AUDIO\_EQ 0x0108**

|      |    |    |    |    |    |   |   |   |   |   |   |   |   |    |   |
|------|----|----|----|----|----|---|---|---|---|---|---|---|---|----|---|
| 15   | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1  | 0 |
| RSVD |    |    |    |    |    |   |   |   |   |   |   |   |   | EN |   |

| BITS | FIELD | RW | RESET | DEFINITION                                                                                                                                                                                                                                                               |
|------|-------|----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:2 | RSVD  | RO | 0x0   | Reserved                                                                                                                                                                                                                                                                 |
| 1:0  | EN    | RW | 0x0   | Selects between PEQ/GEQ/Tone Control and Enables it.<br>0x0 = Disabled.<br>0x1 = Enable PEQ. NOTE: DAP_PEQ->EN bit must also be set to the desired number of filters (bands) in order for the PEQ to be enabled.<br>0x2 = Enable Tone Control<br>0x3 = Enable 5 Band GEQ |

**Table 45. DAP\_SGTL\_SURROUND 0x010A**

| 15   | 14            | 13 | 12 | 11 | 10    | 9                                                                                                                                                      | 8 | 7 | 6             | 5 | 4 | 3    | 2 | 1      | 0 |
|------|---------------|----|----|----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---------------|---|---|------|---|--------|---|
| RSVD |               |    |    |    |       |                                                                                                                                                        |   |   | WIDTH_CONTROL |   |   | RSVD |   | SELECT |   |
| BITS | FIELD         |    |    | RW | RESET | DEFINITION                                                                                                                                             |   |   |               |   |   |      |   |        |   |
| 15:7 | RSVD          |    |    | RO | 0x0   | Reserved                                                                                                                                               |   |   |               |   |   |      |   |        |   |
| 6:4  | WIDTH_CONTROL |    |    | RW | 0x4   | Freescale Surround Width Control - The width control changes the perceived width of the sound field.<br>0x0 = Least Width<br>.....<br>0x7 = Most Width |   |   |               |   |   |      |   |        |   |
| 3:2  | RSVD          |    |    | RO | 0x0   | Reserved                                                                                                                                               |   |   |               |   |   |      |   |        |   |
| 1:0  | SELECT        |    |    | RW | 0x0   | Freescale Surround Selection<br>0x0 = Disabled<br>0x1 = Disabled<br>0x2 = Mono input Enable<br>0x3 = Stereo input Enable                               |   |   |               |   |   |      |   |        |   |

**Table 46. DAP\_FILTER\_COEF\_ACCESS 0x010C**

|      |       |    |    |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |       |   |   |   |   |   |   |   |
|------|-------|----|----|----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---|---|---|---|---|---|---|
| 15   | 14    | 13 | 12 | 11 | 10    | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 8  | 7     | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| RSVD |       |    |    |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | WR | INDEX |   |   |   |   |   |   |   |
| BITS | FIELD |    |    | RW | RESET | DEFINITION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |       |   |   |   |   |   |   |   |
| 15:9 | RSVD  |    |    | RO | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |       |   |   |   |   |   |   |   |
| 8    | WR    |    |    | WO | 0x0   | When set, the coefficients written in the ten coefficient data registers are loaded into the filter specified by INDEX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |       |   |   |   |   |   |   |   |
| 7:0  | INDEX |    |    | RW | 0x0   | <p>Specifies the index for each of the seven bands of the filter coefficient that needs to be written to. Each filter has 5 coefficients that need to be loaded into the 10 coefficient registers (MSB, LSB) before setting the index and WR bit.</p> <p>Steps to write coefficients:</p> <p>1. Write the five 20-bit coefficient values to DAP_COEF_WR_XX_MSB and DAP_COEF_WR_XX_LSB registers (XX= B0,B1,B2,A1,A2)</p> <p>2. Set INDEX of the coefficient from the table below.</p> <p>3. Set the WR bit to load the coefficient.</p> <p>NOTE: Steps 2 and 3 can be performed with a single write to DAP_FILTER_COEF_ACCESS register.</p> <p>Coefficient address:</p> <p>Band 0 = 0x00</p> <p>Band 1 = 0x01</p> <p>Band 2 = 0x02</p> <p>Band 3 = 0x03</p> <p>Band 4 = 0x04</p> <p>...</p> <p>Band 7 = 0x06</p> |    |       |   |   |   |   |   |   |   |

**Table 47. DAP\_COEF\_WR\_B0\_MSB 0x010E**

| 15     | 14     | 13     | 12     | 11                                                                                 | 10     | 9      | 8      | 7      | 6      | 5     | 4     | 3     | 2     | 1     | 0     |
|--------|--------|--------|--------|------------------------------------------------------------------------------------|--------|--------|--------|--------|--------|-------|-------|-------|-------|-------|-------|
| BIT_19 | BIT_18 | BIT_17 | BIT_16 | BIT_15                                                                             | BIT_14 | BIT_13 | BIT_12 | BIT_11 | BIT_10 | BIT_9 | BIT_8 | BIT_7 | BIT_6 | BIT_5 | BIT_4 |
| BITS   | FIELD  | RW     | RESET  | DEFINITION                                                                         |        |        |        |        |        |       |       |       |       |       |       |
| 15     | BIT_19 | WO     | 0x0    | Most significant 16-bits of the 20-bit filter coefficient that needs to be written |        |        |        |        |        |       |       |       |       |       |       |
| 14     | BIT_18 | WO     | 0x0    |                                                                                    |        |        |        |        |        |       |       |       |       |       |       |
| 13     | BIT_17 | WO     | 0x0    |                                                                                    |        |        |        |        |        |       |       |       |       |       |       |
| 12     | BIT_16 | WO     | 0x0    |                                                                                    |        |        |        |        |        |       |       |       |       |       |       |
| 11     | BIT_15 | WO     | 0x0    |                                                                                    |        |        |        |        |        |       |       |       |       |       |       |
| 10     | BIT_14 | WO     | 0x0    |                                                                                    |        |        |        |        |        |       |       |       |       |       |       |
| 9      | BIT_13 | WO     | 0x0    |                                                                                    |        |        |        |        |        |       |       |       |       |       |       |
| 8      | BIT_12 | WO     | 0x0    |                                                                                    |        |        |        |        |        |       |       |       |       |       |       |
| 7      | BIT_11 | WO     | 0x0    |                                                                                    |        |        |        |        |        |       |       |       |       |       |       |
| 6      | BIT_10 | WO     | 0x0    |                                                                                    |        |        |        |        |        |       |       |       |       |       |       |
| 5      | BIT_9  | WO     | 0x0    |                                                                                    |        |        |        |        |        |       |       |       |       |       |       |
| 4      | BIT_8  | WO     | 0x0    |                                                                                    |        |        |        |        |        |       |       |       |       |       |       |
| 3      | BIT_7  | WO     | 0x0    |                                                                                    |        |        |        |        |        |       |       |       |       |       |       |
| 2      | BIT_6  | WO     | 0x0    |                                                                                    |        |        |        |        |        |       |       |       |       |       |       |
| 1      | BIT_5  | WO     | 0x0    |                                                                                    |        |        |        |        |        |       |       |       |       |       |       |
| 0      | BIT_4  | WO     | 0x0    |                                                                                    |        |        |        |        |        |       |       |       |       |       |       |

**Table 48. DAP\_COEF\_WR\_B0\_LSB 0x0110**

| 15   | 14    | 13 | 12    | 11                                                                                  | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3     | 2     | 1     | 0     |
|------|-------|----|-------|-------------------------------------------------------------------------------------|----|---|---|---|---|---|---|-------|-------|-------|-------|
| RSVD |       |    |       |                                                                                     |    |   |   |   |   |   |   | BIT_3 | BIT_2 | BIT_1 | BIT_0 |
| BITS | FIELD | RW | RESET | DEFINITION                                                                          |    |   |   |   |   |   |   |       |       |       |       |
| 15:4 | RSVD  | RO | 0x0   |                                                                                     |    |   |   |   |   |   |   |       |       |       |       |
| 3    | BIT_3 | WO | 0x0   |                                                                                     |    |   |   |   |   |   |   |       |       |       |       |
| 2    | BIT_2 | WO | 0x0   |                                                                                     |    |   |   |   |   |   |   |       |       |       |       |
| 1    | BIT_1 | WO | 0x0   |                                                                                     |    |   |   |   |   |   |   |       |       |       |       |
| 0    | BIT_0 | WO | 0x0   | Least significant 4 bits of the 20-bit filter coefficient that needs to be written. |    |   |   |   |   |   |   |       |       |       |       |

**Table 49. DAP\_AUDIO\_EQ\_BASS\_BAND0 0x0116 115 Hz**

| 15   | 14     | 13 | 12 | 11 | 10    | 9                                                                                                                                  | 8 | 7 | 6      | 5 | 4 | 3 | 2 | 1 | 0 |
|------|--------|----|----|----|-------|------------------------------------------------------------------------------------------------------------------------------------|---|---|--------|---|---|---|---|---|---|
| RSVD |        |    |    |    |       |                                                                                                                                    |   |   | VOLUME |   |   |   |   |   |   |
| BITS | FIELD  |    |    | RW | RESET | DEFINITION                                                                                                                         |   |   |        |   |   |   |   |   |   |
| 15:7 | RSVD   |    |    | RO | 0x0   | Reserved                                                                                                                           |   |   |        |   |   |   |   |   |   |
| 6:0  | VOLUME |    |    | RW | 0x2F  | Sets Tone Control Bass/GEQ Band0<br>0x5F = sets to 12 dB<br>0x2F = sets to 0 dB<br>0x00 = sets to -11.75 dB<br>Each LSB is 0.25 dB |   |   |        |   |   |   |   |   |   |

**Table 50. DAP\_AUDIO\_EQ\_BAND1 0x0118 330 Hz**

| 15   | 14     | 13 | 12 | 11 | 10    | 9                                                                                                                | 8 | 7 | 6      | 5 | 4 | 3 | 2 | 1 | 0 |
|------|--------|----|----|----|-------|------------------------------------------------------------------------------------------------------------------|---|---|--------|---|---|---|---|---|---|
| RSVD |        |    |    |    |       |                                                                                                                  |   |   | VOLUME |   |   |   |   |   |   |
| BITS | FIELD  |    |    | RW | RESET | DEFINITION                                                                                                       |   |   |        |   |   |   |   |   |   |
| 15:7 | RSVD   |    |    | RO | 0x0   | Reserved                                                                                                         |   |   |        |   |   |   |   |   |   |
| 6:0  | VOLUME |    |    | RW | 0x2F  | Sets GEQ Band1<br>0x5F = sets to 12 dB<br>0x2F = sets to 0 dB<br>0x00 = sets to -11.75 dB<br>Each LSB is 0.25 dB |   |   |        |   |   |   |   |   |   |

**Table 51. DAP\_AUDIO\_EQ\_BAND2 0x011A 990 Hz**

| 15   | 14     | 13 | 12 | 11 | 10    | 9                                                                                                                | 8 | 7 | 6      | 5 | 4 | 3 | 2 | 1 | 0 |
|------|--------|----|----|----|-------|------------------------------------------------------------------------------------------------------------------|---|---|--------|---|---|---|---|---|---|
| RSVD |        |    |    |    |       |                                                                                                                  |   |   | VOLUME |   |   |   |   |   |   |
| BITS | FIELD  |    |    | RW | RESET | DEFINITION                                                                                                       |   |   |        |   |   |   |   |   |   |
| 15:7 | RSVD   |    |    | RO | 0x0   | Reserved                                                                                                         |   |   |        |   |   |   |   |   |   |
| 6:0  | VOLUME |    |    | RW | 0x2F  | Sets GEQ Band2<br>0x5F = sets to 12 dB<br>0x2F = sets to 0 dB<br>0x00 = sets to -11.75 dB<br>Each LSB is 0.25 dB |   |   |        |   |   |   |   |   |   |

**Table 52. DAP\_AUDIO\_EQ\_BAND3 0x011C 3000 Hz**

| 15   | 14 | 13     | 12 | 11 | 10 | 9     | 8                                                                                                                | 7 | 6      | 5 | 4 | 3 | 2 | 1 | 0 |
|------|----|--------|----|----|----|-------|------------------------------------------------------------------------------------------------------------------|---|--------|---|---|---|---|---|---|
| RSVD |    |        |    |    |    |       |                                                                                                                  |   | VOLUME |   |   |   |   |   |   |
| BITS |    | FIELD  |    |    | RW | RESET | DEFINITION                                                                                                       |   |        |   |   |   |   |   |   |
| 15:7 |    | RSVD   |    |    | RO | 0x0   | Reserved                                                                                                         |   |        |   |   |   |   |   |   |
| 6:0  |    | VOLUME |    |    | RW | 0x2F  | Sets GEQ Band3<br>0x5F = sets to 12 dB<br>0x2F = sets to 0 dB<br>0x00 = sets to -11.75 dB<br>Each LSB is 0.25 dB |   |        |   |   |   |   |   |   |

**Table 53. DAP\_AUDIO\_EQ\_TREBLE\_BAND4 0x011E 9900 Hz**

|      |        |    |    |    |       |                                                                                                                                      |   |   |        |   |   |   |   |   |   |
|------|--------|----|----|----|-------|--------------------------------------------------------------------------------------------------------------------------------------|---|---|--------|---|---|---|---|---|---|
| 15   | 14     | 13 | 12 | 11 | 10    | 9                                                                                                                                    | 8 | 7 | 6      | 5 | 4 | 3 | 2 | 1 | 0 |
| RSVD |        |    |    |    |       |                                                                                                                                      |   |   | VOLUME |   |   |   |   |   |   |
|      |        |    |    |    |       |                                                                                                                                      |   |   |        |   |   |   |   |   |   |
| BITS | FIELD  |    |    | RW | RESET | DEFINITION                                                                                                                           |   |   |        |   |   |   |   |   |   |
| 15:7 | RSVD   |    |    | RO | 0x0   | Reserved                                                                                                                             |   |   |        |   |   |   |   |   |   |
| 6:0  | VOLUME |    |    | RW | 0x2F  | Sets Tone Control Treble/GEQ Band4<br>0x5F = sets to 12 dB<br>0x2F = sets to 0 dB<br>0x00 = sets to -11.75 dB<br>Each LSB is 0.25 dB |   |   |        |   |   |   |   |   |   |

[Table 54. DAP\\_MAIN\\_CHAN 0x0120](#) sets the main channel volume level

**Table 54. DAP\_MAIN\_CHAN 0x0120**

| 15   | 14    | 13 | 12 | 11 | 10 | 9      | 8                                                                                  | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|------|-------|----|----|----|----|--------|------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|
| VOL  |       |    |    |    |    |        |                                                                                    |   |   |   |   |   |   |   |   |
| BITS | FIELD |    |    |    | RW | RESET  | DEFINITION                                                                         |   |   |   |   |   |   |   |   |
| 15:0 | VOL   |    |    |    | RW | 0x8000 | DAP Main Channel Volume<br>0xFFFF = 200%<br>0x8000 (default) = 100%<br>0x0000 = 0% |   |   |   |   |   |   |   |   |

[Table 55. DAP\\_MIX\\_CHAN 0x0122](#) sets the mix channel volume level

**Table 55. DAP\_MIX\_CHAN 0x0122**

| 15   | 14    | 13 | 12 | 11 | 10 | 9      | 8                                                                                 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|------|-------|----|----|----|----|--------|-----------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|
| VOL  |       |    |    |    |    |        |                                                                                   |   |   |   |   |   |   |   |   |
| BITS | FIELD |    |    |    | RW | RESET  | DEFINITION                                                                        |   |   |   |   |   |   |   |   |
| 15:0 | VOL   |    |    |    | RW | 0x0000 | DAP Mix Channel Volume<br>0xFFFF = 200%<br>0x8000 = 100%<br>0x0000 (default) = 0% |   |   |   |   |   |   |   |   |

**Table 56. DAP\_AVC\_CTRL 0x0124**

| 15    | 14            | 13       | 12 | 11   | 10    | 9                                                                                                                                                                                                              | 8 | 7    | 6 | 5             | 4    | 3 | 2 | 1 | 0  |  |
|-------|---------------|----------|----|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|---|---------------|------|---|---|---|----|--|
| RSVD  | RSVD          | MAX_GAIN |    | RSVD |       | LBI_RESPONSE                                                                                                                                                                                                   |   | RSVD |   | HARD_LIMIT_EN | RSVD |   |   |   | EN |  |
|       |               |          |    |      |       |                                                                                                                                                                                                                |   |      |   |               |      |   |   |   |    |  |
| BITS  | FIELD         |          |    | RW   | RESET | DEFINITION                                                                                                                                                                                                     |   |      |   |               |      |   |   |   |    |  |
| 15    | RSVD          |          |    | RO   | 0x0   | Reserved                                                                                                                                                                                                       |   |      |   |               |      |   |   |   |    |  |
| 14    | RSVD          |          |    | RW   | 0x1   | Reserved.                                                                                                                                                                                                      |   |      |   |               |      |   |   |   |    |  |
| 13:12 | MAX_GAIN      |          |    | RW   | 0x1   | Maximum gain that can be applied by the AVC in expander mode.<br>0x0 = 0 dB gain<br>0x1 = 6.0 dB of gain<br>0x2 = 12 dB of gain                                                                                |   |      |   |               |      |   |   |   |    |  |
| 11:10 | RSVD          |          |    | RO   | 0x0   | Reserved                                                                                                                                                                                                       |   |      |   |               |      |   |   |   |    |  |
| 9:8   | LBI_RESPONSE  |          |    | RW   | 0x1   | Integrator Response<br>0x0 = 0 mS LBI<br>0x1 = 25 mS LBI<br>0x2 = 50 mS LBI<br>0x3 = 100 mS LBI                                                                                                                |   |      |   |               |      |   |   |   |    |  |
| 7:6   | RSVD          |          |    | RO   | 0x0   | Reserved                                                                                                                                                                                                       |   |      |   |               |      |   |   |   |    |  |
| 5     | HARD_LIMIT_EN |          |    | RW   | 0x0   | Enable Hard Limiter Mode<br>0x0 = Hard limit disabled. AVC Compressor/Expander is enabled.<br>0x1 = Hard limit enabled. The signal is limited to the programmed threshold. (Signal saturates at the threshold) |   |      |   |               |      |   |   |   |    |  |
| 4:1   | RSVD          |          |    | RO   | 0x0   | Reserved                                                                                                                                                                                                       |   |      |   |               |      |   |   |   |    |  |
| 0     | EN            |          |    | RW   | 0x0   | Enable/disable AVC<br>0x0 = Disable<br>0x1 = Enable                                                                                                                                                            |   |      |   |               |      |   |   |   |    |  |

**Table 57. DAP\_AVC\_THRESHOLD 0x0126**

| 15     | 14     | 13 | 12 | 11     | 10                                                                                                                                                                                                                                                                                                                  | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|--------|--------|----|----|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|
| THRESH |        |    |    |        |                                                                                                                                                                                                                                                                                                                     |   |   |   |   |   |   |   |   |   |   |
| BITS   | FIELD  |    | RW | RESET  | DEFINITION                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |
| 15:0   | THRESH |    | RW | 0x1473 | AVC Threshold Value<br>Threshold is programmable. Use the following formula to calculate hex value:<br>Hex Value = $((10^{(THRESHOLD\_dB/20)}) * 0.636) * 2^{15}$<br>Threshold can be set in the range of 0 dB to -96 dB<br>Example Values:<br>0x1473 = Set Threshold to -12 dB<br>0x0A40 = Set Threshold to -18 dB |   |   |   |   |   |   |   |   |   |   |

**Table 58. DAP\_AVC\_ATTACK 0x0128**

| 15    | 14    | 13 | 12 | 11   | 10    | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|-------|----|----|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|
| RSVD  |       |    |    | RATE |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |
| BITS  | FIELD |    |    | RW   | RESET | DEFINITION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |   |   |   |   |   |   |   |
| 15:12 | RSVD  |    |    | RO   | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |   |   |   |   |
| 11:0  | RATE  |    |    | RW   | 0x28  | <p>AVC Attack Rate</p> <p>This is the rate at which the AVC applies attenuation to the signal to bring it to the threshold level. AVC Attack Rate is programmable. To use a custom rate, use the formula below to convert from dB/S to hex value:</p> $\text{Hex Value} = (1 - (10^{-(\text{Rate\_dBs}/(20 \cdot \text{SYS\_FS})))) \cdot 2^{19}$ <p>where, SYS_FS is the system sample rate configured in CHIP_CLK_CTRL register.</p> <p>Example values:</p> <p>0x28 = 32 dB/s</p> <p>0x10 = 8.0 dB/s</p> <p>0x05 = 4.0 dB/s</p> <p>0x03 = 2.0 dB/s</p> |   |   |   |   |   |   |   |   |   |

**Table 59. DAP\_AVC\_DECAY 0x012A**

| 15    | 14    | 13 | 12 | 11   | 10    | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|-------|----|----|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|
| RSVD  |       |    |    | RATE |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |   |   |   |   |   |   |   |   |
| BITS  | FIELD |    |    | RW   | RESET | DEFINITION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |   |   |   |   |
| 15:12 | RSVD  |    |    | RO   | 0x0   | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |   |   |
| 11:0  | RATE  |    |    | RW   | 0x50  | <p>AVC Decay Rate</p> <p>This is the rate at which the AVC releases the attenuation previously applied to the signal during attack. AVC Decay Rate is programmable. To use a custom rate, use the formula below to convert from dB/S to hex value:</p> $\text{Hex Value} = (1 - (10^{-(\text{Rate\_dBs}/(20 \cdot \text{SYS\_FS})))) \cdot 2^{23}$ <p>where, SYS_FS is the system sample rate configured in CHIP_CLK_CTRL register.</p> <p>Example values:</p> <p>0x284 = 32 dB/s</p> <p>0xA0 = 8.0 dB/s</p> <p>0x50 = 4.0 dB/s</p> <p>0x28 = 2.0 dB/s</p> |   |   |   |   |   |   |   |   |   |

**Table 60. DAP\_COEF\_WR\_B1\_MSB 0x012C**

| 15   | 14    | 13 | 12 | 11 | 10    | 9                                                                                  | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|------|-------|----|----|----|-------|------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|
| MSB  |       |    |    |    |       |                                                                                    |   |   |   |   |   |   |   |   |   |
| BITS | FIELD |    |    | RW | RESET | DEFINITION                                                                         |   |   |   |   |   |   |   |   |   |
| 15:0 | MSB   |    |    | RW | 0x0   | Most significant 16-bits of the 20-bit filter coefficient that needs to be written |   |   |   |   |   |   |   |   |   |



**Table 61. DAP\_COEF\_WR\_B1\_LSB 0x012E**

|      |    |    |    |    |    |   |   |   |   |   |   |     |   |   |   |
|------|----|----|----|----|----|---|---|---|---|---|---|-----|---|---|---|
| 15   | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3   | 2 | 1 | 0 |
| RSVD |    |    |    |    |    |   |   |   |   |   |   | LSB |   |   |   |

| BITS | FIELD | RW | RESET | DEFINITION                                                                          |
|------|-------|----|-------|-------------------------------------------------------------------------------------|
| 15:4 | RSVD  | RO | 0x0   | Reserved                                                                            |
| 3:0  | LSB   | RW | 0x0   | Least significant 4 bits of the 20-bit filter coefficient that needs to be written. |

**Table 62. DAP\_COEF\_WR\_B2\_MSB 0x0130**

| 15   | 14    | 13 | 12 | 11    | 10                                                                                 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|------|-------|----|----|-------|------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|
| MSB  |       |    |    |       |                                                                                    |   |   |   |   |   |   |   |   |   |   |
| BITS | FIELD |    | RW | RESET | DEFINITION                                                                         |   |   |   |   |   |   |   |   |   |   |
| 15:0 | MSB   |    | RW | 0x0   | Most significant 16-bits of the 20-bit filter coefficient that needs to be written |   |   |   |   |   |   |   |   |   |   |

**Table 63. DAP\_COEF\_WR\_B2\_LSB 0x0132**

|      |       |    |    |    |       |                                                                                     |   |   |   |   |   |     |   |   |   |
|------|-------|----|----|----|-------|-------------------------------------------------------------------------------------|---|---|---|---|---|-----|---|---|---|
| 15   | 14    | 13 | 12 | 11 | 10    | 9                                                                                   | 8 | 7 | 6 | 5 | 4 | 3   | 2 | 1 | 0 |
| RSVD |       |    |    |    |       |                                                                                     |   |   |   |   |   | LSB |   |   |   |
|      |       |    |    |    |       |                                                                                     |   |   |   |   |   |     |   |   |   |
| BITS | FIELD |    |    | RW | RESET | DEFINITION                                                                          |   |   |   |   |   |     |   |   |   |
| 15:4 | RSVD  |    |    | RO | 0x0   | Reserved                                                                            |   |   |   |   |   |     |   |   |   |
| 3:0  | LSB   |    |    | RW | 0x0   | Least significant 4 bits of the 20-bit filter coefficient that needs to be written. |   |   |   |   |   |     |   |   |   |

**Table 64. DAP\_COEF\_WR\_A1\_MSB 0x0134**

| 15   | 14    | 13 | 12 | 11    | 10                                                                                 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|------|-------|----|----|-------|------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|
| MSB  |       |    |    |       |                                                                                    |   |   |   |   |   |   |   |   |   |   |
| BITS | FIELD |    | RW | RESET | DEFINITION                                                                         |   |   |   |   |   |   |   |   |   |   |
| 15:0 | MSB   |    | RW | 0x0   | Most significant 16-bits of the 20-bit filter coefficient that needs to be written |   |   |   |   |   |   |   |   |   |   |

**Table 65. DAP\_COEF\_WR\_A1\_LSB 0x0136**

|      |       |    |    |    |       |                                                                                     |   |   |   |   |   |     |   |   |   |
|------|-------|----|----|----|-------|-------------------------------------------------------------------------------------|---|---|---|---|---|-----|---|---|---|
| 15   | 14    | 13 | 12 | 11 | 10    | 9                                                                                   | 8 | 7 | 6 | 5 | 4 | 3   | 2 | 1 | 0 |
| RSVD |       |    |    |    |       |                                                                                     |   |   |   |   |   | LSB |   |   |   |
|      |       |    |    |    |       |                                                                                     |   |   |   |   |   |     |   |   |   |
| BITS | FIELD |    |    | RW | RESET | DEFINITION                                                                          |   |   |   |   |   |     |   |   |   |
| 15:4 | RSVD  |    |    | RO | 0x0   | Reserved                                                                            |   |   |   |   |   |     |   |   |   |
| 3:0  | LSB   |    |    | RW | 0x0   | Least significant 4 bits of the 20-bit filter coefficient that needs to be written. |   |   |   |   |   |     |   |   |   |

**Table 66. DAP\_COEF\_WR\_A2\_MSB 0x0138**

| 15   | 14    | 13 | 12 | 11    | 10                                                                                 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|------|-------|----|----|-------|------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|
| MSB  |       |    |    |       |                                                                                    |   |   |   |   |   |   |   |   |   |   |
| BITS | FIELD |    | RW | RESET | DEFINITION                                                                         |   |   |   |   |   |   |   |   |   |   |
| 15:0 | MSB   |    | RW | 0x0   | Most significant 16-bits of the 20-bit filter coefficient that needs to be written |   |   |   |   |   |   |   |   |   |   |

**Table 67. DAP\_COEF\_WR\_A2\_LSB 0x013A**

|      |    |       |    |    |       |   |                                                                                     |   |   |   |   |     |   |   |   |
|------|----|-------|----|----|-------|---|-------------------------------------------------------------------------------------|---|---|---|---|-----|---|---|---|
| 15   | 14 | 13    | 12 | 11 | 10    | 9 | 8                                                                                   | 7 | 6 | 5 | 4 | 3   | 2 | 1 | 0 |
| RSVD |    |       |    |    |       |   |                                                                                     |   |   |   |   | LSB |   |   |   |
|      |    |       |    |    |       |   |                                                                                     |   |   |   |   |     |   |   |   |
| BITS |    | FIELD |    | RW | RESET |   | DEFINITION                                                                          |   |   |   |   |     |   |   |   |
| 15:4 |    | RSVD  |    | RO | 0x0   |   | Reserved                                                                            |   |   |   |   |     |   |   |   |
| 3:0  |    | LSB   |    | RW | 0x0   |   | Least significant 4 bits of the 20-bit filter coefficient that needs to be written. |   |   |   |   |     |   |   |   |

## TYPICAL APPLICATIONS

## INTRODUCTION

Typical connections are shown in the following application diagrams. For new designs, and for either the 20 QFN or 32 QFN part, an external VDDD power supply connection is required along with a 0.1  $\mu\text{F}$  cap connection from VDDD to ground.

**CPFILT Note:** The CPFILT cap value is 0.1  $\mu\text{F}$ . If both VDDIO and VDDA are  $\leq 3.0\text{ V}$ , the CPFILT pin must be

connected to a 0.1  $\mu\text{F}$  cap to GND. If either is  $> 3.0\text{ V}$ , the CPFILT cap MUST NOT be placed.

**HP\_VGND Note:** Do not connect HP\_VGND to system ground, even when unused. This is a virtual ground (DC voltage) that should never connect to an actual "0 Volt ground". Use the widest, shortest trace possible for the HP\_VGND.

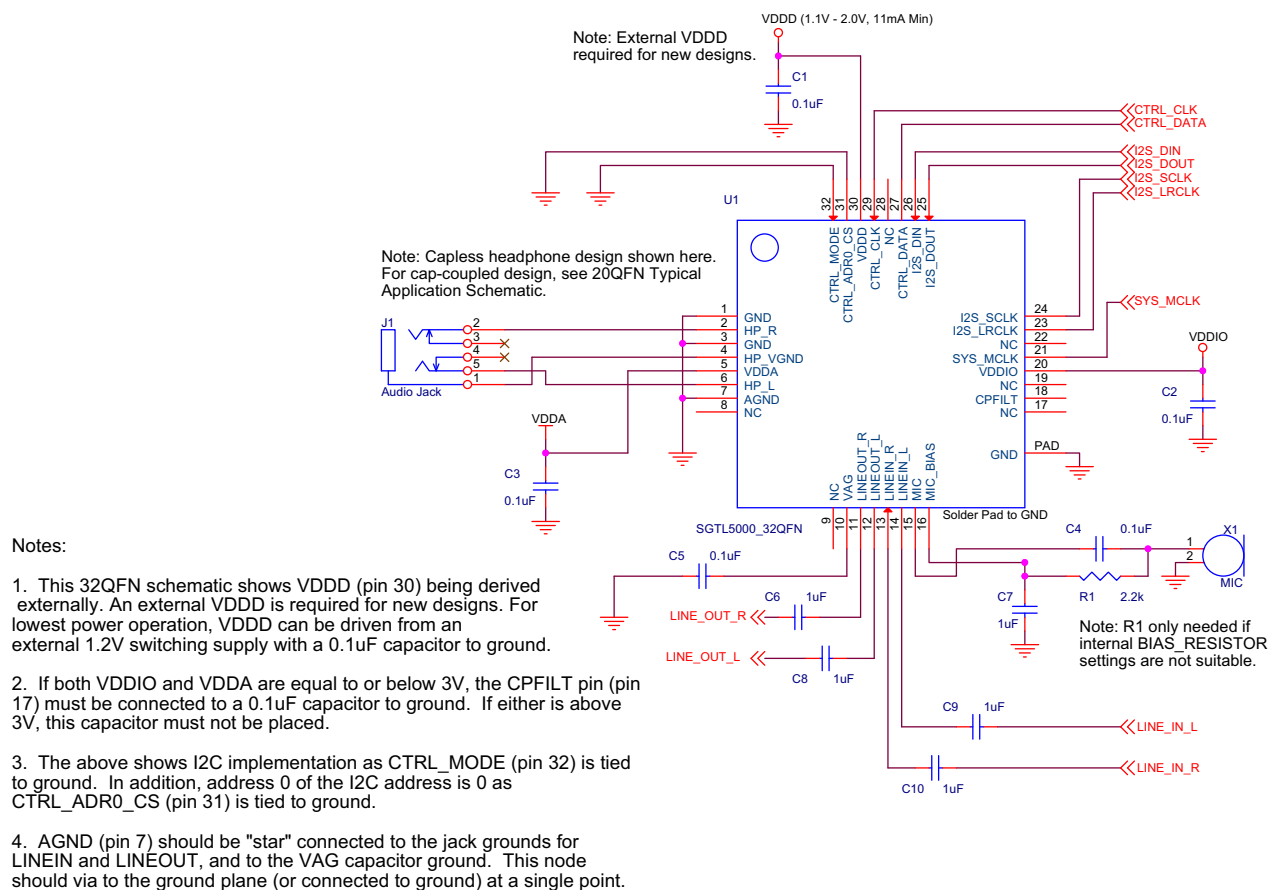
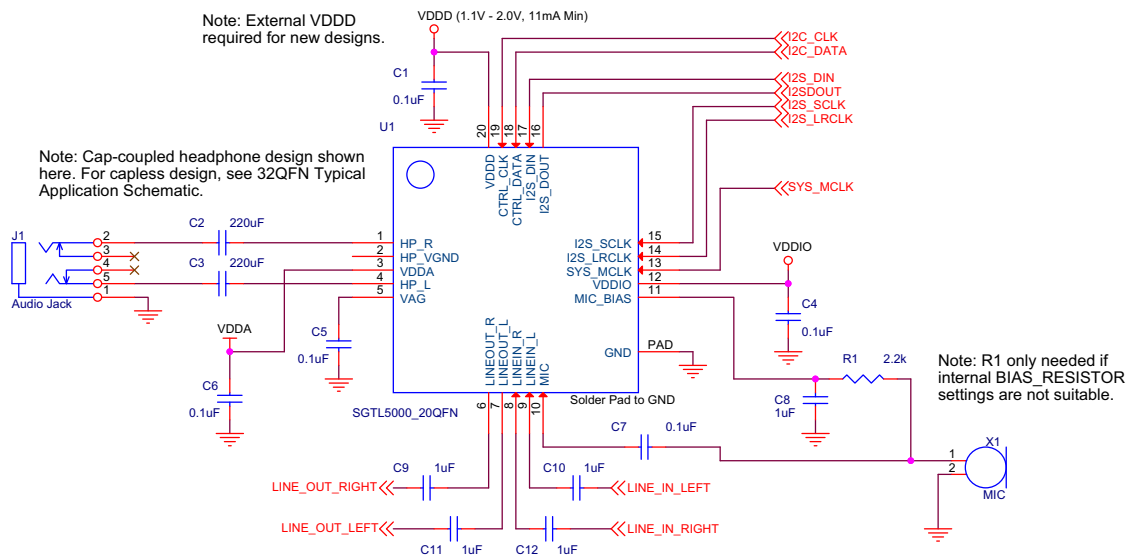


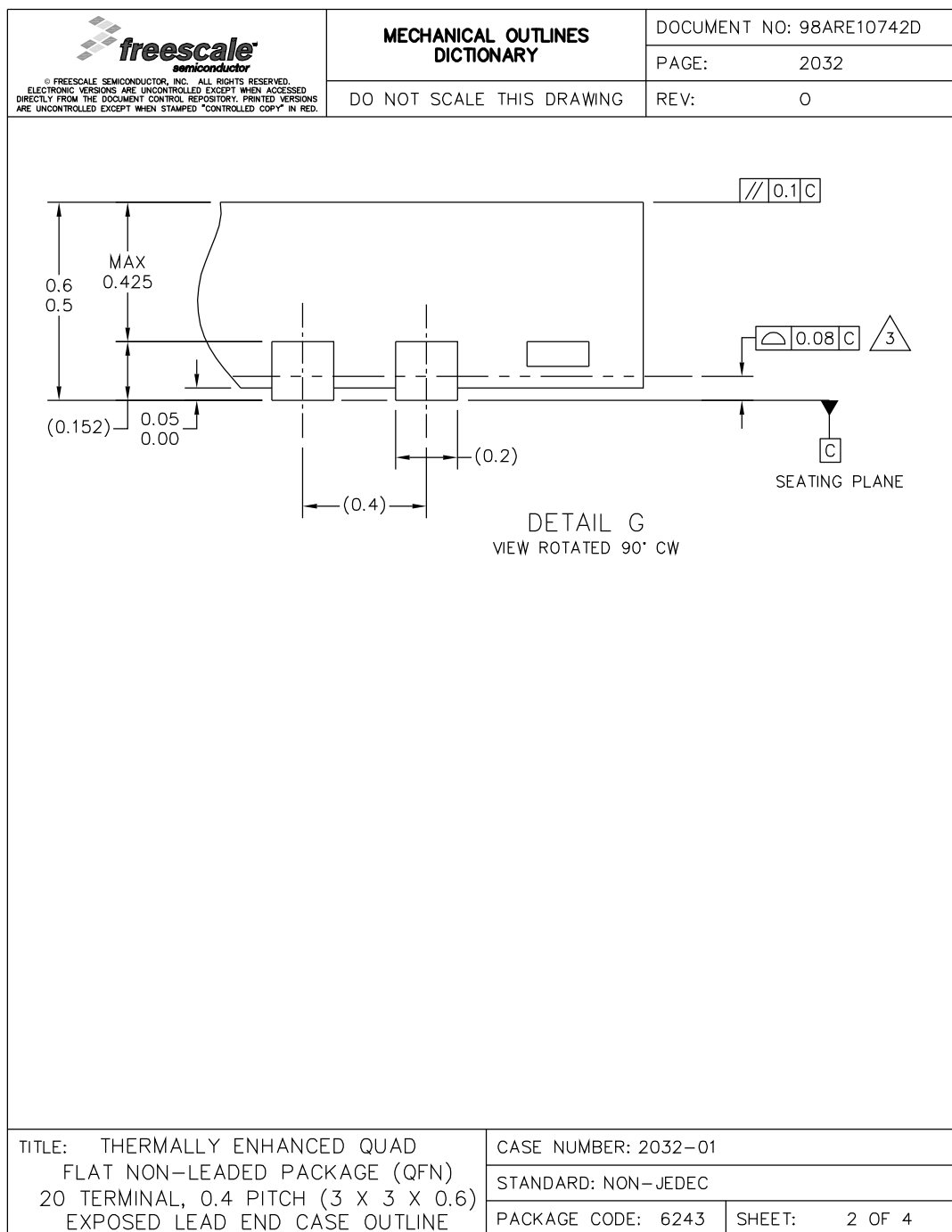
Figure 19. 32 QFN Typical Application Schematic



**Note: Bottom PAD/FLAG/Paddle MUST be connected to ground.**

**Figure 20. 20 QFN Typical Application Schematic**





**EP SUFFIX**  
20-PIN  
98ARE10742D  
REVISION 0

|                                                                                                                                                                                                                                                                                                                                                                                  |                                           |                          |      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------|------|
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|                                                                                                                                                                                                                                                                                                                                                                                  | DO NOT SCALE THIS DRAWING                 | PAGE:                    | 2032 |
|                                                                                                                                                                                                                                                                                                                                                                                  |                                           | REV:                     | 0    |

NOTES:

1.

ALL DIMENSIONS ARE IN MILLIMETERS.

2.

INTERPRET DIMENSIONS AND TOLERANCES PER ASME Y14.5M-1994.

3.

COPLANARITY APPLIES TO LEADS, CORNER LEADS, AND DIE ATTACH PAD.

4.

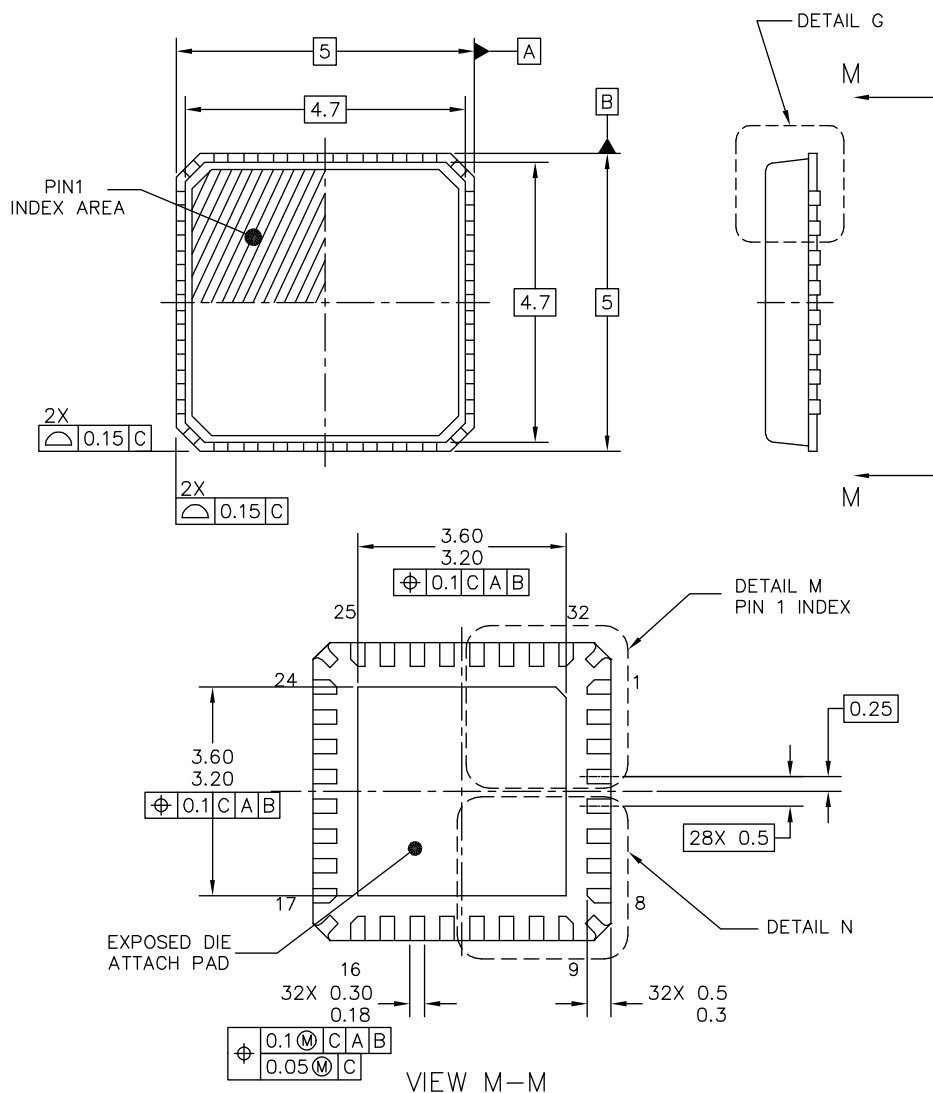
DIMENSION APPLIES TO PLATED TERMINAL AND IS MEASURED BETWEEN 0.20 MM AND 0.25 MM FROM TERMINAL TIP.

5.

MIN. METAL GAP SHOULD BE 0.2MM.

|                                                                                                                                          |                      |               |
|------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------|
| TITLE: THERMALLY ENHANCED QUAD<br>FLAT NON-LEADED PACKAGE (QFN)<br>20 TERMINAL, 0.4 PITCH (3 X 3 X 0.6)<br>EXPOSED LEAD END CASE OUTLINE | CASE NUMBER: 2032-01 |               |
|                                                                                                                                          | STANDARD: NON-JEDEC  |               |
|                                                                                                                                          | PACKAGE CODE: 6243   | SHEET: 3 OF 4 |

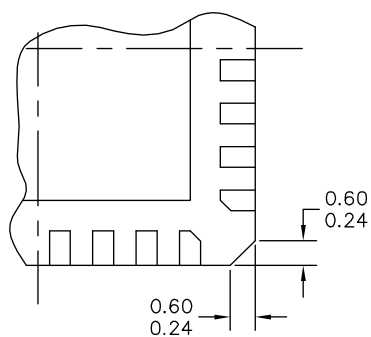
**EP SUFFIX**  
20-PIN  
98ARE10742D  
REVISION 0



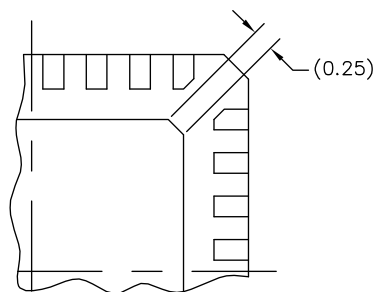
|                                                                                                       |  |                               |  |                            |  |
|-------------------------------------------------------------------------------------------------------|--|-------------------------------|--|----------------------------|--|
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| TITLE: THERMALLY ENHANCED QUAD<br>FLAT NON-LEADED PACKAGE (QFN)<br>32 TERMINAL, 0.5 PITCH (5 X 5 X 1) |  | DOCUMENT NO: 98ARE10739D      |  | REV: 0                     |  |
|                                                                                                       |  | CASE NUMBER: 2029-01          |  | 15 MAY 2008                |  |
|                                                                                                       |  | STANDARD: JEDEC MO-220 VHHD-5 |  |                            |  |

**FC SUFFIX**  
32-PIN  
98ARE10739D  
REVISION 0

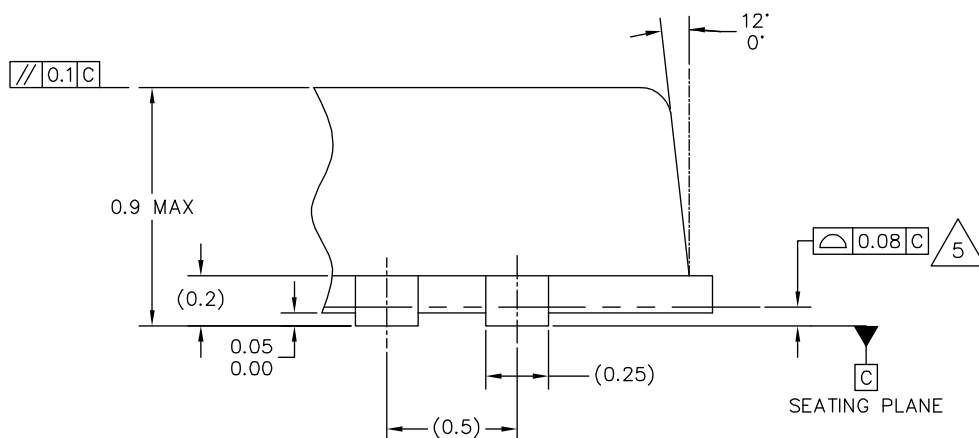




DETAIL N  
CORNER CONFIGURATION OPTION



DETAIL M  
PREFERRED BACKSIDE PIN 1 INDEX



DETAIL G  
VIEW ROTATED 90° CW

|                                                                                                       |  |                               |  |                            |  |
|-------------------------------------------------------------------------------------------------------|--|-------------------------------|--|----------------------------|--|
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| TITLE: THERMALLY ENHANCED QUAD<br>FLAT NON-LEADED PACKAGE (QFN)<br>32 TERMINAL, 0.5 PITCH (5 X 5 X 1) |  | DOCUMENT NO: 98ARE10739D      |  | REV: 0                     |  |
|                                                                                                       |  | CASE NUMBER: 2029-01          |  | 15 MAY 2008                |  |
|                                                                                                       |  | STANDARD: JEDEC MO-220 VHHD-5 |  |                            |  |

**FC SUFFIX**  
32-PIN  
98ARE10739D  
REVISION 0

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994.
3. THE COMPLETE JEDEC DESIGNATOR FOR THIS PACKAGE IS: HF-PQFN.
4. DIMENSIONS OF OPTIONAL FEATURES ARE FOR REFERENCE ONLY.
5. COPLANARITY APPLIES TO LEADS, AND DIE ATTACH PAD.
6. MIN METAL GAP SHOULD BE 0.2MM.

|                                                                                                       |                               |  |                            |  |
|-------------------------------------------------------------------------------------------------------|-------------------------------|--|----------------------------|--|
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| TITLE: THERMALLY ENHANCED QUAD<br>FLAT NON-LEADED PACKAGE (QFN)<br>32 TERMINAL, 0.5 PITCH (5 X 5 X 1) | DOCUMENT NO: 98ARE10739D      |  | REV: 0                     |  |
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|                                                                                                       | STANDARD: JEDEC MO-220 VHDD-5 |  |                            |  |

**FC SUFFIX**  
32-PIN  
98ARE10739D  
REVISION 0

## REVISION HISTORY

| REVISION | DATE   | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.0      | 6/2010 | <ul style="list-style-type: none"> <li>Conversion from the old Freescale form and style to the current version. No existing content has been added, altered, or removed.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 4.0      | 9/2010 | <ul style="list-style-type: none"> <li>Corrected Pin 4 explanation (32-pin package) and added Pin 3 (32-Pin package) to Table 1.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.0      | 5/2013 | <ul style="list-style-type: none"> <li>Corrected <a href="#">LINEOUT - 100 dB SNR (-60 dB input) and -85 dB THD+N (VDDIO = 3.3 V)</a> in features</li> <li>Added note for HP_VGND and CPFILT in pin definition table</li> <li>Moved <a href="#">Recommended Operating Conditions</a> to separate table</li> <li>Added <a href="#">Input/Output Electrical Characteristics</a></li> <li>Corrected <a href="#">LINEIN Input Level</a> from 0.75 to 0.57</li> <li>Corrected Table 7 <a href="#">Test Conditions unless otherwise noted: VDDIO = 1.8 V, VDDA = 1.8 V, TA = 25 °C, Slave mode, fS = 48 kHz, MCLK = 256 fS, 24 bit input.</a></li> <li>Added note for HP_VGND and CPFILT to Typical Applications introduction</li> <li>Corrected pin nomenclature as required for consistency</li> <li>Clarified Bits 3:0 in <a href="#">Figure 27</a></li> <li>Corrected pin name in <a href="#">Figure 3</a> and <a href="#">Table 1</a></li> <li>Corrected address name in <a href="#">Figure 6</a>, I2C, SPI</li> <li>Changed limits on <a href="#">LINEOUT Output level</a></li> <li>Changed 0x00 = sets to 12 dB to 11.75 dB, and deleted "To convert dB to hex value, use Hex Value = 4* dB value + 47" on tables 49, 50, 51, 52 and 53.</li> <li>Revised back page. Updated document properties. Added SMARTMOS sentence to first paragraph.</li> <li>Added comment for "new designs" where applicable</li> <li>Corrected pin designations in the <a href="#">Pin Connections</a> section</li> <li>Changed limits and conditions for <a href="#">LINEOUT Output level</a> and <a href="#">LINEOUT Output level</a></li> <li>Added two new application diagrams in <a href="#">Typical Applications</a> section</li> </ul> |



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